TAMIL NADU AGRICULTURAL UNIVERSITY

B.Sc. (Hons.) Agriculture — 2025 Syllabus — Third Semester

SAC 201

SOIL RESOURCE INVENTORY AND GEOINFORMATICS

Complete Lecture-wise Theory Study Material

All 17 lectures • Definitions and concepts • Tables and classification keys • Worked numerical examples • Text diagrams • Exam-focus sections • Glossary and references

Prepared strictly to the prescribed lecture sequence. Indian and Tamil Nadu examples are used throughout. Sources are listed as plain references at the end of the book.

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How to Use This Study Material

This book follows the SAC 201 syllabus lecture by lecture. Each lecture is one numbered chapter, titled exactly as the syllabus line, and every sub-topic in that line is expanded in the same order. Nothing in the syllabus list has been dropped, merged or reordered.

What every lecture chapter contains

- **Headed sections** that take each sub-topic in turn, with a plain-English definition first, then the detail.
- **At least one table, worked example or diagram** in every lecture, so no lecture is only prose. Classification keys, comparisons and numerical examples are given as tables; processes and structures are given as text diagrams.
- **Indian and Tamil Nadu examples** wherever the concept has a local illustration - Tamil Nadu soil series and soil orders, the Cauvery delta, cadastral maps, Bhuvan and Bhaskara, the IRS satellite series, and the Tamil Nadu soil survey.
- **An Exam focus section** at the end of each lecture: key points to memorise, short-answer questions with model answers, and long-answer questions with model answer outlines.

How to use it for revision

1. Read the definitions in a lecture once. The definition sentence is usually the first line under each heading, and it is what an examiner looks for.
2. Learn the tables by drawing them out from memory. Classification tables (land capability, irrigability, Storie index, suitability, survey types) carry a large share of the marks.
3. Work the numerical examples with a pen. Storie index, productivity potential, scale and area from a toposheet, and Simpson-type area calculations are common short numerical questions.
4. Copy each text diagram once by hand. Diagrams of the soil profile and pedon, catena, the remote sensing process, GIS components and GPS segments are frequently asked as draw-and-explain questions.
5. Answer the Exam focus questions without looking, then check against the model answers and outlines given.

Note on figures. Where a published figure varies between sources, a standard range is given rather than a single false-precise number. Where a value is not fixed by convention, the book says so instead of inventing one.

Syllabus map - the seventeen lectures in order

No.	Lecture title as prescribed	Chapter
1	Soil resource inventory - definition, principles, concepts; soil systematics - pedon and polypedon, control section and soil correlation	1
2	Standard soil survey - scope and objectives; fundamental and applied	2
3	Soil mapping units: soil series, soil association, soil complex, variants, inclusions and miscellaneous land types	3
4	Soil survey concepts - methods of soil survey: base maps, traversing - grid survey and free survey	4
5	Types of soil survey: detailed, reconnaissance, detailed-reconnaissance and semi-detailed soil survey	5
6	Interpretative groupings of soil: Land Capability Classification and Fertility Capability Classification	6

No.	Lecture title as prescribed	Chapter
7	Interpretative groupings of soil: Land Irrigability Classification, Storie index and productivity potential	7
8	Land Suitability Classification for field crops, horticultural crops and forest trees	8
9	Mid Semester Examination	9
10	Land Use Planning: concepts and objectives; land use planning for tropical, subtropical and temperate regions	10
11	Remote Sensing - definition and components; remote sensing platforms and sensors	11
12	Electromagnetic Radiation (EMR) - principle; interaction of EMR in the atmosphere	12
13	Interaction of EMR with earth surface features (soil, water and vegetation)	13
14	Introduction to basemaps and satellite data products (cadastral map, toposheet, aerial photograph and satellite images) and image interpretation elements	14
15	GIS - definition and components; raster and vector data models	15
16	GPS and GNSS - functions, segments, working principle and GPS errors	16
17	Geoinformatics in agricultural applications (soil and crop studies)	17

Table 0.1 The seventeen prescribed lectures, mapped to the chapters of this book.

How the marks are usually distributed

The paper is built from the same seventeen lecture lines. Lectures that carry a classification key, a formula or a process diagram (soil mapping units, methods and types of survey, the interpretative groupings, land suitability, remote sensing, GIS and GPS) are the ones most often set as long-answer questions. Definitions and small comparisons from every lecture are set as short-answer questions. The Exam focus section of each chapter is written to match that pattern.

Everything in this book is written to be reproduced in an answer book: short sentences, one idea per line, and definitions that begin with the term being defined.

Lecture 1 Soil Resource Inventory and Soil Systematics

Soil is a finite, non-renewable resource at human time scales. A soil formed over thousands of years can be destroyed by erosion in a few seasons. Because of this, the first task of any soil programme is to find out what soils exist, where they are, how much of each there is, and how good each is for different uses. That task is soil resource inventory. This lecture defines the subject, states its principles and concepts, and then explains the systematic way in which soils are named and grouped.

1.1 Soil resource inventory - definition

Soil resource inventory is the systematic survey, identification, classification, mapping and evaluation of the soils of an area, carried out to record their kind, extent, distribution, properties and potential for use. It converts the soil body on the ground into information on a map and in a report.

A shorter exam definition: soil resource inventory is the scientific inventory of the soil resources of a region, giving the location and extent of each kind of soil together with its characteristics and its suitability for various land uses.

What the inventory must answer

- **What** soils are present - the kind and taxonomic identity of each soil.
- **Where** each soil occurs - its position on the landscape and on the map.
- **How much** - the area of each soil, usually in hectares and as a percentage of the surveyed area.
- **Why** it occurs there - the soil-forming factors and processes behind its distribution.
- **How good** it is - the capability, irrigability and suitability of the soil for crops, horticulture, forestry and other uses.
- **How to manage** it - the limitations present and the conservation and management measures needed.

1.2 Why soil resource inventory is needed

- **Land use planning.** No plan for crops, irrigation, forestry or industry can be made without knowing the soils of the area.
- **Irrigation and drainage design.** Canal alignment, command area development and drainage design depend on soil depth, texture and slope.
- **Fertiliser and crop advisory.** Soil test and fertiliser recommendation programmes are built on soil maps.
- **Soil and water conservation.** Erosion-prone soils are identified and treated first.
- **Watershed development.** The watershed is the modern unit of planning, and soil maps are its base layer.
- **Taxation and land records.** Soil quality is one basis of land revenue classification in several states.
- **Research and teaching.** Research stations, demonstration plots and student farms are sited on the basis of soil maps.
- **Monitoring change.** A baseline inventory allows later surveys to detect degradation, salinity or waterlogging.

1.3 Principles of soil resource inventory

The principles below are the working rules of every soil survey. They explain why a survey is done the way it is done.

1. **Soils are natural bodies, not random material.** Each soil is a product of climate, organisms, relief, parent material and time. Because the factors repeat, soils repeat in a predictable pattern on the landscape.

2 **Soils occur in an orderly pattern on the landscape.** A soil found on an upper slope in one part of a region will be found on a similar upper slope elsewhere in the same region. This is the basis of free survey.

3 **Soil boundaries are real but gradational.** Most soil boundaries are not sharp lines but zones of change. The mapper draws a line through the zone of greatest change.

4 **The soil individual can be defined and classified.** Soil is studied through a three-dimensional unit, the pedon, and grouped through the polypedon.

5 **The smallest area that can be shown depends on map scale.** At a given scale there is a minimum legible area, so small bodies of soil must be shown as inclusions or complexes.

6 **Survey intensity must match the purpose.** A reconnaissance survey cannot serve as a detailed survey, and the reverse is a waste of money.

7 **Soil properties are inferred from observable features.** Depth, colour, texture, structure, mottles, horizons and boundary are observed; drainage, fertility and behaviour are inferred from them.

8 **Classification precedes mapping.** The kinds of soil are first recognised and defined, then mapped and correlated.

9 **Every survey is a hypothesis tested in the field.** The soil-landscape model built from the first traverses is checked, corrected and refined as the survey proceeds.

10 **The inventory must be usable.** A survey that is accurate but not interpreted for the user has not completed its purpose.

1.4 Concepts used in soil resource inventory

Soil individual, soil body and soilscape

- **Soil individual.** The smallest naturally occurring, three-dimensional body of soil that can be described and classified as one kind of soil. In practice it is studied through the pedon and mapped through the polypedon.
- **Soil body.** A general term for a continuous mass of soil having properties that place it in one taxon.
- **Soilscape.** The pattern of soils across a landscape, considered as a whole. It is the soil equivalent of the word 'landscape'.

Soil profile, solum and horizon

- **Soil profile.** A vertical section of the soil from the surface down to the parent material, showing all its horizons. It is the basic unit of description.
- **Solum.** The upper part of the profile in which soil formation has been most active, usually the A and B horizons together.
- **Horizon.** A layer within the profile, roughly parallel to the surface, differing from the layers above and below in one or more properties such as colour, texture, structure or organic matter.
- **Control section.** The defined depth interval of the profile that is actually used for a given purpose, such as family classification or land evaluation. It is explained in section 1.8.

1.5 Soil systematics - definition

Soil systematics is the branch of soil science that deals with the naming, describing, classifying and correlating of soils. It provides the vocabulary and the hierarchy by which any soil can be placed in a known group and compared with soils elsewhere.

Soil systematics has three parts:

- **Soil taxonomy** - the classification of soils into a hierarchy of categories on the basis of measurable properties.

- **Soil survey and mapping** - the recognition and delineation of soils on the ground and on maps.
- **Soil correlation** - the process of ensuring that soils given the same name in different places really are the same, and that soils given different names really are different.

1.6 The pedon

Definition. A pedon is the smallest three-dimensional body of soil that can be recognised as a soil individual and that contains all the horizons and properties needed to classify the soil. It is the basic sampling and taxonomic unit of soil.

Size. The pedon is commonly described as being about 1 to 10 square metres in cross-sectional area, with a vertical dimension extending from the surface down to the lower limit of the soil or to the base of the solum, usually about 1 to 2 metres. Where the horizon boundaries are irregular, the pedon is large enough that such irregularities are treated as part of the normal variability of the soil.

Shape. The pedon is roughly hexagonal in cross-section - a shape wide enough to include the natural variation of the horizons but small enough to be studied as one unit. Its vertical extent is the full depth of the soil.

Kinds of pedon

- **Mineral pedon.** A pedon in mineral soil, that is, soil whose organic matter content is below the limits set for organic soil. It is the usual case in cultivated land.
- **Organic pedon.** A pedon in organic soil, that is, soil developed mainly from plant remains, such as peat. Such soils occur in the coastal and marshy parts of Tamil Nadu and Kerala.
- **Artificial pedon.** A pedon described in material transported and deposited by human activity, such as an excavated tank bed, a mine spoil or a filled site. It is classified only if the material is thick enough and old enough to behave as soil.

1.7 The polypedon

Definition. A polypedon is a contiguous group of pedons, or a collection of contiguous similar pedons, that together form a body of soil large enough to be shown on a soil map as one mapping unit. It is the soil individual at mapping scale.

Size. A polypedon has a cross-sectional area of roughly 1 to 10 hectares, though the figure varies with the scale and intensity of the survey. Two or more polypedons of the same kind of soil, separated by other soils, together form the area of that soil on the map.

The relationship is simple: many pedons of the same kind make one polypedon, and the polypedon is what is delineated as a soil mapping unit. A polypedon is bounded by other polypedons of different soils, or by non-soil areas such as rock outcrop, water or built-up land.

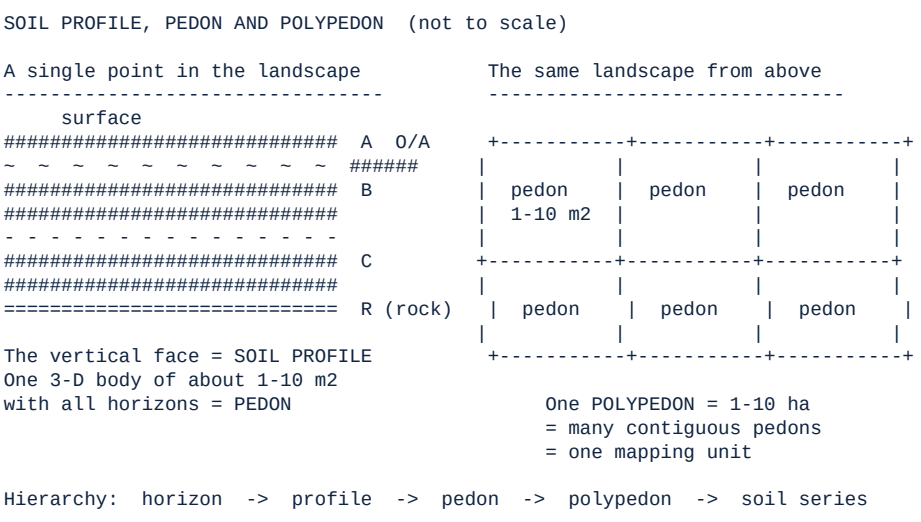


Fig. 1.1 Soil profile, pedon and polypedon, and the hierarchy from horizon to mapping unit.

Feature	Soil profile	Pedon	Polypedon
Nature	A vertical section of soil	A three-dimensional soil body	A group of contiguous similar pedons
Dimensions	Depth only; width not defined	About 1-10 m2 area; depth to base of solum	About 1-10 ha area
Role	The unit of description	The unit of sampling and classification	The unit of mapping
Boundaries	Not applicable	Set by the extent of horizon variation	Set by the boundary with a different soil or non-soil area
Use in survey	Described at a pit or auger point	Defines the taxon of the soil	Delineated as a mapping unit

Table 1.1 Comparison of soil profile, pedon and polypedon.

1.8 The control section

Definition. The control section is the specific depth interval of the soil profile that is used as the standard basis for a particular classification or evaluation decision. Only the properties within that depth interval are considered for that decision.

The control section exists because different purposes need different depths. A shallow-rooted crop and a deep-rooted tree are affected by different parts of the profile. To make classification consistent between one soil and another, the depth interval used for each decision is fixed in advance. This is what the control section does.

Commonly used control sections

Purpose	Control section (depth)	Properties considered
Particle-size class (texture) of a soil family	25 cm to 100 cm below the surface	Weighted average of clay, silt and sand
Mineralogy class of a soil family	25 cm to 100 cm	Mineral composition of the fine-earth fraction
Cation-exchange activity class (CEC)	25 cm to 100 cm	Cation-exchange capacity per 100 g clay
Reaction class (pH) of a family	25 cm to 100 cm	Weighted average pH
Depth class of a soil (very shallow to very deep)	Surface to the root-limiting layer	Depth to hard rock, dense layer or water table
Land capability and land suitability for crops	Surface to about 100-150 cm, depending on the crop	Depth, texture, slope, erosion, drainage, salinity

Table 1.2 Control sections used for common soil family and land evaluation decisions. Depth limits are the standard published ranges; exact limits vary slightly between classification systems.

Remember the key pair. The control section for texture, mineralogy, CEC and reaction of a soil family is 25-100 cm. Depth class, by contrast, is judged from the surface down to the root-limiting layer. A soil can be deep for one purpose and shallow for another.

1.9 Soil correlation

Definition. Soil correlation is the process of comparing soils described in different places, or by different surveyors, and deciding whether they are the same soil or different soils, so that each kind of soil carries one agreed name everywhere it occurs.

Correlation protects the whole classification. Without it, two different soils may be given the same series name in two districts, or one soil may be given two names. Either error makes the soil map unreliable and the interpretation wrong.

Levels of correlation

- **Correlation within a survey area.** The surveyor checks that the delineations in different parts of the same survey area are consistent, usually before the map is finalised.
- **Correlation between survey areas within a state.** State-level correlation committees ensure that series defined in one district are named and defined consistently with series in another district. In Tamil Nadu this work is coordinated by the state soil survey organisation.
- **Correlation between states.** Regional correlation committees compare series across state boundaries so that a soil crossing a boundary keeps one name.
- **National correlation.** The national correlation committee examines new series proposed by states and accepts, modifies or rejects the names, so that the national series register stays consistent.
- **International correlation.** At the level of soil orders and great groups, Indian soils are correlated with the international systems such as Soil Taxonomy and the World Reference Base, so that Indian soils can be compared with soils in other countries.

What correlation actually compares

1. The **horizon sequence** - whether the same horizons occur in the same order and thickness.
2. The **diagnostic horizons and properties** - such as an argillic horizon, a mollic epipedon or a cambic horizon, and their depth limits.
3. The **control-section properties** - texture, mineralogy, CEC and reaction over 25-100 cm.
4. The **depth class** and the nature of the root-limiting layer.
5. The **colour, structure, mottling and drainage** class.
6. The **soil moisture and temperature regimes** of the area.
7. The **landscape position and parent material**, which explain why the soil is where it is.

1.10 Tamil Nadu context

Tamil Nadu has been surveyed systematically since the 1950s, first by the state Department of Agriculture and later through the National Bureau of Soil Survey and Land Use Planning and the Tamil Nadu Agricultural University. Well-known soil series that appear in Tamil Nadu soil survey literature and in TNAU teaching material include Coimbatore, Palladam, Peelamedu, Irugur, Vellore, Hosur and Anamalai series, among many others. Each is a polypedon-scale body of soil defined by its horizon sequence, texture, colour and depth, and each is used as a mapping unit in the districts where it occurs.

At the order level, the soils of Tamil Nadu fall mainly in the orders Alfisols, Inceptisols, Entisols and Vertisols, with smaller areas of Mollisols and Ultisols, and Aridisols in the drier rain-shadow tracts. The red soils of the uplands are largely Alfisols and Ultisols; the delta and valley alluvium are largely Inceptisols and Entisols; the black cotton soils of the Coimbatore and Madurai plains are Vertisols.

Tamil Nadu soil group	Typical order	Where it occurs	Common use
Red sandy and red loamy soils	Alfisols, Ultisols	Uplands and plateaus of the interior districts	Millets, pulses, groundnut
Black cotton soils	Vertisols	Coimbatore, Madurai, Tirunelveli plains	Cotton, sorghum, pulses
Delta alluvial soils	Inceptisols, Entisols	Cauvery delta of Thanjavur, Thiruvarur, Nagapattinam	Paddy, sugarcane
Coastal and saline soils	Entisols, some Aridisols	Coastal tracts of Ramanathapuram, Nagapattinam	Paddy, salt-tolerant crops
Lateritic and hilly soils	Ultisols, Inceptisols	Nilgiris, Shevaroys, Javadhu hills	Tea, coffee, horticulture, forestry

Table 1.3 Broad soil groups of Tamil Nadu and their typical soil orders. The correspondence is typical, not exclusive; local exceptions occur.

Exam focus

Key points to remember

- Soil resource inventory = systematic survey, identification, classification, mapping and evaluation of the soils of an area, giving kind, extent, distribution, properties and potential.
- Its principles rest on soil being a natural body that repeats in an orderly way on the landscape, on boundaries being gradational, and on survey intensity matching purpose.
- A pedon is the smallest 3-D soil individual, about 1-10 m2, containing all horizons needed for classification.
- A polypedon is a contiguous group of similar pedons, about 1-10 ha, and is the mapping unit.
- The control section is the standard depth interval used for a classification decision; for texture, mineralogy, CEC and reaction of a soil family it is 25-100 cm.
- Soil correlation is the process of deciding whether soils in different places are the same or different, and giving each kind of soil one agreed name.
- Correlation operates at survey, state, interstate, national and international levels.

Short-answer questions (with model answers)

Q1. Define soil resource inventory.

Model answer. Soil resource inventory is the systematic survey, identification, classification, mapping and evaluation of the soils of an area to record their kind, extent, distribution, properties and potential for use.

Q2. Distinguish between a pedon and a polypedon.

Model answer. A pedon is the smallest three-dimensional body of soil that can be classified, about 1-10 m2 in area, and is the unit of sampling and classification. A polypedon is a group of contiguous similar pedons, about 1-10 ha in area, and is the unit of mapping. Many pedons of the same kind make one polypedon.

Q3. What is a control section and why is it needed?

Model answer. The control section is the fixed depth interval of the profile used for a particular classification or evaluation decision. It is needed so that the same depth range is compared in every soil, making classification consistent. For texture, mineralogy, CEC and reaction of a soil family the control section is 25-100 cm.

Q4. Why is soil correlation necessary?

Model answer. Soil correlation ensures that soils given the same name in different places are genuinely the same soil and that different soils are not given the same name. Without it, the same series name may be used for two different soils or one soil may carry two names, which makes the soil map and its interpretations unreliable.

Long-answer questions (with model answer outlines)

Q1. Explain the principles of soil resource inventory.

Outline.

- Define soil resource inventory in one sentence.
- Principle 1: soil is a natural body produced by climate, organisms, relief, parent material and time, so it is predictable.
- Principle 2: soils repeat in an orderly pattern on the landscape, which is the basis of free survey.
- Principle 3: soil boundaries are gradational, not sharp lines.
- Principle 4: the soil individual is definable through the pedon and groupable through the polypedon.

- Principle 5: the smallest mappable area depends on map scale, so small bodies become inclusions or complexes.
- Principle 6: survey intensity must match the purpose.
- Principle 7: classification precedes mapping.
- Principle 8: the survey model is a hypothesis, tested and refined in the field.
- Principle 9: the inventory must be interpreted before it is useful.
- Close with one Tamil Nadu illustration, e.g. the delta survey.

Q2. Describe soil systematics, its components, and the place of the pedon and control section in it. Outline.

- Define soil systematics: naming, describing, classifying and correlating soils.
- Component 1: soil taxonomy - hierarchy of categories on measurable properties.
- Component 2: soil survey and mapping - delineation on the ground and on maps.
- Component 3: soil correlation - ensuring names mean the same thing everywhere.
- Explain the pedon: definition, size, shape, kinds (mineral, organic, artificial).
- Explain the polypedon and the pedon-to-polypedon relationship.
- Explain the control section, its depth ranges and its purpose.
- Explain the levels of correlation.
- Conclude with a Tamil Nadu example such as a red soil series on an upland and the delta alluvium.

Lecture 2 Standard Soil Survey - Scope and Objectives

Lecture 1 defined what a soil inventory is. This lecture explains the standard soil survey through which that inventory is actually made - what it covers, what it is meant to achieve, and how its objectives are divided into fundamental (knowledge for its own sake) and applied (knowledge for a use).

2.1 Standard soil survey - definition

Standard soil survey is the systematic examination, description, classification and mapping of soils of an area according to a uniform, accepted procedure, so that the resulting soil map and report describe the soils in a form that can be compared with surveys of other areas and used for land use planning.

The word **standard** is important. It means the survey follows an agreed national procedure - the same kinds of observation, the same depth limits, the same definition of a mapping unit, the same classification and correlation rules - whether the survey is done in Tamil Nadu or in Punjab. Only then are the maps comparable and the national soil resource data consistent.

2.2 Scope of a standard soil survey

The scope of a soil survey is the range of ground it covers - the features it studies and the outputs it delivers.

Features studied

- **Morphology of the profile** - horizon sequence, depth, colour, texture, structure, consistence, mottles, pores, roots and boundary.
- **Physical properties** - texture, structure, bulk density, infiltration, water-holding capacity, depth to a root-limiting layer.
- **Chemical properties** - pH, electrical conductivity, organic carbon, cation-exchange capacity, base saturation, available nutrients.
- **Site and land features** - slope, erosion, drainage, water table, flooding, stoniness, rockiness, land use and vegetation.
- **Distribution of soils** - the pattern of soils on the landscape and their relative extent.

Outputs delivered

- A **soil map** at a stated scale, showing the soil mapping units.
- A **soil survey report** describing each mapping unit, its properties and its distribution.
- A **soil legend** that names each mapping unit and links it to the classification and to the mapping symbols.
- An **interpretation report** giving capability, irrigability, suitability and management recommendations.
- Supporting **data sheets**, profile descriptions and analytical results.

Scope in one line. A standard soil survey covers the whole area, studies the soil body from the surface to the parent material, records both soil and site features, and delivers a map, a legend, a report and an interpretation.

2.3 Objectives of a standard soil survey

Objectives are grouped in two ways in the syllabus: **fundamental** objectives, which build knowledge, and **applied** objectives, which serve a use. Both are present in every standard survey, but the weight given to each depends on the purpose for which the survey was ordered.

Fundamental objectives

- **To know the soils of the area.** To identify and describe every kind of soil present, and to record its properties accurately.

- **To classify the soils.** To place each soil in the national classification at the appropriate category, so that it can be named and compared.
- **To map the distribution.** To show the location, pattern and extent of each soil on a map at a stated scale.
- **To understand soil genesis.** To explain why each soil occurs where it does, in terms of parent material, relief, climate, organisms and time.
- **To establish soil-landscape relationships.** To build the model that lets a surveyor predict soil from landscape position, which makes later surveys faster and cheaper.
- **To provide a permanent record.** To leave a baseline description that later surveys can compare against to detect change.

Applied objectives

- **Land use planning.** To guide which crop, forest or other use is best suited to each soil.
- **Irrigation and drainage planning.** To identify areas fit for irrigation, the type of irrigation suited, and areas needing drainage.
- **Soil and water conservation.** To locate erodible soils and prescribe conservation measures.
- **Fertiliser and nutrient management.** To base fertiliser recommendations on soil properties.
- **Crop suitability and diversification.** To identify soils suited to specific crops, including horticultural crops and plantation crops.
- **Watershed and command area development.** To provide the soil base for watershed and canal command planning.
- **Land valuation, taxation and land records.** To support revenue and land-use classification.
- **Non-agricultural uses.** To advise on sites for roads, buildings, reservoirs, effluent disposal and other engineering uses.
- **Environmental protection.** To identify soils vulnerable to salinity, waterlogging, erosion or contamination.

Point of comparison	Fundamental objectives	Applied objectives
Purpose	To build knowledge of the soils and their genesis	To serve a specific use of the land
Question answered	What soil is here, and why?	What is the best use of this soil, and how should it be managed?
Output	Classification, genesis, soil-landscape model, baseline record	Capability, irrigability, suitability, management recommendations
Typical user	Soil scientist, researcher, teacher	Planner, extension officer, farmer, engineer, administrator
Example	Establishing that a Coimbatore-series soil is a Vertisol formed on calcareous parent material	Advising cotton and sorghum with irrigation and drainage on that same soil

Table 2.1 Fundamental and applied objectives of a standard soil survey.

2.4 The soil survey process

A standard soil survey follows a fixed sequence. Each step feeds the next, and the last step is what makes the survey useful.

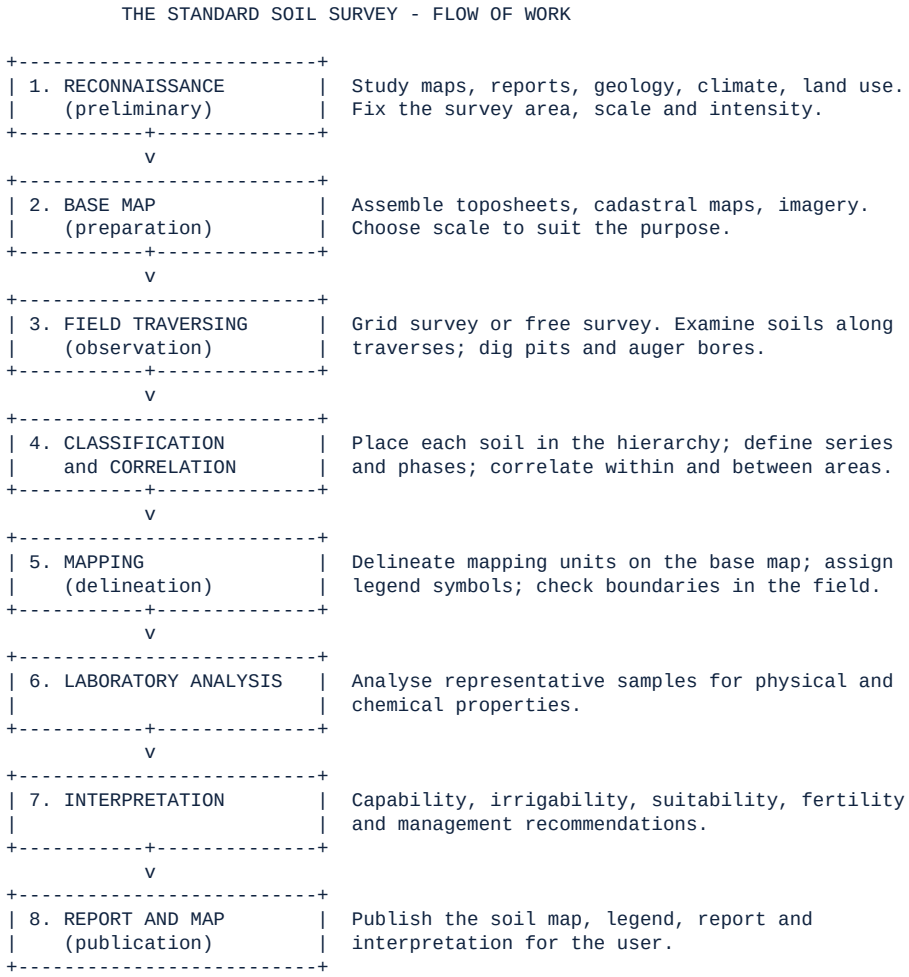


Fig. 2.1 The sequence of work in a standard soil survey.

2.5 What makes a survey 'standard'

- **Uniform procedure.** The same field methods, depth limits and definitions are used throughout.
- **Uniform classification.** Soils are classified in the accepted national system at the accepted category.
- **A defined mapping unit.** Each mapping unit is defined by a rule, usually the soil series or its phases, or a named association or complex.
- **A stated scale and intensity.** The scale and the observation density are stated, so the user knows the level of detail.
- **Correlation.** The results are correlated with the surveys of adjoining areas.
- **A legend.** A written legend names every symbol on the map.
- **Interpretation.** The map is accompanied by interpretations for stated uses.
- **Publication.** The map and report are published, so the work is permanent and open to use and review.

2.6 Scales and the level of detail

The scale of a soil map fixes how much detail can be shown and, therefore, which objectives the survey can meet. A larger-scale map (for example 1:10,000) shows small soil bodies and serves farm-level planning. A smaller-scale map (for example 1:250,000) shows only broad soil groups and serves regional planning.

Map scale	Minimum legible area (approx.)	Detail shown	Suited objective
1:5,000 to 1:10,000	About 0.1-0.4 ha	Individual soil series and phases	Farm and watershed planning
1:12,500 to 1:25,000	About 0.6-2.5 ha	Soil series and complexes	Detailed agricultural planning, command areas
1:50,000 to 1:100,000	About 10-40 ha	Associations of series	District and block planning
1:250,000 and smaller	About 250 ha and above	Broad soil groups and orders	State and national planning

Table 2.2 Map scale, minimum legible area and the level of detail. Areas are the usual rule-of-thumb ranges and vary with the map specification.

Rule of thumb for the minimum legible area: it is commonly taken as about 0.25 square centimetres on the map, which at a scale of 1:50,000 works out to about 6.25 hectares on the ground. The exact figure is fixed by the survey specification, so always quote the specification rather than a single number.

2.7 Tamil Nadu illustration

In Tamil Nadu, standard soil surveys have been carried out at several scales. The detailed surveys of the Cauvery delta command areas, needed for canal and drainage design, were done at larger scales so that the deltaic alluvium could be separated into its constituent series and phases. District-level surveys were done at smaller scales for block and district planning. Both are standard surveys: they use the same classification and correlation rules, but at different scales and intensities, because their objectives differ.

Exam focus

Key points to remember

- A standard soil survey is the systematic examination, description, classification and mapping of soils by a uniform accepted procedure, so that maps are comparable between areas.
- Scope = features studied (profile morphology, physical and chemical properties, site features, distribution) plus outputs (map, legend, report, interpretation).
- Fundamental objectives build knowledge: identify, classify, map, explain genesis, establish soil-landscape relations, keep a baseline.
- Applied objectives serve use: land use planning, irrigation and drainage, conservation, fertiliser management, crop suitability, watershed development, taxation, engineering and environment.
- The survey process runs reconnaissance, base map, traversing, classification and correlation, mapping, laboratory analysis, interpretation, publication.
- Map scale fixes the minimum legible area and therefore the level of detail and the objectives that can be met.
- A survey is 'standard' because of uniform procedure, uniform classification, a defined mapping unit, a stated scale, correlation, a legend, interpretation and publication.

Short-answer questions (with model answers)

Q1. Define a standard soil survey.

Model answer. A standard soil survey is the systematic examination, description, classification and mapping of the soils of an area by a uniform and accepted procedure, producing a soil map and report that can be compared with surveys of other areas and used for land use planning.

Q2. Give the scope of a standard soil survey.

Model answer. Its scope is to study the soil profile morphology, the physical and chemical properties, and the site features such as slope, erosion and drainage, and to record the distribution and extent of the soils. Its outputs are a soil map, a legend, a soil survey report and an interpretation report.

Q3. Distinguish fundamental from applied objectives of soil survey.

Model answer. Fundamental objectives aim at knowledge - identifying, describing, classifying and mapping soils, explaining their genesis and establishing soil-landscape relationships. Applied objectives aim at use - land use planning, irrigation and drainage design, conservation, fertiliser management, crop suitability and watershed development.

Q4. Why does map scale matter in a soil survey?

Model answer. Map scale fixes the minimum legible area, that is, the smallest soil body that can be shown. A larger scale shows small bodies and suits farm and watershed planning; a smaller scale shows only broad groups and suits district, state or national planning.

Long-answer questions (with model answer outlines)

Q1. State the objectives of a standard soil survey and explain each.

Outline.

- Define standard soil survey.
- State that objectives fall into fundamental and applied groups.
- Fundamental: identify and describe the soils; classify them; map their distribution; explain genesis; establish soil-landscape relationships; provide a permanent baseline record.
- Applied: land use planning; irrigation and drainage; soil and water conservation; fertiliser and nutrient management; crop suitability and diversification; watershed and command area development; land valuation and records; non-agricultural uses; environmental protection.
- Explain with a Tamil Nadu example, e.g. delta survey for irrigation and drainage versus district survey for block planning.
- Conclude that both sets of objectives are present in a standard survey and their relative weight depends on the purpose ordered.

Q2. Describe the scope and the step-by-step process of a standard soil survey.

Outline.

- Define standard soil survey and explain the word 'standard'.
- Scope - features studied: morphology, physical properties, chemical properties, site features, distribution.
- Scope - outputs: map, report, legend, interpretation, data sheets.
- Step 1 reconnaissance; step 2 base map preparation; step 3 field traversing; step 4 classification and correlation; step 5 mapping; step 6 laboratory analysis; step 7 interpretation; step 8 report and publication.
- Explain the role of scale and minimum legible area.
- Conclude with the eight features that make a survey standard.

Lecture 3 Soil Mapping Units

A soil map is not made of soil names alone; it is made of areas. Each area drawn on the map is a **soil mapping unit**. This lecture explains the kinds of mapping unit - soil series, soil association, soil complex, variants, inclusions and miscellaneous land types - and the rule that decides when each is used.

3.1 What a soil mapping unit is

Definition. A soil mapping unit is a delineated area on a soil map that is identified by a symbol and described in the legend, and that contains one or more kinds of soil, or non-soil area, within stated limits of purity.

The mapping unit is the bridge between the soil on the ground and the soil on the map. Its name tells the user what to expect inside the boundary. The choice of mapping unit depends on the scale of the survey and on the complexity of the soil pattern.

Consociations and compound units

- **Consociation.** A mapping unit that is dominantly one kind of soil, with only minor inclusions. It is used where the soil pattern is simple and the scale is large enough to separate each soil.
- **Compound unit.** A mapping unit that contains two or more kinds of soil, or soil and non-soil area, in a pattern too fine or too intricate to be separated at the survey scale. Soil associations and soil complexes are compound units.

3.2 The soil series

Definition. A soil series is a group of soils having horizons that are similar in kind, thickness and arrangement, and having a similar range of properties in the control section, developed from a similar parent material. The series is the basic taxonomic and mapping unit of the soil survey.

The soil series is named after a place - usually the town, village or locality where it was first described. This is why a soil series name such as Coimbatore, Palladam or Peelamedu looks like a place name. The name carries no other meaning; it is simply a label for a defined range of soil properties.

Defining features of a soil series

- Similar **horizon sequence** - the same horizons in the same order.
- Similar **kind and thickness** of each horizon.
- Similar **control-section properties** - texture, mineralogy, cation-exchange activity and reaction over the 25-100 cm interval.
- Similar **parent material** and mode of deposition.
- Similar **depth class** and nature of the root-limiting layer.
- Similar **colour, structure and mottling**, hence a similar drainage class.

Soil phase

A **phase** is a subdivision of a soil series or other unit made on a feature that is not part of the soil itself but affects its use - for example slope, erosion, stoniness, depth, salinity or flooding. A phase is written after the series name: 'Coimbatore series, sloping phase'. Phases are used because two areas of the same series can need very different management if one is level and the other is steep.

- **Slope phases** - level to gently sloping, moderately sloping, steeply sloping.
- **Erosion phases** - slightly eroded, moderately eroded, severely eroded.
- **Depth phases** - shallow, deep, very deep.
- **Stoniness and rockiness phases** - stony, very stony, rocky.
- **Salinity and alkalinity phases** - saline, sodic, saline-sodic.

- **Drainage and flooding phases** - poorly drained, frequently flooded.

3.3 The soil association

Definition. A soil association is a mapping unit that consists of two or more defined soil series or other soil bodies occurring together in a recognisable and repeating pattern on the landscape, in proportions that are too intricate to be separated at the survey scale.

In an association, the constituent soils occur in a **regular, repeating pattern** related to the landscape - for example a summit-shoulder-backslope pattern, or a ridge-slope-valley pattern. The pattern is predictable, so the surveyor can describe it and the user can find each soil by knowing its landscape position.

- The soils occur in a definite, mappable pattern.
- The pattern repeats wherever the association occurs.
- The individual soils can be separated only by going to a larger scale.
- The association is named after its dominant or most characteristic series, or after the locality.
- An association is used mainly at medium and small scales.

3.4 The soil complex

Definition. A soil complex is a mapping unit that consists of two or more kinds of soil occurring together in a pattern that is so intricate and so irregular that the individual soils cannot be separated on the map at any practical scale, and the pattern is not predictable from the landscape.

The difference between an association and a complex is the **regularity of the pattern**. In an association the pattern is regular and related to landscape position, so the user can predict where each soil is. In a complex the pattern is irregular and not predictable, so the user must expect any of the soils anywhere inside the unit.

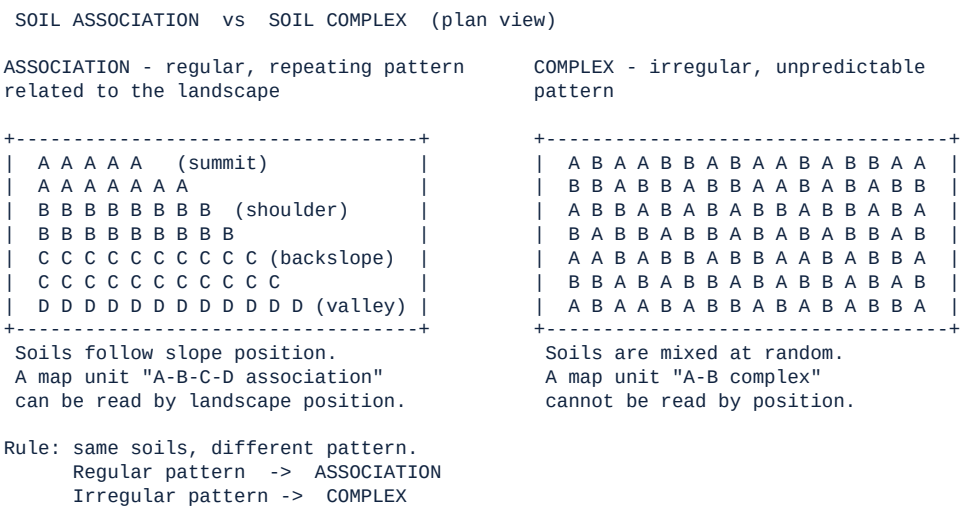


Fig. 3.1 Soil association and soil complex - the difference is the regularity of the pattern.

Point	Soil association	Soil complex
Number of soils	Two or more	Two or more
Pattern	Regular and repeating	Irregular and unpredictable
Relation to landscape	Pattern follows landscape position	No relation to landscape position
Predictability for the user	Position of each soil can be predicted	Any of the soils may occur anywhere in the unit
Usual survey scale	Medium and small scales	Any scale, where the pattern is intricate
Example	Red loamy soil on the slope with black soil in the valley	Patches of calcareous and non-calcareous soil mixed on a level plain

Table 3.1 Soil association versus soil complex.

3.5 Variants

Definition. A variant is a subdivision of a soil series, or of another mapping unit, created when the soil differs from the defined series in some property that is important for use but is not important enough for the soil to be given a new series name.

A variant is therefore a local, practical refinement. It is used when the surveyor recognises a consistent difference - for example a soil that is like the named series but shallower, or sandier, or more eroded - and needs to show that difference on the map without redefining the series.

- A variant is defined **within** an existing series, not outside it.
- The difference is usually in a phase-type feature: depth, texture, erosion, stoniness or salinity.
- A variant is given a symbol in the legend and described in the report.
- It is used mainly in detailed surveys where the user needs the finer distinction.

3.6 Inclusions

Definition. An inclusion is a soil, or an area of non-soil, that occurs within a mapping unit but is not shown separately on the map because it is too small to be delineated at the survey scale.

- **Why inclusions occur.** At any given scale there is a minimum legible area. Bodies smaller than that cannot be drawn, so they are absorbed into the surrounding mapping unit.
- **Effect on the unit.** A mapping unit is therefore never perfectly pure; it contains a stated dominant soil plus minor inclusions.
- **Purity.** The purity of a mapping unit is the proportion of the unit occupied by the named soil or soils. Detailed surveys aim at higher purity than reconnaissance surveys.
- **Why they matter.** Inclusions can affect management - a small patch of saline soil inside a normal field still harms the crop. The report therefore warns the user which inclusions to expect.

Association, complex and inclusion - keep them apart. An association is a mapped unit of two or more soils in a regular pattern; a complex is a mapped unit of two or more soils in an irregular pattern; an inclusion is an unmapped minor body inside a unit. Association and complex are on the map; the inclusion is not.

3.7 Miscellaneous land types

Definition. Miscellaneous land types are areas shown on a soil map that are not soil in the ordinary sense, or are so altered or so unproductive that they are treated as separate units rather than as soils.

Miscellaneous land type	Description	Usual symbol
Rock outcrop	Exposed bedrock covering a large part of the surface	R
Stony land	Surface strewn with stones and boulders, no continuous soil	S
Gullied land	Land cut by gullies, mostly lost to agriculture	G
Rough broken land	Steep, dissected, non-arable terrain	B
Riverwash	Unstable stream-bed deposits, frequently reworked	W
Marsh and swamp	Waterlogged organic or mineral areas, often under water	M
Beaches and sand dunes	Coastal sand, unstable or saline	D
Made land and mine spoil	Artificially filled, excavated or dumped land	ML
Urban and built-up land	Land covered by buildings, roads and industry	U
Water bodies	Tanks, reservoirs, canals, rivers and sea	Wt

Table 3.2 Common miscellaneous land types and their usual map symbols. Symbols vary between survey organisations; the legend of the survey is always authoritative.

3.8 Choosing the right mapping unit

The choice is governed by two things: the **complexity of the soil pattern** and the **scale of the survey**.

Situation	Mapping unit used	Reason
Simple pattern, large scale	Soil series (consociation)	Each soil can be delineated separately
Soil differs only in a phase feature	Soil series with phase or variant	Difference matters for use but not for taxonomy
Two or more soils in a regular landscape pattern	Soil association	Pattern is predictable; separation needs a larger scale
Two or more soils in an intricate irregular pattern	Soil complex	Pattern is not predictable; separation is impractical
Small bodies below the minimum legible area	Shown as inclusions	Too small to delineate at the survey scale
Non-soil or unproductive areas	Miscellaneous land type	Not soil, or soil so altered that it is treated separately

Table 3.3 Rule for selecting the mapping unit.

3.9 Tamil Nadu illustration

Tamil Nadu soil survey reports use all these units. In the uplands of the interior districts, red loamy soils on the upper slopes and black soils in the lower positions occur in a repeating catenary pattern; where the scale does not allow separation, the two are mapped as an association. On the level deltaic plains, where calcareous and non-calcareous alluvium occur in small irregular patches, a complex is used. Coastal sand, salt flats and tank beds are shown as miscellaneous land types.

Exam focus

Key points to remember

- A soil mapping unit is a delineated area on a soil map identified by a symbol and described in the legend, containing one or more soils within stated limits of purity.
- A consociation is dominantly one soil; a compound unit contains two or more soils.
- A soil series is a group of soils with similar horizons, similar control-section properties and similar parent material; it is the basic mapping unit and is named after a place.
- A phase is a subdivision of a series based on a non-soil feature such as slope, erosion, depth, stoniness or salinity.
- A soil association is two or more soils in a regular, landscape-related repeating pattern.
- A soil complex is two or more soils in an irregular, unpredictable pattern.
- A variant is a local subdivision of a series based on a use-related difference too small to define a new series.
- An inclusion is a minor soil or non-soil body too small to map separately, absorbed into the surrounding unit.
- Miscellaneous land types are non-soil or unproductive areas such as rock outcrop, gullied land, marsh and built-up land.

Short-answer questions (with model answers)

Q1. Define a soil series.

Model answer. A soil series is a group of soils having horizons similar in kind, thickness and arrangement and a similar range of properties in the control section, developed from a similar parent material. It is the basic taxonomic and mapping unit of the soil survey and is named after a locality.

Q2. Distinguish a soil association from a soil complex.

Model answer. Both contain two or more soils. In a soil association the soils occur in a regular, repeating pattern related to landscape position, so the position of each soil can be predicted. In a soil complex the pattern is irregular and unrelated to landscape position, so any of the soils may occur anywhere in the unit.

Q3. What is a soil phase? Give two examples.

Model answer. A soil phase is a subdivision of a soil series made on a feature that is not part of the soil itself but affects its use. Examples are the slope phase (level, moderately sloping, steep) and the erosion phase (slightly, moderately or severely eroded).

Q4. What is an inclusion and why does it occur?

Model answer. An inclusion is a soil or non-soil body inside a mapping unit that is too small to be shown separately at the survey scale. It occurs because every map scale has a minimum legible area, so bodies smaller than that cannot be delineated and are absorbed into the surrounding unit.

Q5. Name four miscellaneous land types.

Model answer. Any four of: rock outcrop, stony land, gullied land, rough broken land, riverwash, marsh and swamp, beaches and sand dunes, made land and mine spoil, urban and built-up land, and water bodies.

Long-answer questions (with model answer outlines)

Q1. Explain the soil mapping units used in soil survey, with the rule for choosing each.

Outline.

- Define a soil mapping unit and explain consociation and compound unit.
- Soil series: definition, defining features, naming after a place.
- Soil phase and variant: definition and examples.
- Soil association: definition, regular pattern, use at medium and small scales.
- Soil complex: definition, irregular pattern, contrast with association.
- Inclusions: definition, reason (minimum legible area), effect on purity.
- Miscellaneous land types: definition and examples.
- State the selection rule based on pattern complexity and scale.
- Give a Tamil Nadu illustration from the uplands and the delta.

Q2. Compare and contrast soil association, soil complex and inclusion.

Outline.

- Define all three terms first.
- Number of soils in each.
- Pattern: regular for association, irregular for complex, not applicable for inclusion.
- Relation to landscape position.
- Whether the unit appears on the map: association and complex do, inclusion does not.
- Predictability for the user.
- Usual survey scale at which each is used.
- A Tamil Nadu example of each.
- Conclude with the one-line distinguishing rule.

Lecture 4 Soil Survey Concepts - Base Maps and Traversing

Every soil survey needs two things before the first pit is dug: a map to draw on, and a method of moving across the land to find the soil boundaries. The map is the **base map**. The method of movement is **traversing**, and it takes one of two forms - grid survey or free survey. This lecture explains both.

4.1 The base map

Definition. A base map is the map on which the soil boundaries and mapping-unit symbols are drawn. It carries the reference features - roads, streams, settlements, field boundaries, contours or survey numbers - that allow the surveyor to locate a point on the ground accurately and to plot it on the map.

The base map must be at the same scale as the intended soil map, and must have enough reference detail to fix positions. Without a good base map, soil boundaries cannot be plotted accurately, and the resulting soil map will not match the ground.

Kinds of base map used in India

- **Topographic sheets (toposheets).** Published by the Survey of India at scales of 1:50,000, 1:25,000 and 1:250,000, showing contours, drainage, roads, settlements and land use. They are the standard base for reconnaissance and medium-scale surveys.
- **Cadastral maps.** Village-level maps showing survey numbers and field boundaries, at scales of about 1:4,000 to 1:10,000. They are the standard base for detailed farm-level soil surveys, because every field can be identified by its survey number.
- **Aerial photographs.** Vertical photographs at scales of 1:10,000 to 1:50,000, used as base maps because soil and land boundaries are often visible on them as tonal or textural changes.
- **Satellite images.** Digital images from the IRS and other satellites, used as base maps for large-area surveys and for updating earlier surveys.
- **Plane-table sheets and existing soil maps.** Prepared locally or carried forward from an earlier survey, used where a fresh base map is not available.

Base map	Usual scale	Best suited survey	Key advantage
Cadastral map	1:4,000 to 1:10,000	Detailed farm-level survey	Every field identified by survey number
Toposheet 1:25,000	1:25,000	Detailed and semi-detailed survey	Contours and drainage available
Toposheet 1:50,000	1:50,000	Reconnaissance and medium-scale survey	Complete national coverage, uniform standard
Aerial photograph	1:10,000 to 1:50,000	Detailed and reconnaissance survey	Soil and land boundaries visible as tones
Satellite image (IRS)	Wide range, 1:25,000 to 1:250,000	Reconnaissance and updating	Large area covered in one scene

Table 4.1 Common base maps used in Indian soil survey and the surveys they suit.

4.2 Traversing - the idea

Definition. Traversing is the movement of the surveyor across the survey area along planned lines, examining the soil at intervals by auger bore or pit, so that the sequence of soils and the position of boundaries between them can be established.

The surveyor does not examine the whole area. Instead the surveyor walks along lines that cross the main soil-landscape pattern, examines the soil at points along the line, and uses the soil-landscape model to

predict what lies between the observation points. The denser the traverses, the more confident the prediction.

Observations made along a traverse

1. **Auger bores** - a hand auger is used to bring up soil from successive depths, giving a quick check of texture, colour and depth at many points.
2. **Pits** - a pit of about 1 to 2 metres depth is dug at representative points to describe the full profile, sample horizons and classify the soil.
3. **Road cuts and stream banks** - natural or man-made exposures that give a free profile description at low cost.
4. **Boundary checking** - walking across the boundary of a mapping unit to confirm where the change occurs and to draw the line.
5. **Site observations** - slope, erosion, drainage, water table, stoniness and present land use at every observation point.

4.3 Grid survey

Definition. Grid survey is a method of soil survey in which the area is divided into a regular grid of squares and the soil is examined at the corners, or at the centre, of every grid square, so that the observations are spread uniformly over the whole area.

The grid is laid out on the base map first and then located on the ground. The spacing of the grid is decided from the scale of the survey and the expected complexity of the soils. The observations are plotted on the map, the soils at neighbouring points are joined, and the boundaries are drawn between them.

Advantages of grid survey

- The observations are **uniformly distributed**, so no part of the area is left unchecked.
- The method is **objective and repeatable**; any other surveyor would make observations at the same points.
- It is **statistically sound**, so the data can be used for quantitative work such as estimating the proportion of each soil.
- It is **easy to supervise**, because the plan of work is fixed in advance.
- It is well suited to **level or gently undulating land**, and to areas where the soil pattern is not obvious from the landscape.

Limitations of grid survey

- It is **wasteful where the soil pattern is simple**, because many grid points fall inside the same soil.
- It is **inefficient in dissected or hilly terrain**, where a grid line may cross a cliff or a stream and cannot be walked.
- It **ignores the soil-landscape relationship**, so a boundary may fall between two grid points and be missed.
- It **takes longer** for the same number of useful observations, because walking to fixed grid points is slower than walking to the places where a change is expected.
- A **coarse grid may miss small but important soil bodies** altogether.

4.4 Free survey

Definition. Free survey is a method of soil survey in which the surveyor, after a preliminary study of the area, chooses observation points freely along lines that are expected to cross the main soil boundaries, using the soil-landscape relationship to decide where to look and where to place boundaries.

In free survey the surveyor first builds a mental model of how soils are related to the landscape - which soil is on the summit, which on the slope, which in the valley. Then the surveyor walks the landscape, examines the soil where the model predicts a change, and checks the prediction. Free survey is therefore an **interpretive** method: the surveyor's judgement does most of the work.

Advantages of free survey

- It is **efficient in areas with a clear soil-landscape relationship**, because observations are made only where a change is expected.
- It **finds boundaries more accurately**, because the surveyor goes to the boundary and examines it.
- It **covers difficult terrain** better than a grid, since the surveyor can follow the land.
- It is **faster and cheaper** for the same accuracy in simple or well-understood landscapes.
- It **uses the surveyor's skill**, which improves as the survey proceeds.

Limitations of free survey

- It depends heavily on the **experience and judgement** of the surveyor; a less experienced surveyor may produce a poorer map.
- It is **less objective**, because the observation points are not fixed in advance, so two surveyors may place them differently.
- It is **harder to supervise and to check**.
- It may **leave gaps** in parts of the area that the surveyor judged uninteresting but which in fact contain a soil body.
- It is **less suitable for statistical work**, because the observations are not evenly spread.

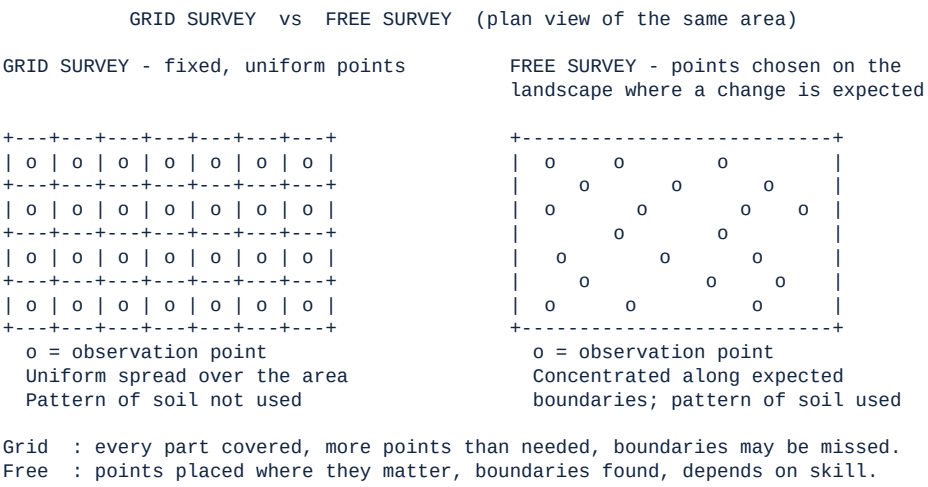


Fig. 4.1 Grid survey and free survey - the distribution of observation points.

Point of comparison	Grid survey	Free survey
Basis of point selection	A fixed regular grid	Surveyor's judgement using the soil-landscape model
Distribution of observations	Uniform over the whole area	Concentrated where a change is expected
Objectivity	High; repeatable by any surveyor	Lower; depends on the surveyor
Boundary accuracy	Boundaries may fall between points and be missed	Boundaries examined directly, so more accurate
Terrain suited	Level to gently undulating land	All terrain, including dissected and hilly land
Efficiency in simple landscapes	Low; many redundant points	High; few redundant points

Point of comparison	Grid survey	Free survey
Suitability for statistical work	Good	Poorer
Skill required	Moderate	High

Table 4.2 Comparison of grid survey and free survey.

Which to use in practice. Grid survey is preferred where the soils are not obviously related to the landscape, or where statistical reliability is wanted. Free survey is preferred where the soil-landscape relationship is clear and the surveyor is experienced. In practice, most Indian soil surveys use free survey as the main method, with a few grid or check traverses to test the boundaries.

4.5 Other traversing patterns

- **Free traverse.** The surveyor follows the natural lines of the landscape - ridge, slope, valley - so that the catenary sequence of soils is crossed in one walk.
- **Parallel traverse.** Straight parallel lines are run across the area at a fixed interval; this is a grid survey with only one direction of lines.
- **Contour traverse.** Lines are run along contours, useful for mapping soils that change with elevation.
- **Check traverse.** A short traverse run at right angles to the main traverses to test the boundaries already drawn.

4.6 Worked example - measuring the area of a soil delineation

The area of each mapping unit is measured from the soil map. A common manual method is to draw parallel lines across the delineation at a fixed spacing and measure the length of each line within the delineation. The areas between successive lines are then summed. **Simpson's rule** gives a good estimate by treating the shape as a series of strips with a parabolic top.

The rule

For an odd number of equally spaced ordinates y0, y1, y2, ... yn, with a constant spacing d between them, the area is approximately:

```
Simpson's rule (n must be even, i.e. an odd number of ordinates):

   A  =  (d / 3)  x  [ (y0 + yn)  +  4 (y1 + y3 + ... + y(n-1))
                                  +  2 (y2 + y4 + ... + y(n-2)) ]

   d  = spacing between ordinates (in map units)
   y  = length of each ordinate inside the delineation (in map units)
```

Fig. 4.2 Simpson's rule for the area of an irregular delineation.

Worked example 4.1

A soil delineation on a 1:25,000 soil map is measured by drawing seven equally spaced ordinates 2.0 cm apart on the map. The lengths of the ordinates inside the delineation are 0, 1.2, 3.0, 3.6, 3.0, 1.4 and 0 cm. Find the area of the delineation on the ground in hectares.

- **Step 1 - area on the map.** Here d = 2.0 cm and n = 6, so there are seven ordinates (y0 to y6).
- **Odd-position ordinates** (y1, y3, y5) = 1.2 + 3.6 + 1.4 = 6.2 cm.
- **Even-position ordinates** (y2, y4) = 3.0 + 3.0 = 6.0 cm.
- **End ordinates** (y0 + y6) = 0 + 0 = 0 cm.
- **Simpson's rule:** A = (2.0 / 3) x [0 + 4(6.2) + 2(6.0)]
- A = 0.6667 x [0 + 24.8 + 12.0] = 0.6667 x 36.8 = 24.53 cm2 on the map.

- **Step 2 - map area to ground area.** Scale 1:25,000 means 1 cm on the map represents 25,000 cm = 250 m on the ground.
- So 1 cm2 on the map = 250 m x 250 m = 62,500 m2 = 6.25 ha on the ground.
- **Step 3 - ground area.** Area = 24.53 x 6.25 = **153.3 ha**.

Remember the conversion. At 1:25,000, one square centimetre on the map is 6.25 hectares on the ground. At 1:50,000 it is 25 hectares. At 1:10,000 it is 1 hectare. Learn these three and the area calculation becomes a single multiplication.

Exam focus

Key points to remember

- A base map is the map on which soil boundaries are plotted; it must be at the soil-map scale and carry enough reference detail to fix positions.
- Common Indian base maps: cadastral maps, Survey of India toposheets, aerial photographs and satellite images.
- Traversing is planned movement across the area, examining soil by auger bore or pit at intervals to fix soil boundaries.
- Grid survey uses a fixed regular grid of observation points, giving uniform, objective and statistically usable data, but is wasteful in simple terrain and may miss boundaries.
- Free survey places observation points by judgement along expected boundaries, using the soil-landscape relationship; it is faster and more accurate but depends on surveyor skill.
- Most Indian surveys use free survey with check traverses.
- Simpson's rule gives the area of an irregular delineation from equally spaced ordinates.

Short-answer questions (with model answers)

Q1. What is a base map and why is it needed?

Model answer. A base map is the map on which soil boundaries and mapping-unit symbols are drawn. It is needed because it carries the reference features - roads, streams, field boundaries, contours and survey numbers - that let the surveyor locate a point on the ground and plot it accurately, so that the soil map matches the ground.

Q2. Define traversing.

Model answer. Traversing is the movement of the surveyor across the survey area along planned lines, examining the soil at intervals by auger bore or pit, so that the sequence of soils and the position of boundaries between them can be established.

Q3. State two advantages and two limitations of grid survey.

Model answer. Advantages: observations are uniformly distributed over the whole area; the method is objective and repeatable. Limitations: it is wasteful where the soil pattern is simple, and it may miss a boundary that falls between two grid points.

Q4. State two advantages and two limitations of free survey.

Model answer. Advantages: it is efficient where the soil-landscape relationship is clear, and it finds boundaries more accurately because the surveyor examines them directly. Limitations: it depends heavily on the surveyor's skill and judgement, and it is harder to supervise and check.

Q5. Which base map would be used for a detailed farm-level soil survey and why?

Model answer. A cadastral map, at about 1:4,000 to 1:10,000, because it shows field boundaries and survey numbers, so every field can be identified on the ground and the soil boundaries can be plotted field by field.

Long-answer questions (with model answer outlines)

Q1. Explain the methods of traversing used in soil survey.

Outline.

- Define traversing and state its purpose.
- Describe the observations made: auger bores, pits, road cuts and stream banks, boundary checking, site observations.
- Grid survey: definition, how the grid is laid out, spacing.
- Advantages of grid survey - uniform, objective, statistical, easy to supervise, suits level land.
- Limitations of grid survey - wasteful in simple terrain, unsuitable in dissected land, ignores the soil-landscape relation, slower, may miss small bodies.
- Free survey: definition, preliminary study, use of the soil-landscape model.
- Advantages of free survey - efficient, accurate boundaries, suits difficult terrain, faster.
- Limitations of free survey - depends on skill, less objective, harder to check, may leave gaps, poorer for statistics.
- Other patterns: free, parallel, contour and check traverses.
- Conclude that Indian surveys use free survey with check traverses.

Q2. Describe the base maps used in soil survey and explain how the area of a mapping unit is measured.

Outline.

- Define base map and state its requirements - same scale, reference detail.
- Cadastral maps: scale, content, use.
- Toposheets: scales, content, use.
- Aerial photographs: scales, why boundaries are visible.
- Satellite images: use for large areas and updating.
- Explain how area is measured: parallel lines and ordinates.
- State Simpson's rule and define each symbol.
- Work the numerical example with d = 2.0 cm, seven ordinates, scale 1:25,000, giving about 153 ha.
- Conclude with the three useful scale conversions.

Lecture 5 Types of Soil Survey

The same land can be surveyed in several ways, and the survey that is right for one purpose is wrong for another. The types of soil survey differ chiefly in **scale**, **intensity of observation**, **kind of mapping unit** and **purpose**. This lecture explains the four types named in the syllabus: detailed, reconnaissance, detailed-reconnaissance and semi-detailed.

5.1 What distinguishes one type of survey from another

- **Scale of the map** - the larger the scale, the more detail that can be shown.
- **Intensity of observation** - the number of observations per unit area, or the observation density, expressed for example as the number of observations per square kilometre or the spacing of traverses.
- **Kind of mapping unit** - detailed surveys separate soil series and phases; reconnaissance surveys map associations and broad groups.
- **Purity of the mapping unit** - the proportion of the unit occupied by the named soil.
- **Purpose** - farm planning, project planning, district planning or national planning.
- **Time and cost** - a detailed survey takes far longer and costs far more per hectare than a reconnaissance survey.

The trade-off in one line. Greater detail costs more time and money per hectare and covers less area; less detail is cheaper and faster but cannot support field-level decisions. The survey type is chosen to match the decision it must support.

5.2 Detailed soil survey

Definition. A detailed soil survey is a survey carried out at a large scale, usually 1:5,000 to 1:10,000 and sometimes up to 1:15,000, in which the soils are examined at close intervals and mapped as soil series and phases, so that the map is suitable for farm-level and watershed-level planning.

Features of a detailed soil survey

- **Scale.** Usually 1:5,000 to 1:10,000.
- **Mapping unit.** Soil series and their phases; the mapping unit is usually a consociation with high purity.
- **Minimum legible area.** About 0.1 to 0.4 hectare.
- **Observation density.** High - a pit or a closely spaced set of auger bores at short intervals, often one observation per 1 to 2 hectares or closer.
- **Base map.** Cadastral map or a large-scale toposheet or a large-scale aerial photograph.
- **Traversing.** Mainly free survey, checked by close auger bores along field boundaries and bunds.
- **Output.** A farm-level map showing each field's soil, with detailed management recommendations.

Where a detailed soil survey is used

- Farm-level land use planning and crop planning.
- Watershed development and soil conservation planning at micro-watershed level.
- Command area development of an irrigation project, for the layout of field channels and drains.
- Site selection for research stations, orchards, plantations and experimental plots.
- Reclamation planning for saline, sodic or waterlogged land.
- High-value horticulture, floriculture and protected cultivation.

5.3 Reconnaissance soil survey

Definition. A reconnaissance soil survey is a rapid survey carried out at a small scale, usually 1:100,000 to 1:250,000 or smaller, in which the soils are examined at widely spaced intervals and mapped as broad soil groups, associations or even great groups and suborders, so that the map is suitable for regional and national planning.

Features of a reconnaissance soil survey

- **Scale.** Usually 1:100,000 to 1:250,000, and smaller for national work.
- **Mapping unit.** Soil associations, broad groups, great groups or suborders; not individual series.
- **Minimum legible area.** About 25 to 250 hectares or more.
- **Observation density.** Low - observations at widely spaced intervals, often one observation per several hundred hectares, using mainly auger bores and road cuts, with few pits.
- **Base map.** Small-scale toposheets or satellite images.
- **Traversing.** Free survey along roads, rivers and other accessible lines, supported by satellite image interpretation.
- **Output.** A regional map of broad soil groups with general recommendations.

Where a reconnaissance soil survey is used

- State-level and national-level resource assessment.
- Identification of areas for irrigation, drainage or reclamation projects, before a detailed survey is ordered.
- Selection of areas for detailed survey - reconnaissance precedes detail.
- Catchment and river basin planning.
- Broad land use and agricultural development planning.

5.4 Detailed-reconnaissance soil survey

Definition. A detailed-reconnaissance soil survey is an intermediate survey, of medium scale and medium intensity, in which the soil pattern is studied in some detail along selected traverses but the whole area is not examined closely, so that the map shows soil associations and the more important series but is not suitable for field-level planning.

The name describes the method: the survey is **detailed along the traverses** and **reconnaissance between them**. The surveyor examines the soils carefully where the traverses cross the area, and then interpolates across the gaps. This gives a faster result than a full detailed survey while retaining more information than a pure reconnaissance survey.

Features of a detailed-reconnaissance soil survey

- **Scale.** Usually about 1:50,000 to 1:100,000.
- **Mapping unit.** Soil associations and, where the pattern allows, the more important soil series.
- **Minimum legible area.** About 10 to 40 hectares.
- **Observation density.** Moderate - closely spaced observations along selected traverses, sparser between them.
- **Base map.** 1:50,000 toposheet, sometimes with aerial photographs.
- **Traversing.** Free survey along chosen traverses, with pits at representative points and auger bores in between.
- **Output.** A block- or taluk-level map of associations and important series.

Where a detailed-reconnaissance soil survey is used

- Block-level and taluk-level agricultural planning.
- Planning of medium irrigation and minor irrigation schemes.
- Planning of watershed development at the block level.
- Agricultural development programmes at district sub-unit level.
- Pre-project studies where a full detailed survey is not yet justified.

5.5 Semi-detailed soil survey

Definition. A semi-detailed soil survey is a survey of medium to large scale, usually 1:20,000 to 1:50,000, in which the soils are examined at moderately close intervals and mapped as soil series, associations and complexes, so that the map is suitable for project planning and command area planning.

A semi-detailed survey sits between the detailed and the detailed-reconnaissance surveys. Its observation density is higher than a detailed-reconnaissance survey but lower than a full detailed survey. It is the survey usually ordered for an irrigation project command area, because it separates the soils of the command into units that can be given different irrigation and drainage treatment.

Features of a semi-detailed soil survey

- **Scale.** Usually 1:20,000 to 1:50,000.
- **Mapping unit.** Soil series, associations and complexes.
- **Minimum legible area.** About 0.6 to 10 hectares.
- **Observation density.** Moderate to high - regularly spaced observations, with pits at representative sites.
- **Base map.** 1:25,000 or 1:50,000 toposheet, or aerial photographs, or a cadastral map enlarged.
- **Traversing.** Free survey with a systematic check grid.
- **Output.** A project-level map of series and associations with irrigation and drainage recommendations.

Where a semi-detailed soil survey is used

- Irrigation project command area planning.
- Drainage and salinity management planning.
- Land reclamation project planning.
- Soil and water conservation planning at project level.
- Planning of plantations and large orchards.
- Detailed planning for a watershed or a command block before field-level design.

Point	Reconnaissance	Detailed-reconnaissance	Semi-detailed	Detailed
Usual scale	1:100,000 to 1:250,000 and smaller	About 1:50,000 to 1:100,000	About 1:20,000 to 1:50,000	1:5,000 to 1:10,000 (to 1:15,000)
Mapping unit	Associations, broad groups, great groups	Associations and important series	Series, associations and complexes	Series and phases
Minimum legible area	About 25-250 ha	About 10-40 ha	About 0.6-10 ha	About 0.1-0.4 ha
Observation density	Low; widely spaced	Moderate along traverses	Moderate to high; regularly spaced	High; closely spaced
Usual base map	Small toposheet, satellite image	1:50,000 toposheet	1:25,000 or 1:50,000 toposheet, aerial photo	Cadastral map, large-scale photo
Purity of mapping unit	Low	Low to moderate	Moderate	High

Point	Reconnaissance	Detailed-reconnaissance	Semi-detailed	Detailed
Pits vs auger bores	Few pits, many auger bores	Pits on traverses	Regular pits and auger bores	Many pits
Main purpose	Regional and national planning	Block and taluk planning	Project and command area planning	Farm and watershed planning
Relative cost per hectare	Lowest	Low	Moderate	Highest

Table 5.1 Comparison of the four types of soil survey. Scales and areas are the usual published ranges; the survey specification is authoritative for any particular project.

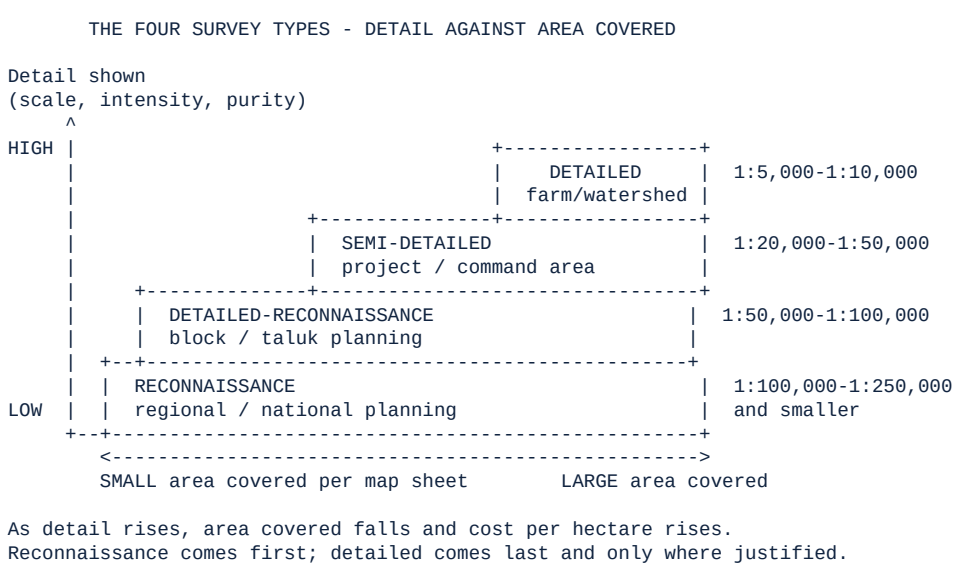


Fig. 5.1 The four survey types - detail, area covered and cost move in opposite directions.

5.6 Choosing the type of survey

1. State the **decision the map must support**. A farm-level decision needs a detailed survey; a district decision does not.
2. Match the **scale** to the size of the smallest management unit - the field, the command block, the watershed or the district.
3. Match the **mapping unit** to the scale: series for large scales, associations for medium scales, broad groups for small scales.
4. Set the **observation density** from the expected complexity of the soil pattern.
5. Choose a **base map** at the same scale and with enough reference detail.
6. Consider **time and cost**: never order a more detailed survey than the decision needs, and never accept a less detailed one than the decision requires.
7. Where the area is large, do a **reconnaissance survey first** and a detailed survey only of the priority pockets.

5.7 Tamil Nadu illustration

Tamil Nadu soil survey work shows all four types in use. Broad reconnaissance surveys have covered large parts of the state to give the district-level picture of soil groups. Semi-detailed surveys have been used for the command areas of major irrigation systems, including the Cauvery delta, where the deltaic alluvium had to be separated into units needing different irrigation and drainage treatment. Detailed surveys at cadastral scale have been done for research farms, horticultural blocks and micro-watersheds, where the soil of each field had to be known. The principle followed is the same everywhere: the survey type is chosen to match the decision it must support.

Exam focus

Key points to remember

- Survey types differ in scale, observation density, kind of mapping unit, purity, purpose, and time and cost.
- Detailed survey: scale 1:5,000 to 1:10,000; series and phases; minimum legible area about 0.1-0.4 ha; for farm and watershed planning.
- Reconnaissance survey: scale 1:100,000 to 1:250,000 and smaller; associations and broad groups; for regional and national planning.
- Detailed-reconnaissance survey: scale about 1:50,000 to 1:100,000; associations and important series; for block and taluk planning; detailed along traverses, reconnaissance between them.
- Semi-detailed survey: scale about 1:20,000 to 1:50,000; series, associations and complexes; for project and command area planning.
- Reconnaissance precedes detail; the survey type is chosen to match the decision it must support.
- As detail rises, area covered falls and cost per hectare rises.

Short-answer questions (with model answers)

Q1. Define a detailed soil survey and state its usual scale.

Model answer. A detailed soil survey is a large-scale survey, usually at 1:5,000 to 1:10,000, in which soils are examined at close intervals and mapped as soil series and phases, so that the map suits farm-level and watershed-level planning.

Q2. Define a reconnaissance soil survey and state its purpose.

Model answer. A reconnaissance soil survey is a rapid small-scale survey, usually 1:100,000 to 1:250,000 or smaller, in which soils are examined at widely spaced intervals and mapped as associations or broad groups. It is used for regional and national resource assessment and planning, and to select areas for detailed survey.

Q3. What is a detailed-reconnaissance soil survey?

Model answer. It is an intermediate survey, usually at about 1:50,000 to 1:100,000, in which the soils are studied in detail along selected traverses but the whole area is not examined closely. It maps associations and the more important series and is used for block and taluk planning.

Q4. Which type of survey would be ordered for an irrigation command area and why?

Model answer. A semi-detailed soil survey, at about 1:20,000 to 1:50,000, because it separates the command area into soil series, associations and complexes at a scale fine enough to plan irrigation and drainage treatment, without the cost of a full field-by-field detailed survey.

Q5. Why is a reconnaissance survey done before a detailed survey?

Model answer. Because it is rapid and cheap, it gives a broad picture of the soils of a large area and identifies the pockets that most need detailed attention. The detailed survey is then ordered only for those pockets, saving time and money.

Long-answer questions (with model answer outlines)

Q1. Describe the four types of soil survey and compare them.

Outline.

- State that survey types differ in scale, intensity, mapping unit, purity, purpose and cost.
- Detailed survey: definition, scale, mapping unit, minimum legible area, observation density, base map, uses.
- Reconnaissance survey: the same features.
- Detailed-reconnaissance survey: the same features, and explain the 'detailed along traverses, reconnaissance between them' idea.
- Semi-detailed survey: the same features.
- Draw or reproduce the comparison table.
- State the rule for choosing a survey type from the decision it must support.
- Give a Tamil Nadu illustration from reconnaissance, semi-detailed and detailed work.

Q2. Explain how the type of soil survey is chosen for a project.

Outline.

- List the factors that distinguish survey types.
- Factor 1: the decision the map must support.
- Factor 2: scale, matched to the smallest management unit.
- Factor 3: mapping unit, matched to the scale.
- Factor 4: observation density, from the complexity of the soil pattern.
- Factor 5: base map choice.
- Factor 6: time and cost - never over-survey or under-survey.
- Factor 7: staged approach - reconnaissance first, detail only where justified.
- Illustrate with the Cauvery delta command area and a micro-watershed.
- Conclude with the one-line principle that survey type matches the decision.

Lecture 6 Interpretative Groupings - Land Capability and Fertility Capability Classification

A soil map shows what soils are present. It does not, by itself, say what to do with them. **Interpretative groupings** translate soil properties into groups that answer a practical question - how good is this land for sustained agricultural use, and what limits it? This lecture covers the two most widely used groupings: Land Capability Classification and Fertility Capability Classification.

6.1 What an interpretative grouping is

Definition. An interpretative grouping is a classification of soils or land into groups on the basis of their behaviour under a stated kind of use, so that each group carries a similar set of limitations and a similar management requirement.

- It is based on the **use**, not on the taxonomy. Two soils of the same order can fall in different capability classes.
- It groups land by its **limitations** - slope, erosion, depth, drainage, salinity, flooding, climate.
- It is **relative**, not absolute: capability classes describe land in relation to a stated use and a stated level of management.
- It is **general purpose** (land capability) or **specific purpose** (irrigability, crop suitability, fertility capability).

6.2 Land Capability Classification - concept

Definition. Land Capability Classification is a system of grouping land into eight classes according to the degree of limitation on its sustained use for agriculture, together with the risk of damage and the intensity of management required.

The system was developed by the United States Soil Conservation Service and is widely used in India, with Indian adaptations. Its purpose is soil and water conservation planning: it tells the planner which land can be cultivated safely, how intensively, and which land must be kept under grass, trees or permanent vegetation.

The three assumptions of the system

1. The land is used for **sustained production** of common cultivated crops, pasture or forest, without damage to the resource.
2. **Good management** is assumed, at the level normal for the area. Poor management can ruin even Class I land.
3. The **climate is not limiting** in the sense of a desert or a permanent frost; capability measures soil and land limitations, not rainfall, unless the system is applied to a specific climatic zone.

6.3 The structure of the system

The system has four levels. From general to specific:

- **Capability class** - eight classes, I to VIII. The broadest grouping, showing the degree of limitation.
- **Capability subclass** - within a class, a letter showing the kind of limitation: e - erosion, w - wetness or drainage, s - soil or rooting depth, c - climate. Written as IIe, IIIs, IVw and so on.
- **Capability unit** - a group of soils within a subclass that need the same management and have the same responses.
- **Capability phase** - a further subdivision for a specific local problem, such as a particular slope or flooding frequency.

Read the symbol in two parts. The Roman numeral gives the degree of limitation; the small letter gives the kind. 'IIIe' means moderate limitation, and the limitation is erosion. 'IIw' means slight limitation, and the limitation is wetness or drainage.

6.4 Land capability classes I to VIII

The eight classes are arranged so that the limitations increase and the choice of safe uses narrows from Class I to Class VIII. Classes I to IV can be cultivated, with increasing care; Classes V to VII are not suitable for cultivation but are suited to grazing, forestry or wildlife; Class VIII is not suited to any productive use and is kept for recreation, wildlife or water supply.

Class	Degree of limitation	Suitability for cultivation	Chief limitations	Conservation need
I	None to very slight	Very good; safe for all common crops	None of significance; deep, level, well drained, fertile	Ordinary good farming practice
II	Slight	Good; safe for a wide range of crops	Gentle slope, slight erosion, moderate depth, occasional wetness	Simple practices: contour farming, cover crops, simple waterways
III	Moderate	Moderate; safe for a restricted range of crops	Moderate slope, moderate erosion, shallow depth, poor drainage, salinity	Complex practices: terracing, strip cropping, drainage, grassed waterways
IV	Severe	Limited; safe for a few crops with careful management	Steep slope, severe erosion, very shallow depth, frequent flooding, strong salinity	Intensive practices; often best kept in grass or trees with occasional cultivation
V	Very severe, not erodible	Not suitable for cultivation	Wetness, frequent flooding, stoniness, but nearly level and not erodible	Grazing, forestry or wildlife; drainage may allow limited use
VI	Very severe	Not suitable for cultivation	Steep slope, severe erosion, shallow depth, low rainfall	Grazing and forestry with careful management
VII	Very severe, more restricted than VI	Not suitable for cultivation	Very steep slope, very severe erosion, very shallow, rough stony land	Grazing and forestry only, with strict control; watershed protection
VIII	Extreme	Not suitable for any productive use	Rock outcrop, cliffs, marsh, sand dunes, very severe limitation	Recreation, wildlife, water supply, watershed protection only

Table 6.1 Land capability classes I to VIII with their limitations and conservation needs.

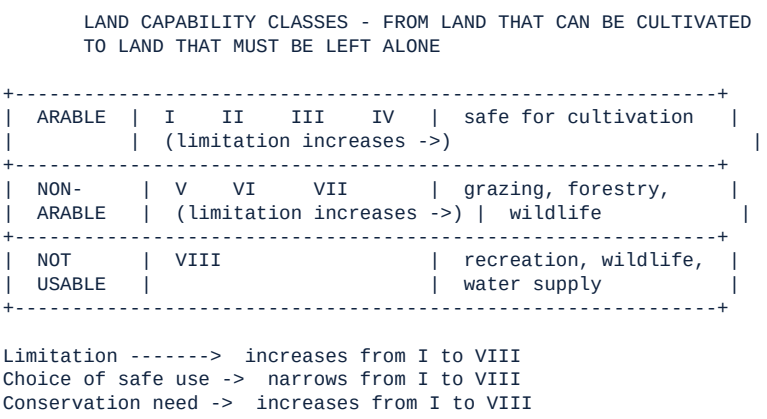


Fig. 6.1 The eight land capability classes grouped by the choice of safe use.

6.5 Worked example - assigning a land capability class

A block of land in an upland part of Tamil Nadu has the following features. Assign the capability class and subclass, and justify it.

Land feature	Value observed
Slope	8 to 10 per cent (moderately sloping)
Erosion	Moderate sheet erosion; some rills; no gullies
Soil depth	60 to 90 cm over weathered rock
Texture	Sandy loam surface, sandy clay loam subsoil
Drainage	Well drained
Salinity or alkalinity	None
Flooding	None
Stoniness and rockiness	Slight stoniness, no rock outcrop
Present land use	Rainfed millets and pulses

Table 6.2 Field data for worked example 6.1.

Worked example 6.1 - reasoning

1. **Is the land cultivable at all?** Yes - slope is moderate, depth is adequate, drainage is good, no salinity and no flooding. So the class is within I to IV.
2. **Which limitation dominates?** Slope of 8 to 10 per cent with moderate sheet erosion is the dominant limitation. The kind of limitation is therefore **erosion**, so the subclass letter is **e**.
3. **How severe is it?** A slope of 8 to 10 per cent with moderate erosion is a moderate limitation. It does not force the land out of cultivation, but it restricts the choice of crops and demands conservation measures. That is the definition of Class III.
4. **Is it worse than III?** No - depth is more than 50 cm, drainage is good, and erosion has not formed gullies. A severe limitation (Class IV) would need a steeper slope, deeper erosion, shallower soil or poor drainage.
5. **Is it better than III?** No - a slight limitation (Class II) would need a gentler slope, about 2 to 5 per cent, with only slight erosion and deeper soil.
6. **Assignment: Class IIIe** - moderate limitation, erosion.
7. **Management:** contour bunding or graded bunding, contour cultivation, strip cropping with a cover crop, grassed waterways along the natural drains, and a crop choice of millets, pulses and other drought-tolerant crops suited to rainfed uplands.

> **How to write the answer.** State the class, then the subclass letter, then the reason in one line, then the management measures. The examiner looks for the class symbol and the reason, not for a long description.

6.6 Fertility Capability Classification

Definition. Fertility Capability Classification (FCC) is a technical classification of soils for fertility management, in which soils are grouped by a small number of properties that directly control their fertility and fertiliser requirement - chiefly surface texture, subsoil texture and a set of fertility-limiting modifiers.

FCC was developed by Buol, Sanchez and others for tropical soils, where the usual soil survey data were not enough for fertility advice. It uses only properties that a soil surveyor can observe or measure easily, and it produces a compact symbol that tells the agronomist what fertility problem to expect.

The FCC symbol

An FCC symbol has two texture parts followed by modifiers:

- **First letter - surface texture** (topsoil, about 0 to 20 cm). S = sandy, L = loamy, C = clayey, O = organic.
- **Second letter - subsoil texture** (about 20 to 50 cm or more). S = sandy, L = loamy, C = clayey, R = rock or hardpan within 50 cm.
- **Modifiers** - small letters that state the fertility limitations.

Example: the symbol **LSe** means a loamy surface, a sandy subsoil, and the modifier 'e' for a low cation-exchange capacity (leaching-prone, low nutrient retention).

Common FCC modifiers

Modifier	Meaning	Fertility consequence
g	Gleying, or a wet, poorly drained subsoil	Reduced aeration; nitrogen loss; need for drainage
e	Low cation-exchange capacity, leaching-prone	Poor nutrient retention; split nitrogen application
a	Aluminium toxicity (very acid soil)	Root damage in acid soils; need for liming
h	Acid, low pH	Phosphorus fixation; need for lime
i	High phosphorus fixation (iron-rich, acid soils)	Phosphorus unavailable; need for higher or banded phosphorus
k	Potassium deficiency	Need for potassic fertiliser
n	High nitrate or salinity from groundwater	Salt injury; need for drainage and leaching
s	Salinity, high soluble salts	Salt injury to crops; need for leaching and drainage
d	Droughtiness, low water-holding capacity	Moisture stress; need for mulching, organic matter, irrigation
r	Shallow depth, rock or hardpan near the surface	Restricted rooting; lower nutrient and water reserve
c	Cat clay, or sulphidic material	Extreme acidity on drying; special reclamation
b	Basic reaction, high pH with free lime	Micronutrient deficiency, especially iron and zinc

Table 6.3 Common Fertility Capability Classification modifiers and their fertility consequences.

FCC compared with land capability

Point	Land Capability Classification	Fertility Capability Classification
Purpose	Conservation and general land use planning	Fertility management and fertiliser advice
Basis	Limitations to sustained use - slope, erosion, drainage, depth, climate	Surface texture, subsoil texture and fertility modifiers
Number of groups	Eight classes, with subclasses and units	A compact symbol: two texture letters plus modifiers
Scale of use	Broad - a region, a watershed, a farm	Detailed - a field or a soil mapping unit
Main output	Capability class and conservation measures	Fertility limitations and nutrient management
Typical symbol	IIIe, IVw, VIIs	LSe, CSg, LCs

Table 6.4 Land Capability Classification compared with Fertility Capability Classification.

6.7 Tamil Nadu illustration

The upland red loamy soils of the interior districts, on slopes of about 5 to 10 per cent with moderate erosion, commonly fall in capability Class IIIe or IVe and need contour bunding and strip cropping. The deep, level, well-drained deltaic alluvium of the Cauvery delta falls largely in Class I or II and can carry paddy and sugarcane with ordinary good practice. Coastal saline patches are limited by salinity and fall in Classes III to VI depending on the degree of salinity and the drainage. In FCC terms, the red sandy upland soils commonly carry the droughtiness and low-exchange modifiers, while the delta soils usually need only routine nutrient management.

Exam focus

Key points to remember

- An interpretative grouping classifies land by its behaviour under a stated use, on the basis of limitations.
- Land Capability Classification has eight classes, I to VIII, with limitation increasing and choice of safe use narrowing from I to VIII.
- Classes I to IV are arable, V to VII are for grazing and forestry, and VIII is for recreation, wildlife and water supply only.
- The subclass letter states the kind of limitation: e erosion, w wetness, s soil or depth, c climate.
- The three assumptions are sustained production, good management, and climate not limiting.
- FCC groups soils by surface texture, subsoil texture and fertility modifiers for fertility management.
- An FCC symbol is two texture letters (S, L, C, O for surface; S, L, C, R for subsoil) plus modifiers such as g, e, a, h, i, k, n, s, d, r, c, b.
- Land capability answers 'what use?'; FCC answers 'what fertility problem and what fertiliser?'.

Short-answer questions (with model answers)

Q1. Define Land Capability Classification.

Model answer. Land Capability Classification is a system of grouping land into eight classes according to the degree of limitation on its sustained use for agriculture, along with the risk of damage and the intensity of management required.

Q2. Name the eight land capability classes and state which are arable.

Model answer. The classes are I, II, III, IV, V, VI, VII and VIII. Classes I to IV are arable, that is, safe for cultivation with increasing care. Classes V to VII are suited to grazing, forestry and wildlife, and Class VIII is suited only to recreation, wildlife and water supply.

Q3. What does the subclass letter indicate? Give two examples.

Model answer. The subclass letter indicates the kind of limitation. The letter e means erosion, w means wetness or drainage, s means soil or rooting depth, and c means climate. Thus IIe is Class II limited by erosion, and IVw is Class IV limited by wetness.

Q4. What is Fertility Capability Classification?

Model answer. Fertility Capability Classification is a technical classification of soils for fertility management, in which soils are grouped by surface texture, subsoil texture and a set of fertility-limiting modifiers, giving a compact symbol that states the fertility problem to expect.

Q5. Decode the FCC symbol LSe.

Model answer. L means a loamy surface texture, S means a sandy subsoil, and e is the modifier for low cation-exchange capacity, that is, a leaching-prone soil with poor nutrient retention.

Long-answer questions (with model answer outlines)

Q1. Explain Land Capability Classification with its structure and the meaning of each class.

Outline.

- Define land capability and state its purpose - conservation and land use planning.
- State the three assumptions: sustained production, good management, climate not limiting.
- Describe the four levels: class, subclass, unit, phase.
- Explain the subclass letters e, w, s and c with examples.
- Give the eight classes in a table with limitation, cultivation suitability and conservation need.
- Explain the three groups: arable (I-IV), grazing and forestry (V-VII), and no productive use (VIII).
- Work the numerical or descriptive example giving Class IIIe.
- Give a Tamil Nadu illustration: uplands, delta, coastal saline.
- Conclude with the principle that limitation rises and choice of safe use narrows from I to VIII.

Q2. Compare Land Capability Classification and Fertility Capability Classification.

Outline.

- Define both systems.
- Purpose: conservation and general land use planning versus fertility management.
- Basis: limitations such as slope, erosion, drainage, depth and climate versus surface texture, subsoil texture and fertility modifiers.
- Structure: eight classes with subclasses and units versus a compact two-letter-plus-modifier symbol.
- Scale of use: broad (region, watershed, farm) versus detailed (field, mapping unit).
- Output: capability class and conservation measures versus fertility limitations and nutrient management.
- Give example symbols for each.
- Give a Tamil Nadu illustration of each.
- Conclude with the one-line distinction between the two questions they answer.

Lecture 7 Interpretative Groupings - Land Irrigability, Storie Index and Productivity Potential

Land capability says how good the land is for agriculture in general. It does not say whether the land can be irrigated. That is the job of **Land Irrigability Classification**. Alongside it stand two numerical ratings that give a single number for soil quality - the **Storie index** and the **productivity potential**. This lecture covers all three.

7.1 Land Irrigability Classification - concept

Definition. Land Irrigability Classification is a system of grouping land according to its suitability for sustained irrigation development, on the basis of soil properties and site features that affect irrigation, such as soil depth, texture, permeability, drainage, slope, salinity and water table.

The system is used mainly in irrigation project planning. It answers three questions: can this land be irrigated at all, how good will the irrigated land be, and what problem must be corrected before irrigation is safe? Land that is good for rainfed agriculture may still be poor for irrigation - for example a level, deep, well-drained soil with a very sandy texture loses water and nutrients too fast, while a heavy clay with slow permeability waterlogs under irrigation.

The two groups

- **Arable irrigable land** - land that can be irrigated safely and brought under cultivation, at Classes 1, 2 and 3.
- **Non-arable land** - land that cannot be irrigated for cultivation, at Classes 4, 5 and 6. It may still be used for grazing, forestry, wildlife or recreation.

7.2 The land irrigability classes

Class	Suitability for irrigation	Limitations	Usual management need
1	Excellent; highly suitable for irrigation	No significant limitation; deep, well-drained, level, good texture	Normal irrigation practice and routine management
2	Good; suitable for irrigation	Slight limitation - slightly shallow, slightly coarse or fine texture, gentle slope	Simple corrective measures: levelling, slight drainage, careful water control
3	Fair or moderate; suitable with special measures	Moderate limitation - moderate slope, moderate salinity, slow or rapid permeability, high water table	Costly or complex measures: land levelling, drainage, gypsum or leaching, careful design
4	Poor; temporarily non-irrigable	Severe limitation that may be corrected by major works - deep drainage needed, strong salinity, heavy clay	Reclamation before irrigation; irrigation only after the limitation is removed
5	Non-arable, but usable for grazing, forestry or wildlife	Very severe limitation for irrigation - very steep, very shallow, rocky or permanently wet	No irrigation for cultivation; keep under grass, trees or wildlife
6	Non-arable and not usable	Extreme limitation - rock outcrop, marsh, sand dunes, cliff	No productive use; protect for water supply, wildlife or recreation

Table 7.1 Land irrigability classes 1 to 6 with their limitations and management needs. The class limits are the standard published ranges and are fixed in detail by the project specification.

Subclasses

As in land capability, a small letter states the kind of limitation within the class:

- **s** - soil limitation: texture, depth, permeability, salinity, alkalinity.
- **t** - topographic limitation: slope, relief, erosion.

- **d** - drainage limitation: water table, flooding, waterlogging.

So Class 3s means fair irrigability limited by a soil property, and Class 4d means poor irrigability limited by drainage.

> **Irrigability is not the same as capability.** A soil can be Class I in land capability and only Class 3 in irrigability, for example a deep, level, well-drained sandy soil that is easy to farm but loses irrigation water and nutrients too fast. The two classifications ask different questions and can give different answers for the same land.

7.3 The Storie index

Definition. The Storie index is a numerical rating of the degree of suitability of a soil for general agricultural use, obtained by multiplying together the percentage ratings of a few key soil and site factors and expressing the product as a percentage.

The index was developed by R. E. Storie in California. Its logic is that the factors act together, not separately: a soil that is only 50 per cent good in depth and 50 per cent good in slope is not 100 per cent good overall but 25 per cent, because both limits operate at once. Hence the factors are **multiplied**, not added.

The formula

```
          THE STORIE INDEX

Storie index  =  A  x  B  x  C  x  X  x  100

A =  rating for PROFILE CHARACTER  (depth, texture of the profile,
     structure, permeability, drainage, salinity, stoniness)
B =  rating for SURFACE TEXTURE
C =  rating for SLOPE
X =  rating for OTHER factors (drainage, erosion, salinity, fertility,
     micro-relief) - may be split into X1, X2, X3 ...

Each rating is a decimal fraction between 0 and 1 (or a percentage
between 0 and 100). The product x 100 gives the index as a percentage.

The factor that is worst limits the result, because it multiplies
the whole product down.
```

Fig. 7.1 The Storie index and its four factor groups.

Standard Storie index ratings

Factor and description	Rating (per cent)
A - Profile character	
Deep, uniform, friable, well drained, no salinity	100
Slightly limited depth or texture, slight restriction	80
Moderately limited depth or texture, moderate restriction	60
Severely limited depth or texture, severe restriction	40
Very severely limited, very shallow or very heavy	20
B - Surface texture	
Loam, silt loam, sandy loam	100
Sandy clay loam, silty clay loam, clay loam	85
Sandy clay, silty clay, clay	70
Loamy sand, sandy loam with heavy sand	60
Sand, coarse sand	30
C - Slope	
Nearly level, 0 to 2 per cent	100

Factor and description	Rating (per cent)
Gently sloping, 2 to 5 per cent	95
Moderately sloping, 5 to 9 per cent	80
Strongly sloping, 9 to 15 per cent	60
Steep, 15 to 30 per cent	40
Very steep, above 30 per cent	20
X - Other factors (each rated separately)	
Drainage - good to excessive	100
Drainage - imperfect	80
Drainage - poor	50
Erosion - none to slight	100
Erosion - moderate	80
Erosion - severe	50
Salinity - none	100
Salinity - slight	80
Salinity - moderate	60

Table 7.2 Standard Storie index ratings for the factor groups. The precise rating within a group is fixed by the published Storie tables and varies slightly between adaptations; the values shown are the standard published ratings.

Interpreting the Storie index

Storie index (per cent)	Grade	Interpretation
80 to 100	1	Excellent land; very good for general agricultural use
60 to 79	2	Good land; suitable for most crops with ordinary management
40 to 59	3	Fair land; restricted use, needs careful management
20 to 39	4	Poor land; limited to a few uses, often best in pasture or trees
Below 20	5	Very poor land; not suited to cultivation

Table 7.3 Interpretation of the Storie index. The grade boundaries are the standard published ranges.

7.4 Worked example - computing a Storie index

A soil in an upland block has the following properties. Compute the Storie index and state its grade.

- Profile: moderately limited depth (60 to 90 cm over weathered rock) and a slightly heavy subsoil. **A = 60 per cent.**
- Surface texture: sandy loam. **B = 100 per cent.**
- Slope: 8 per cent (moderately sloping). **C = 80 per cent.**
- Drainage: good. **X1 = 100 per cent.**
- Erosion: moderate sheet erosion. **X2 = 80 per cent.**
- Salinity: none. **X3 = 100 per cent.**

Worked example 7.1 - solution

1 Write the formula: Storie index = A x B x C x X1 x X2 x X3 x 100.

2 Convert each rating to a decimal: A = 0.60, B = 1.00, C = 0.80, X1 = 1.00, X2 = 0.80, X3 = 1.00.

3 Multiply: 0.60 x 1.00 = 0.60; 0.60 x 0.80 = 0.48; 0.48 x 1.00 = 0.48; 0.48 x 0.80 = 0.384; 0.384 x 1.00 = 0.384.

4 Multiply by 100: Storie index = 0.384 x 100 = **38.4 per cent**.

5 Interpret: 38.4 falls in the range 20 to 39, so the **grade is 4, poor land**, limited to a few uses and often best kept in pasture or trees.

6 **Check the reasoning:** the two limiting factors are profile character (0.60) and erosion (0.80). If the erosion were controlled and the soil deepened by management, the index would rise: with A = 0.80 and X2 = 1.00 the index becomes 0.80 x 1.00 x 0.80 x 1.00 x 1.00 x 1.00 x 100 = **64 per cent**, which is grade 2, good land. This shows how strongly a single poor factor pulls the index down.

Common mistake. Do not add the ratings. Adding gives 0.60 + 1.00 + 0.80 + 1.00 + 0.80 + 1.00 = 5.20, which is meaningless. The Storie index multiplies, because the limitations act together.

7.5 Productivity potential

Definition. Productivity potential is an index that expresses the capacity of a soil to produce crops, as a single number or percentage, obtained by rating a set of soil properties that control plant growth and combining the ratings.

Several methods exist, and the examiner should be given the method actually used. Two families of method are common:

- **Multiplicative methods** - of which the Storie index is the best known. Ratings are multiplied, so the worst factor limits the result.
- **Weighted-sum methods** - the soil properties that control growth are each given a rating and a weight reflecting its importance, and the weighted ratings are added to give a percentage. The Riquier productivity index is a well-known example of this family.

The weighted-sum method is easy to work and easy to defend, because the weights are stated openly. The worked example below uses it.

The weighted-sum method

```
      PRODUCTIVITY POTENTIAL BY THE WEIGHTED-SUM METHOD

Productivity potential (%)  =  SUM ( weight x rating )  for all properties

Each property has:
   - a RATING from 0 to 100, from how favourable it is for crop growth
   - a WEIGHT, the sum of all weights being 1.00 (or 100 per cent)

Properties commonly used, with typical weights:
   rooting depth ............ 0.20        drainage ......... 0.10
   texture .................. 0.15        organic matter ... 0.10
   slope .................... 0.15        available nutrients 0.10
   erosion .................. 0.10        water-holding capacity 0.10

The weighted ratings are added; the result is the productivity
potential as a percentage of the best possible soil.
```

Fig. 7.2 The weighted-sum method for productivity potential.

7.6 Worked example - computing productivity potential

The same upland soil is rated for eight properties. Compute the productivity potential.

Property	Rating (0-100)	Weight	Weight x rating
Rooting depth (60-90 cm)	70	0.20	14.0
Texture (sandy loam)	90	0.15	13.5
Slope (8 per cent)	70	0.15	10.5

Property	Rating (0-100)	Weight	Weight x rating
Erosion (moderate)	60	0.10	6.0
Drainage (good)	100	0.10	10.0
Organic matter (low, 0.4 per cent)	50	0.10	5.0
Available nutrients (low)	60	0.10	6.0
Water-holding capacity (medium)	75	0.10	7.5
Total		**1.00**	**72.5**

Table 7.4 Property ratings, weights and weighted ratings for worked example 7.2.

Worked example 7.2 - solution

1. Check that the weights sum to 1.00: 0.20 + 0.15 + 0.15 + 0.10 + 0.10 + 0.10 + 0.10 + 0.10 = 1.00. Correct.
2. Multiply each rating by its weight and write the product in the last column, as shown in the table.
3. Add the weighted ratings: 14.0 + 13.5 + 10.5 + 6.0 + 10.0 + 5.0 + 6.0 + 7.5 = **72.5**.
4. The productivity potential is therefore **72.5 per cent** - about three-quarters of the potential of an ideal soil for the same crop.
5. **Interpretation.** The soil is moderately productive. The weakest contributions are organic matter (5.0), available nutrients (6.0) and erosion (6.0), so the best returns to management would come from adding organic matter and fertiliser and from controlling erosion.
6. **Check against the Storie index.** The Storie index for the same soil was 38.4 per cent (grade 4). The two numbers differ because the two methods weight the factors differently and combine them differently. Both are correct for their own purpose; they must never be compared as if they were the same scale.

7.7 Tamil Nadu illustration

In Tamil Nadu irrigation planning, the command areas of the Cauvery delta have large stretches of deep, level, well-drained alluvium that rate as irrigability Class 1 or 2 and carry paddy and sugarcane safely. Towards the tail ends and the coastal margins, salinity, a high water table and slow drainage reduce large areas to Class 3 or 4, where drainage, gypsum application or leaching is needed before irrigation is safe. In the uplands, red loamy soils with moderate slope and erosion rate as irrigability Class 3t or 3s, and irrigation there needs land levelling and erosion control along with water. The Storie index and productivity potential are used in the same areas to rank soils for crop planning and to compare the value of alternative land improvements.

Exam focus

Key points to remember

- Land Irrigability Classification groups land by its suitability for sustained irrigation, on soil and site features that affect irrigation.
- There are six irrigability classes: 1 excellent, 2 good, 3 fair or moderate, 4 poor or temporarily non-irrigable, 5 non-arable but usable for grazing or forestry, 6 non-arable and not usable.
- Subclass letters are s for soil, t for topography and d for drainage.
- Irrigability is not the same as capability; the same land can differ in the two systems.
- Storie index = A x B x C x X x 100, where A is profile character, B surface texture, C slope and X other factors such as drainage, erosion and salinity.
- The Storie index multiplies its ratings because the limitations act together; the worst factor limits the result.

- Storie index grades: 80-100 grade 1 excellent, 60-79 grade 2 good, 40-59 grade 3 fair, 20-39 grade 4 poor, below 20 grade 5 very poor.
- Productivity potential combines property ratings, either by multiplication (Storie-type) or by a weighted sum, to give a percentage of the best possible soil.

Short-answer questions (with model answers)

Q1. Define Land Irrigability Classification.

Model answer. Land Irrigability Classification is a system of grouping land according to its suitability for sustained irrigation development, on the basis of soil properties and site features such as depth, texture, permeability, drainage, slope, salinity and water table.

Q2. Name the six land irrigability classes.

Model answer. Class 1 excellent, Class 2 good, Class 3 fair or moderate, Class 4 poor or temporarily non-irrigable, Class 5 non-arable but usable for grazing, forestry or wildlife, and Class 6 non-arable and not usable.

Q3. What do the letters s, t and d mean in irrigability subclasses?

Model answer. s means a soil limitation such as texture, depth, permeability, salinity or alkalinity; t means a topographic limitation such as slope, relief or erosion; and d means a drainage limitation such as a high water table, flooding or waterlogging.

Q4. Write the formula of the Storie index and explain each term.

Model answer. Storie index = A x B x C x X x 100. A is the rating for profile character, B the rating for surface texture, C the rating for slope, and X the rating for other factors such as drainage, erosion, salinity and fertility. Each rating is a fraction between 0 and 1.

Q5. Why are the factors of the Storie index multiplied and not added?

Model answer. Because the limitations act together. A soil that is only half good in depth and half good in slope is not fully good overall, because both limits operate at the same time. Multiplication makes the worst factor limit the result, which is the correct behaviour.

Long-answer questions (with model answer outlines)

Q1. Explain Land Irrigability Classification and distinguish it from Land Capability Classification.

Outline.

- Define land irrigability and state its purpose in project planning.
- State the three questions it answers.
- List the six classes with their suitability and limitations.
- Explain the subclass letters s, t and d.
- Define land capability and recall its eight classes briefly.
- Compare the two: purpose, basis, number of classes, scale of use, and the kind of decision each supports.
- Give an example where the two systems differ for the same land - a level sandy soil.
- Give a Tamil Nadu illustration from the delta command area and the uplands.
- Conclude with the one-line distinction.

Q2. Explain the Storie index, its rating factors, and work an example.

Outline.

- Define the Storie index and state its purpose.
- Write the formula and explain A, B, C and X.
- Explain why the factors are multiplied.

- Give the standard ratings for profile character, surface texture, slope, drainage, erosion and salinity.
- Give the interpretation grades and their index ranges.
- Work the numerical example: A 0.60, B 1.00, C 0.80, X1 1.00, X2 0.80, X3 1.00, giving 38.4 per cent, grade 4.
- Show how improving the two limiting factors raises the index to 64 per cent.
- Explain productivity potential briefly and the difference from the Storie index.
- Give a Tamil Nadu illustration.

Lecture 8 Land Suitability Classification

Land capability asks how good the land is for agriculture in general. Land suitability asks a narrower and more useful question: how good is this land for **this particular crop or tree**? The same soil can be highly suitable for paddy and poorly suitable for cotton. This lecture explains the framework of land suitability classification and applies it to field crops, horticultural crops and forest trees.

8.1 What land suitability classification is

Definition. Land suitability classification is the process of assessing and grouping land according to its suitability for a specified kind of use, by comparing the requirements of that use with the properties of the land.

The framework used in India is the FAO framework for land evaluation. It is a **specific-purpose** classification: the assessment is made for one stated use, and it is valid only for that use. Land is compared against the crop's own requirements, so a soil that fails for one crop may pass for another.

The two sides of the comparison

- **Land qualities and land characteristics** - the properties of the land that affect the crop: climate, slope, drainage, soil depth, texture, fertility, salinity, flooding, erosion and so on.
- **Land use requirements** - the conditions the crop needs: its water requirement, rooting depth, tolerance of salinity and waterlogging, temperature range and length of growing season.
- The assessment is the match between the two. Where the land meets the requirement, the land is suitable; where it fails, the land is limited for that use.

8.2 The structure of the FAO suitability classification

Orders

- **Order S - Suitable.** Land on which the use can be sustained without damage to the land resource, and with benefits that justify the input.
- **Order N - Not suitable.** Land on which the use cannot be sustained without damage, or which cannot produce a worthwhile return.

Classes

Class	Name	Meaning	Inputs needed
S1	Highly suitable	Land with no significant limitation for the specified use, or with only minor limitations that do not reduce yield or raise inputs appreciably	Normal management for the crop
S2	Moderately suitable	Land with limitations that moderately reduce productivity or increase inputs, but that can be corrected by accepted practice	Moderate additional inputs and management
S3	Marginally suitable	Land with severe limitations that substantially reduce productivity or increase inputs, so that the use is only just worthwhile	Major inputs and careful management; the return is marginal
N1	Currently not suitable	Land with limitations that cannot be corrected with present knowledge or at present cost, but that could be corrected in future	Reclamation needed before use
N2	Permanently not suitable	Land with limitations so severe that they cannot be corrected, for example rock outcrop or a cliff	No use possible for this crop

Table 8.1 Land suitability classes S1, S2, S3, N1 and N2.

Subclasses

Within a class, a small letter states the kind of limitation. The common ones are:

- **t** - topographic limitation: slope, relief.
- **w** - wetness limitation: drainage, water table, flooding.
- **s** - soil limitation: depth, texture, stoniness, salinity, alkalinity, fertility.
- **r** - rooting or depth limitation.
- **c** - climatic limitation: rainfall, temperature, length of season.
- **e** - erosion limitation.

So S2s means moderately suitable with a soil limitation, and S3w means marginally suitable with a wetness limitation.

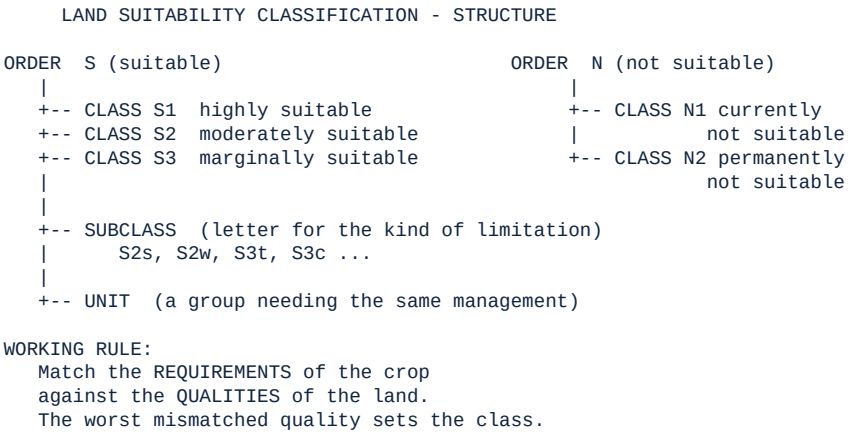


Fig. 8.1 The structure of land suitability classification - order, class, subclass and unit.

8.3 Land suitability for field crops

Field crops are annual or seasonal crops grown on arable land - cereals, pulses, oilseeds, fibre crops and fodder. Their requirements differ widely: paddy needs standing water and heavy soil, cotton needs deep well-drained soil with a long season, groundnut needs a light friable soil that lifts cleanly at harvest.

Crop	Soil depth	Texture	Drainage	Slope	pH	Key sensitivity
Paddy (rice)	More than 50 cm	Clay loam to clay	Poor to imperfect (retains water)	Level, 0-2%	5.5-8.0	Needs water retention; sensitive to salinity
Sorghum	More than 60 cm	Sandy loam to clay loam	Well drained	0-5%	6.0-8.5	Tolerant of moderate drought and shallow soil
Maize	More than 60 cm	Sandy loam to clay loam	Well drained	0-8%	5.5-8.0	Needs good fertility and moisture
Groundnut	More than 60 cm	Sandy loam, loose, friable	Well drained	0-5%	6.0-7.5	Needs free soil for pod development; sensitive to calcium deficiency
Cotton	More than 90 cm	Clay loam to clay (black soils favoured)	Well drained	0-5%	6.5-8.5	Long growing season; sensitive to waterlogging
Sugarcane	More than 100 cm	Loam to clay loam	Well drained	0-3%	6.0-8.0	Very high water requirement; long season
Pulses (redgram, blackgram)	More than 45 cm	Sandy loam to clay loam	Well drained	0-8%	6.0-8.0	Tolerant of poor soils; sensitive to waterlogging

Crop	Soil depth	Texture	Drainage	Slope	pH	Key sensitivity
Millets (ragi, cumbu)	More than 30 cm	Sandy loam to loam	Well drained	0-10%	5.5-8.0	Hardy; suited to rainfed uplands

Table 8.2 Typical requirement ranges of common field crops. Ranges are the commonly used published values and vary with variety and climate; a specific crop requirement table is always used in a formal evaluation.

Worked example 8.1 - suitability of a soil for groundnut

A red loamy upland soil has the following properties. Assess its suitability for groundnut.

- Depth: 75 cm over weathered rock.
- Surface texture: sandy loam, friable.
- Drainage: well drained.
- Slope: 6 per cent.
- pH: 6.5.
- Erosion: slight to moderate.
- Salinity: none.

Assessment.

1 **Depth.** More than 60 cm is required; 75 cm is met, though not generously. This is a slight limitation.

2 **Texture.** Sandy loam, friable, is ideal for pod development. No limitation.

3 **Drainage.** Well drained is required and present. No limitation.

4 **Slope.** Groundnut is best on 0 to 5 per cent; 6 per cent is slightly above the ideal, and with slight to moderate erosion it is a slight to moderate topographic and erosion limitation.

5 **pH.** 6.5 is within the ideal 6.0 to 7.5 range. No limitation.

6 **Worst limiting quality.** The combination of slope (6 per cent) and slight to moderate erosion gives a moderate limitation of the topographic and erosion kind.

7 **Class.** With one moderate limitation that can be corrected by contour bunding and contour cultivation, the land is **S2t** - moderately suitable with a topographic limitation.

8 **Management to reach S1.** Contour bunding, contour cultivation and a cover crop would remove the slope limitation and raise the land towards S1 for groundnut.

8.4 Land suitability for horticultural crops

Horticultural crops are perennial or long-duration crops - fruit trees, plantation crops, vegetables and flowers. Their requirements differ from field crops in three ways that matter for suitability:

- **Depth.** A fruit tree needs a deep soil, usually more than 100 cm, because its permanent root system must support the tree for decades.
- **Drainage.** Most fruit and plantation crops are very sensitive to waterlogging; even short periods of standing water can kill the roots.
- **Quality factors.** For horticultural crops, suitability is judged not only by yield but by quality - size, colour, taste, keeping quality and market value - and quality is often controlled by soil and climate.

Crop	Soil depth	Texture	Drainage	Slope	pH	Key sensitivity
Mango	More than 150 cm	Loam to clay loam, deep and friable	Well drained	0-10%	5.5-7.5	Very sensitive to waterlogging; needs deep soil

Crop	Soil depth	Texture	Drainage	Slope	pH	Key sensitivity
Banana	More than 100 cm	Loam to silt loam, rich	Well drained	0-5%	6.0-7.5	High water and nutrient demand; sensitive to drainage
Coconut	More than 100 cm	Sandy loam to loam	Well drained	0-8%	5.5-8.0	Tolerant of coastal conditions; needs good drainage
Citrus	More than 100 cm	Sandy loam to loam	Well drained	0-8%	5.5-7.0	Sensitive to salinity and to poor drainage
Grapes	More than 100 cm	Sandy loam to loam, well aerated	Well drained	0-8%	6.5-8.0	Needs good drainage and a dry ripening period
Tomato	More than 60 cm	Sandy loam to loam	Well drained	0-5%	6.0-7.0	Sensitive to waterlogging and to salinity
Tea	More than 100 cm	Deep, friable, acidic	Well drained	0-25% (hill slopes)	4.5-5.5	Needs acid soil, high rainfall, good drainage
Coffee	More than 100 cm	Deep loam, rich in organic matter	Well drained	0-20%	5.5-6.5	Needs shade and well-drained upland soil

Table 8.3 Typical requirement ranges of common horticultural and plantation crops. Ranges are the commonly used published values and vary with variety, rootstock and climate.

Worked example 8.2 - suitability for mango

A shallow, imperfectly drained soil on a gentle slope has a depth of 80 cm, a clay loam texture, imperfect drainage and a pH of 7.0. Assess its suitability for mango.

1. **Depth.** Mango needs more than 150 cm; 80 cm is about half the requirement. This is a severe rooting limitation, so the subclass letter is r.
2. **Drainage.** Mango is very sensitive to waterlogging; imperfect drainage is a severe wetness limitation, subclass letter w.
3. **Texture and pH.** Clay loam and pH 7.0 are acceptable. No limitation.
4. **Class.** Two severe limitations that can be corrected only by costly measures - deep drainage and possibly deep ripping or mounding - place the land in **N1**, currently not suitable, or at best **S3wr** if drainage is installed and the trees are planted on mounds.
5. **Better crop choice.** Because the limitation is crop-specific, the same land may be **S2** for banana, which is shallower-rooted, or **S1** for a shallow-rooted vegetable. This is the practical value of suitability classification: it steers the choice of crop to the land.

8.5 Land suitability for forest trees

Forest trees are grown for timber, fuel, pulpwood and protection. Their requirements differ from those of crops:

- **Tolerance.** Forest species tolerate shallower, stonier, steeper and poorer soils than crops. Land in capability Classes VI and VII, which cannot be cultivated, is often suitable for forest trees.
- **Depth.** Deep-rooted timber species need more than 100 cm; hardy species such as casuarina and eucalyptus tolerate 50 to 75 cm.
- **Slope.** Forest trees are the correct use for steep land, where cultivation would cause erosion.
- **Purpose.** Suitability is judged by growth rate and by the intended product - timber, fuel, pulp or protection.

Species	Soil depth	Texture	Drainage	Slope	pH	Key sensitivity
Teak	More than 100 cm	Loam to clay loam, well aerated	Well drained	0-25%	6.0-7.5	Needs deep, well-drained soil; sensitive to waterlogging
Eucalyptus	More than 75 cm	Sandy loam to loam	Well drained	0-20%	5.5-8.0	Hardy; high water use; tolerant of poor soil
Casuarina	More than 50 cm	Sandy loam, including coastal sand	Well drained	0-10%	6.0-8.5	Tolerant of coastal sand and salinity; sensitive to waterlogging
Acacia (e.g. Acacia nilotica)	More than 60 cm	Loam to clay loam	Moderate to well drained	0-10%	6.0-9.0	Tolerant of salinity and drought
Bamboo	More than 60 cm	Loam, rich in organic matter	Well drained	0-15%	5.5-7.5	Needs moisture; sensitive to waterlogging
Silver oak (Grevillea robusta)	More than 75 cm	Loam to clay loam	Well drained	0-25%	5.5-7.5	Good for shade and timber on upland slopes

Table 8.4 Typical requirement ranges of common forest and farm-forestry tree species. Ranges are the commonly used published values and vary with provenance and site.

Why the same land can be S1 for one crop and N1 for another. Suitability is judged against the requirements of the crop, not the land in the abstract. A shallow, imperfectly drained clay loam is N1 for mango but may be S1 for paddy. This is exactly why a suitability classification must always state the crop and the level of management assumed.

8.6 Steps in a land suitability assessment

1. **State the land use** being assessed - the crop, the cropping system or the tree species, and the level of inputs assumed.
2. **List the land use requirements** of that use - water, depth, drainage, slope, pH, salinity tolerance and climate.
3. **List the land qualities and characteristics** of each mapping unit from the soil survey data.
4. **Match** the two, quality by quality, and rate each as highly suitable, moderately suitable, marginally suitable or not suitable.
5. **Take the worst limiting quality** to set the class of the unit.
6. **Assign the subclass letter** for the kind of limitation.
7. **State the management** needed to move the land to a higher class, and whether it is feasible and economic.

8.7 Tamil Nadu illustration

In the Cauvery delta, deep level alluvium with assured irrigation is highly suitable (S1) for paddy and sugarcane, and moderately suitable (S2) for banana where drainage is imperfect. On the red loamy uplands of the interior districts, soils of 60 to 90 cm depth on gentle slopes are moderately suitable (S2) for groundnut, millets and pulses, and marginally suitable (S3) for mango because of depth. The hilly tracts of the Nilgiris, Shevaroys and Javadhu hills, too steep and shallow for cultivation, are suitable for tea and coffee where the slope is moderate and well drained, and for forest trees and farm forestry where the slope is steep. Coastal sandy tracts suit casuarina and coconut better than most field crops.

Exam focus

Key points to remember

- Land suitability classification groups land by its suitability for a specified use, by matching the crop's requirements against the land's qualities.
- Orders: S suitable and N not suitable.
- Classes: S1 highly suitable, S2 moderately suitable, S3 marginally suitable, N1 currently not suitable, N2 permanently not suitable.
- Subclass letters state the kind of limitation: t topography, w wetness, s soil, r rooting, c climate, e erosion.
- The worst limiting quality sets the class of the land.
- Field crops vary widely: paddy needs water retention, groundnut a friable well-drained soil, cotton depth and a long season.
- Horticultural crops need deeper soil and better drainage than field crops and are judged also on quality.
- Forest trees tolerate shallower, stonier and steeper land and are the correct use for land in capability Classes VI and VII.

Short-answer questions (with model answers)

Q1. Define land suitability classification.

Model answer. Land suitability classification is the process of assessing and grouping land according to its suitability for a specified kind of use, by comparing the requirements of that use with the properties of the land.

Q2. Name the land suitability orders and classes.

Model answer. The orders are S, suitable, and N, not suitable. The classes are S1 highly suitable, S2 moderately suitable, S3 marginally suitable, N1 currently not suitable, and N2 permanently not suitable.

Q3. What do the subclass letters t, w, s and c mean?

Model answer. t means a topographic limitation such as slope or relief; w means a wetness limitation such as drainage, water table or flooding; s means a soil limitation such as depth, texture, stoniness or salinity; and c means a climatic limitation such as rainfall, temperature or length of growing season.

Q4. Why do horticultural crops need deeper soil than field crops?

Model answer. Because fruit and plantation crops are perennial. Their permanent root system must anchor and feed the tree for many years, so they need a deep soil, usually more than 100 cm, and good drainage. Most are also very sensitive to waterlogging.

Q5. Why are forest trees suited to land that cannot be cultivated?

Model answer. Because forest species tolerate shallower, stonier, steeper and poorer soils than crops do. Steep land in capability Classes VI and VII, where cultivation would cause erosion, can carry forest trees, which protect the soil while producing timber, fuel or pulp.

Long-answer questions (with model answer outlines)

Q1. Explain land suitability classification and apply it to field crops.

Outline.

- Define land suitability classification and state that it is specific-purpose.
- Explain the comparison between land qualities and land use requirements.
- Give the orders S and N.
- Give the classes S1, S2, S3, N1, N2 with meaning and inputs needed.
- Explain the subclass letters t, w, s, r, c, e.
- Give the requirement table for field crops such as paddy, cotton, groundnut and sugarcane.

- Work the groundnut example, ending in S2t.
- State the rule that the worst limiting quality sets the class.
- Give a Tamil Nadu illustration from the delta and the uplands.

Q2. Compare land suitability for field crops, horticultural crops and forest trees.

Outline.

- Define land suitability and the FAO framework briefly.
- Field crops: annual, need moderate depth, wide range of soil tolerance, judged mainly on yield.
- Horticultural crops: perennial, need depth more than 100 cm and good drainage, judged also on quality, very sensitive to waterlogging.
- Forest trees: perennial, tolerate shallow, stony, steep and poor soil, suited to non-arable capability classes, judged by growth and product.
- Give one requirement table or example for each group.
- Work the mango example showing N1 or S3wr.
- Explain why the same land can be S1 for one crop and N1 for another.
- Give a Tamil Nadu illustration covering all three groups.
- Conclude with the steps of a suitability assessment.

Lecture 9 Mid Semester Examination

The ninth lecture in the syllabus is the mid semester examination. It carries no new theory. It tests Lectures 1 to 8 - soil resource inventory and soil systematics, standard soil survey, soil mapping units, methods and types of soil survey, and the interpretative groupings of soil. This chapter is a revision aid: a one-page summary of the first eight lectures, a model mid semester question paper, and model answers and answer outlines.

9.1 Revision summary of Lectures 1 to 8

Lecture	Core definition	Must-remember key or table
1	Soil resource inventory is the systematic survey, identification, classification, mapping and evaluation of the soils of an area	Pedon (1-10 m2) vs polypedon (1-10 ha); control section 25-100 cm for a soil family; correlation levels
2	Standard soil survey is the systematic examination, description, classification and mapping of soils by a uniform procedure	Fundamental vs applied objectives; the eight-step survey process; minimum legible area
3	A soil mapping unit is a delineated area on the soil map identified by a symbol and described in the legend	Series, association (regular pattern), complex (irregular pattern), phase, variant, inclusion, miscellaneous land types
4	A base map is the map on which soil boundaries are drawn; traversing is the planned movement across the area to fix boundaries	Grid survey vs free survey; Simpson's rule for area
5	The four survey types differ in scale, intensity, mapping unit and purpose	Detailed 1:5,000-1:10,000; semi-detailed 1:20,000-1:50,000; detailed-reconnaissance 1:50,000-1:100,000; reconnaissance 1:100,000-1:250,000
6	An interpretative grouping classifies land by its behaviour under a stated use	Land capability classes I-VIII with subclasses e, w, s, c; FCC symbol of two texture letters plus modifiers
7	Land irrigability groups land by suitability for sustained irrigation; Storie index is a numerical rating of soil quality	Irrigability classes 1-6 with s, t, d; Storie index = A x B x C x X x 100; grades 1 to 5
8	Land suitability groups land by suitability for a specified use, matching crop requirements against land qualities	Orders S and N; classes S1, S2, S3, N1, N2; subclasses t, w, s, r, c, e

Table 9.1 One-line revision summary of Lectures 1 to 8.

9.2 Model mid semester question paper

Time: three hours. Maximum marks: 100. The pattern below matches the usual university pattern: objective and short-answer questions in Part A, medium-length questions in Part B, and long-answer questions with a choice in Part C.

Part A - Answer all questions (10 x 2 = 20 marks)

1 Define soil resource inventory.

2 Distinguish between a pedon and a polypedon.

3 What is a control section? State the control section used for the texture of a soil family.

4 Name any four miscellaneous land types.

5 Distinguish a soil association from a soil complex.

6 What is the minimum legible area and how does it depend on map scale?

7 Name the eight land capability classes.

8 Write the formula of the Storie index and explain the term A.

9 Decode the Fertility Capability Classification symbol LSe.

10 Name the five land suitability classes.

Part B - Answer any five questions (5 x 6 = 30 marks)

1 Explain the principles of soil resource inventory.

2 Explain grid survey and free survey, and state when each is preferred.

3 Describe the four types of soil survey and give the scale of each.

4 Explain land capability classes I to IV with their limitations and conservation needs.

5 Explain the six land irrigability classes and the subclass letters s, t and d.

6 Explain why horticultural crops need deeper soil than field crops, with examples.

Part C - Answer any two questions (2 x 25 = 50 marks)

1 Define standard soil survey. State its scope and objectives, and describe the step-by-step process of a standard soil survey.

2 Explain soil mapping units - soil series, soil association, soil complex, variants, inclusions and miscellaneous land types - and give the rule for choosing each.

3 Explain Land Capability Classification and Land Suitability Classification, and distinguish between them.

4 Explain the Storie index and productivity potential, and compute a Storie index for a soil with A = 0.60, B = 1.00, C = 0.80, X1 = 1.00, X2 = 0.80 and X3 = 1.00.

9.3 Model answers - Part A

1 **Soil resource inventory** is the systematic survey, identification, classification, mapping and evaluation of the soils of an area to record their kind, extent, distribution, properties and potential for use.

2 **Pedon** is the smallest three-dimensional soil individual, about 1-10 m2, the unit of classification. **Polypedon** is a group of contiguous similar pedons, about 1-10 ha, the unit of mapping.

3 The **control section** is the fixed depth interval used for a classification decision. For the texture of a soil family it is 25-100 cm.

4 Any four miscellaneous land types: rock outcrop, stony land, gullied land, marsh and swamp, riverwash, beaches and sand dunes, made land and mine spoil, urban and built-up land, water bodies.

5 A **soil association** has two or more soils in a regular, landscape-related repeating pattern. A **soil complex** has two or more soils in an irregular, unpredictable pattern.

6 The **minimum legible area** is the smallest area that can be shown on a map at a given scale. The larger the scale, the smaller the minimum legible area, so more detail can be shown.

7 Land capability classes: I, II, III, IV, V, VI, VII and VIII.

8 **Storie index = A x B x C x X x 100.** A is the rating for profile character - depth, texture, structure, permeability, drainage, salinity and stoniness.

9 **LSe** means a loamy surface texture, a sandy subsoil, and the modifier e for low cation-exchange capacity, that is, a leaching-prone soil with poor nutrient retention.

10 Land suitability classes: S1 highly suitable, S2 moderately suitable, S3 marginally suitable, N1 currently not suitable, N2 permanently not suitable.

9.4 Model answers - Part B

Q1. Principles of soil resource inventory

- Soil is a natural body produced by climate, organisms, relief, parent material and time, so it is predictable.

- Soils repeat in an orderly pattern on the landscape, which is the basis of free survey.
- Soil boundaries are gradational, not sharp lines; the line is drawn through the zone of greatest change.
- The soil individual is definable through the pedon and groupable through the polypedon.
- The smallest mappable area depends on map scale, so small bodies become inclusions or complexes.
- Survey intensity must match the purpose.
- Classification precedes mapping.
- The survey model is a hypothesis, tested and refined in the field.
- The inventory must be interpreted before it is useful.

Q2. Grid survey and free survey

- Define both methods.
- Grid survey uses a fixed regular grid of observation points; free survey places points by judgement along expected boundaries.
- Grid survey is uniform, objective, statistically usable and easy to supervise, but wasteful in simple terrain, unsuitable in dissected land, and may miss a boundary between two points.
- Free survey is efficient where the soil-landscape relationship is clear and finds boundaries accurately, but depends on the surveyor's skill, is less objective and is harder to check.
- Grid survey suits level to gently undulating land and statistical work; free survey suits all terrain and is the main method in Indian surveys, with check traverses.

Q3. The four types of soil survey

- **Detailed:** scale 1:5,000 to 1:10,000; mapping unit is series and phases; minimum legible area about 0.1-0.4 ha; for farm and watershed planning.
- **Semi-detailed:** scale about 1:20,000 to 1:50,000; series, associations and complexes; for project and command area planning.
- **Detailed-reconnaissance:** scale about 1:50,000 to 1:100,000; associations and important series; for block and taluk planning; detailed along traverses, reconnaissance between them.
- **Reconnaissance:** scale 1:100,000 to 1:250,000 and smaller; associations and broad groups; for regional and national planning.
- Conclude that reconnaissance precedes detail and that the survey type is chosen to match the decision it must support.

Q4. Land capability classes I to IV

- **Class I:** no significant limitation; safe for all common crops; ordinary good farming practice.
- **Class II:** slight limitation - gentle slope, slight erosion, moderate depth, occasional wetness; simple practices such as contour farming and cover crops.
- **Class III:** moderate limitation - moderate slope and erosion, shallower depth, poor drainage or salinity; complex practices such as terracing, strip cropping, drainage and grassed waterways.
- **Class IV:** severe limitation - steep slope, severe erosion, very shallow depth, frequent flooding or strong salinity; intensive practices, and the land is often best kept in grass or trees.
- Conclude that limitation increases and choice of safe use narrows from I to IV.

Q5. Land irrigability classes and subclasses

- Define land irrigability classification.
- **Class 1** excellent, **Class 2** good, **Class 3** fair or moderate - all arable irrigable.

- **Class 4** poor or temporarily non-irrigable, **Class 5** non-arable but usable for grazing or forestry, **Class 6** non-arable and not usable.
- **s** is a soil limitation such as texture, depth, permeability or salinity; **t** is a topographic limitation such as slope or relief; **d** is a drainage limitation such as a high water table or flooding.
- Give an example such as Class 3s for a fair land limited by salinity, and note that irrigability is not the same as capability.

Q6. Why horticultural crops need deeper soil

- Horticultural crops are perennial; their root system must support the plant for many years, so they need depth, usually more than 100 cm.
- Most fruit and plantation crops are very sensitive to waterlogging, so they also need good drainage.
- Suitability is judged not only on yield but on quality - size, colour, taste and keeping quality.
- Examples: mango needs more than 150 cm and good drainage; banana more than 100 cm; tea needs deep, friable, acid, well-drained hill soil.
- Contrast with field crops: paddy does well on shallow, poorly drained clay because it needs water retention.

9.5 Model answer outlines - Part C

Q1. Standard soil survey - scope, objectives and process

- Define standard soil survey and explain the word 'standard'.
- Scope - features studied: morphology, physical properties, chemical properties, site features, distribution.
- Scope - outputs: map, legend, report, interpretation, data sheets.
- Fundamental objectives: identify and describe, classify, map, explain genesis, establish soil-landscape relationships, keep a baseline record.
- Applied objectives: land use planning, irrigation and drainage, conservation, fertiliser management, crop suitability, watershed development, valuation and records, engineering uses, environment.
- Process: reconnaissance, base map, traversing, classification and correlation, mapping, laboratory analysis, interpretation, publication.
- Explain scale and minimum legible area.
- Give a Tamil Nadu illustration from the delta and district surveys.

Q2. Soil mapping units and the rule for choosing each

- Define a soil mapping unit and explain consociation and compound unit.
- Soil series: definition and defining features.
- Soil phase and variant: definitions and examples.
- Soil association: two or more soils, regular landscape-related pattern.
- Soil complex: two or more soils, irregular pattern.
- Inclusions: minor bodies below the minimum legible area; effect on purity.
- Miscellaneous land types: definition and examples.
- Rule: choice depends on the complexity of the soil pattern and the survey scale.
- Tamil Nadu illustration from the uplands and the delta.

Q3. Land Capability Classification and Land Suitability Classification

- Define both systems.
- Capability: general purpose, eight classes I to VIII, subclasses e, w, s, c; assumes sustained production, good management and climate not limiting.
- Suitability: specific purpose, orders S and N, classes S1, S2, S3, N1, N2, subclasses t, w, s, r, c, e; matched to one stated crop.
- Compare purpose, basis, number of groups, scale of use and output.
- Give an example where they differ: a level, deep sandy soil is capability Class I but may be only irrigability Class 3 and only marginally suitable for a crop needing heavy soil.
- Tamil Nadu illustration.
- Conclude with the one-line distinction: capability answers 'what use in general?', suitability answers 'how good for this crop?'.

Q4. Storie index and productivity potential, with computation

- Define the Storie index and state its purpose.
- Write the formula A x B x C x X x 100 and explain each factor.
- Explain why the factors are multiplied.
- Give the standard ratings and the grade table.
- Compute: A 0.60, B 1.00, C 0.80, X1 1.00, X2 0.80, X3 1.00.
- Product = 0.60 x 1.00 x 0.80 x 1.00 x 0.80 x 1.00 = 0.384, so the index is **38.4 per cent**, grade 4, poor land.
- Show the improvement: with A 0.80 and X2 1.00 the index is 64 per cent, grade 2.
- Define productivity potential and give the weighted-sum method with the worked figure of 72.5 per cent.
- State the warning that the two numbers are not on the same scale and must not be compared directly.

Exam focus

Key points to remember

- The mid semester examination tests Lectures 1 to 8 only.
- Definitions carry easy marks: soil resource inventory, standard soil survey, soil mapping unit, base map, traversing, the interpretative groupings.
- The classification keys - capability I-VIII, irrigability 1-6, suitability S1-N2, the Storie grade table - are the most heavily examined items.
- Pedon and polypedon, and the control section of 25-100 cm, are the most frequently asked distinctions from Lecture 1.
- Grid survey versus free survey, and the four survey types with their scales, are the most frequently asked comparisons.
- The Storie index is computed by multiplication; the worst factor limits the result.
- Every long answer should end with a Tamil Nadu illustration.

Short-answer questions (with model answers)

Q1. How many marks does the mid semester examination carry and what does it cover?

Model answer. It is usually a three-hour paper of 100 marks covering Lectures 1 to 8: soil resource inventory and soil systematics, standard soil survey, soil mapping units, methods and types of soil survey, and the interpretative groupings of soil.

Q2. Name the two most commonly asked distinctions from Lecture 1.

Model answer. The pedon versus the polypedon, and the soil profile versus the control section. The pedon is about 1-10 m2 and the polypedon about 1-10 ha; the control section for a soil family is 25-100 cm.

Q3. Which three tables must be memorised for the mid semester examination?

Model answer. The land capability classes I to VIII with their limitations, the land irrigability classes 1 to 6, and the land suitability classes S1 to N2. The four survey types comparison and the Storie index grade table are also frequently asked.

Long-answer questions (with model answer outlines)

Q1. Prepare a revision plan for the mid semester examination.

Outline.

- List the eight lectures covered.
- Lecture 1: definitions, pedon, polypedon, control section, correlation.
- Lecture 2: standard survey, scope, fundamental and applied objectives, process.
- Lecture 3: mapping units and the rule for choosing each.
- Lecture 4: base maps, traversing, grid and free survey, Simpson's rule.
- Lecture 5: the four survey types and their scales.
- Lecture 6: land capability classes and FCC.
- Lecture 7: irrigability classes, Storie index, productivity potential.
- Lecture 8: land suitability classes and crop requirement tables.
- State a time allocation and the order of revision, ending with the numerical and table items which carry the most marks.

Lecture 10 Land Use Planning

The soil survey, the capability classification and the suitability classification all exist to serve one purpose: to decide, on evidence, how each piece of land should be used. That decision-making process is **land use planning**. This lecture explains its concepts and objectives, and then the differences in planning for tropical, subtropical and temperate regions.

10.1 Land use planning - definition

Definition. Land use planning is the systematic assessment of land and its resources, and the choice and design of land uses and management practices that will best meet the needs of the people while conserving the land resource for the future.

A shorter exam definition: land use planning is the process of deciding the best use of each area of land, on the basis of its capability and suitability, in the interest of the people and of sustained productivity.

10.2 Concepts of land use planning

1. **Land is a resource with limits.** Every piece of land can support only a certain intensity of use without damage. Planning works within those limits.
2. **Land has many possible uses.** The same land can be used for crops, grazing, forestry, horticulture, industry, housing or recreation. Planning chooses among them.
3. **The best use is the one that fits the land.** The correct use is the one whose requirements the land can meet without degradation. This is the principle of **land suitability** applied at the planning scale.
4. **Land use must be matched to land capability.** Land that is not capable of cultivation must not be cultivated, however great the demand.
5. **Planning is for people.** Land use must meet human needs - food, fodder, fuel, fibre, income and living space - and must be socially acceptable.
6. **Planning is for the long term.** Sustained productivity, not a single season's return, is the test.
7. **Land use is a system.** Crops, livestock, forestry, water and human settlement are linked; a change in one affects the others. Planning therefore works on the whole system, usually the **watershed**.
8. **Planning needs alternatives.** Good planning compares two or more possible uses and chooses on stated criteria, not on habit.
9. **Planning is participatory.** The people who use the land must be part of the decision, or the plan will not be followed.
10. **Planning is iterative.** A plan is monitored, evaluated and revised as conditions change.

10.3 Objectives of land use planning

- **To select the best use for each area of land**, on the evidence of soil survey, capability and suitability.
- **To increase and sustain production** of food, fodder, fuel, fibre and timber to meet present and future needs.
- **To conserve the soil and water resource** - to control erosion, salinity, waterlogging and depletion of fertility.
- **To protect the environment** - to maintain watersheds, groundwater, biodiversity and air and water quality.
- **To raise the standard of living** of the people of the area, especially small and marginal farmers.
- **To reduce risk** - to reduce the vulnerability of farming to drought, flood and market failure.

- **To resolve conflicting demands** for land between agriculture, forestry, industry, housing and infrastructure, on stated criteria.
- **To provide a basis for investment** - to guide public and private investment in irrigation, roads, markets and credit.
- **To ensure equity** - to protect the interests of the weaker sections who depend most directly on land.
- **To provide a framework for monitoring** change in land use and land condition over time.

The test of a land use plan. A good plan raises production, protects the resource, is acceptable to the people who must carry it out, and can be paid for. A plan that meets only the first of these four is not a land use plan but a production target.

10.4 The steps of land use planning

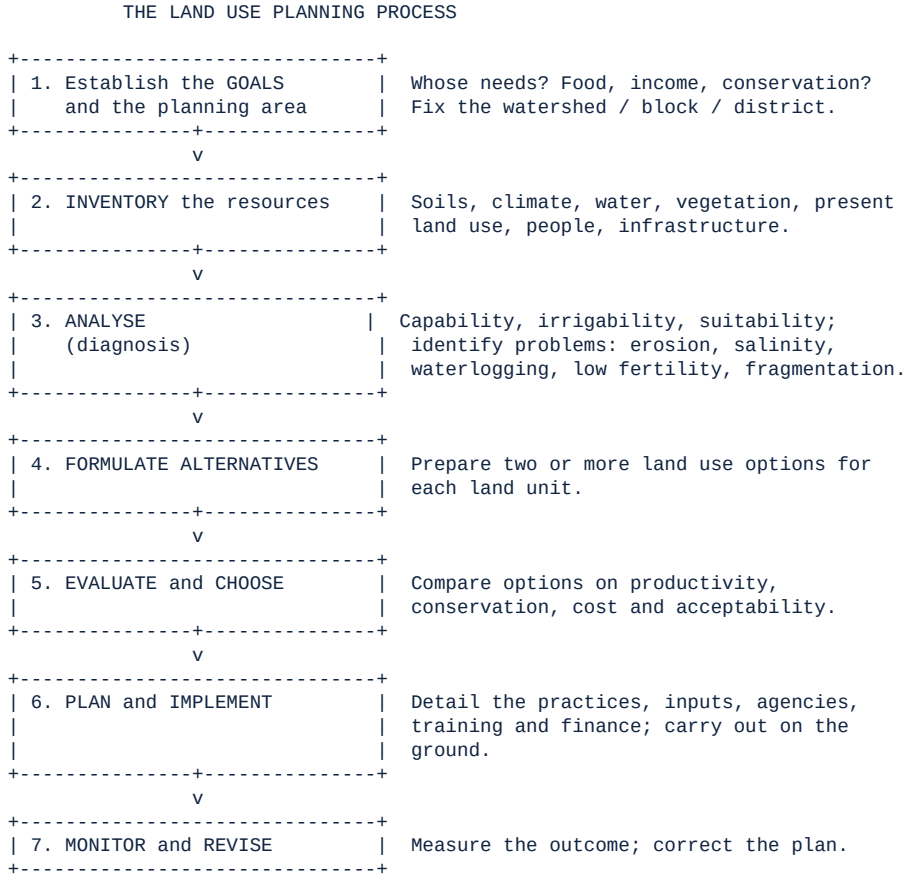


Fig. 10.1 The seven steps of the land use planning process.

10.5 Land use planning in the tropics

Definition of the tropics. The tropics are the region between the Tropic of Cancer (about 23.5 degrees N) and the Tropic of Capricorn (about 23.5 degrees S). Most of peninsular India, including Tamil Nadu, lies in the tropics.

Characteristics that shape planning in the tropics

- **High temperature throughout the year**, with a small seasonal range; rapid organic matter decomposition and low organic matter accumulation.
- **High and often intense rainfall** in the wet season, causing severe erosion on bare soil; or a long dry season in the semi-arid tropics.

- **Deeply weathered soils** - laterites, red soils and acid soils, often low in available phosphorus and in base saturation.
- **Rapid nutrient cycling** but a small nutrient reserve; fertility falls quickly when the forest or grass cover is removed.
- **High biological diversity and pest and disease pressure.**
- **High population pressure on land**, small holdings and cultivation of marginal land.

Planning responses in the tropics

- **Keep the soil covered.** Mulching, cover crops, crop residue retention and agroforestry are the first defence against erosion and heat.
- **Contour and bund.** Contour cultivation, contour bunding and graded bunding on sloping land.
- **Maintain organic matter.** Farmyard manure, green manure, compost and residue incorporation.
- **Diversify.** Mixed cropping, intercropping, crop rotation and integration of crops with livestock and trees.
- **Use the watershed as the unit.** Ridge-to-valley treatment with trenches, check dams, percolation ponds and gully control.
- **Protect the remaining forest** and bring steep and shallow land under trees and grass.
- **Provide irrigation where it is safe**, with drainage to prevent waterlogging and salinity.

10.6 Land use planning in the subtropics

Definition of the subtropics. The subtropics lie roughly between 23.5 and 35 degrees latitude in each hemisphere. In India, this corresponds broadly to the northern plains and parts of the north-western and north-eastern regions.

Characteristics that shape planning in the subtropics

- **A distinct cool season**, which allows temperate crops such as wheat, mustard, potato and some fruits, and a hot summer that allows tropical crops.
- **Marked seasonality of rainfall**, much of it from a single monsoon, with a long dry period.
- **Soils** range from alluvial and fertile to saline and alkaline in the drier tracts.
- **Two or three cropping seasons** are possible where irrigation is available, so irrigation and drainage planning are central.
- **Salinity and alkalinity** are major problems in the semi-arid subtropics.

Planning responses in the subtropics

- **Match crops to season.** Plan a crop calendar that uses the cool season for temperate crops and the warm season for tropical crops.
- **Plan irrigation and drainage together.** Canal and tube-well irrigation must be accompanied by drainage to prevent waterlogging and secondary salinity.
- **Reclaim saline and alkali soils** by gypsum, leaching and drainage, with a suitable cropping sequence.
- **Conserve moisture** by mulching, deep ploughing and water harvesting in the dry period.
- **Prevent wind erosion** by shelter belts and stubble mulching in the dry, windy tracts.
- **Diversify into horticulture, dairy and agro-processing** where irrigation is assured.

10.7 Land use planning in the temperate regions

Definition of the temperate regions. The temperate regions lie roughly between 35 and 60-65 degrees latitude, with a cool to cold climate, a distinct winter and a moderate growing season. In India, temperate conditions occur in the high hills - the Himalayas, the Nilgiris, the Palani hills and similar high ranges.

Characteristics that shape planning in the temperate regions

- **A cold winter**, which stops growth and sets a frost limit on crops; the growing season is short.
- **Snow and frost** affect land preparation, grazing and infrastructure.
- **Soils** are often acid, with high organic matter and slow decomposition in the cold season.
- **Sloping land** is common in the hill regions, so erosion control and forest cover are central.
- **Low population density** in the hill tracts, with a high dependence on horticulture, plantation crops, livestock and forestry.

Planning responses in the temperate regions

- **Match crops to the short season.** Choose short-duration varieties and cold-tolerant crops; plan for a single main crop where the season allows only one.
- **Protect steep land.** Keep steep slopes under forest, grass and plantation crops; terrace and bench where cultivation continues.
- **Manage acid soils.** Liming, and fertiliser and manure management suited to acid soils.
- **Develop horticulture and plantation crops** - apple, pear, plum, potato, tea and coffee - where the climate suits.
- **Manage grazing carefully** to prevent overgrazing and trampling of steep pasture.
- **Protect the watershed**, since hill watersheds feed the rivers and canals of the plains below.

Point	Tropics	Subtropics	Temperate
Latitude (approx.)	23.5 N to 23.5 S	23.5 to 35 N and S	35 to 60-65 N and S
Temperature	High all year, small seasonal range	Hot summer, distinct cool season	Cool to cold, cold winter, short growing season
Rainfall	High and intense, or a long dry season	Marked monsoon seasonality	Moderate, well distributed, snow in winter
Dominant soils	Deeply weathered red, lateritic and acid soils	Alluvial, and saline and alkaline soils	Acid soils with high organic matter
Chief problems	Erosion, loss of organic matter, low nutrient reserve, marginal land cultivation	Salinity, alkalinity, waterlogging, moisture stress	Short season, frost, acid soils, erosion on steep slopes
Main planning response	Soil cover, contour bunding, organic matter, diversification, watershed treatment	Season-based crop calendar, irrigation with drainage, reclamation, moisture conservation	Short-duration crops, terracing, liming, horticulture and plantation, forest protection
Indian example	Most of Tamil Nadu and peninsular India	Northern plains and parts of the north-west	Nilgiris, Palani hills and the Himalayan hill tracts

Table 10.1 Land use planning in the tropics, subtropics and temperate regions compared.

10.8 Tamil Nadu illustration - the Cauvery delta

The Cauvery delta is a classic example of land use planning in the tropics. The delta is a level, deep, alluvial tract, mostly irrigated by the Cauvery system, with paddy as the dominant crop and sugarcane, banana and pulses in rotation. Land use planning there has four main tasks.

- **Match the cropping pattern to water availability.** The delta has three seasons - Kuruvai, Samba and Thaladi. In years of low Cauvery release, the plan shifts area from paddy to short-duration pulses and millets, and to systems of rice intensification.
- **Manage drainage and salinity.** The tail-end and coastal parts of the delta suffer waterlogging and salinity; planning provides field drains, removes excess water and guides leaching.
- **Protect the coastal zone.** Mangrove and casuarina planting and regulation of land use near the coast protect against storm surge and salt-water intrusion.
- **Plan for the uplands of the delta margin.** The red loamy uplands around the delta are planned for millets, pulses, groundnut and horticulture with contour bunding and moisture conservation.

The delta plan illustrates the general principle: the correct use is the one the land can sustain, and the plan must be adjusted as water, market and climate change.

Exam focus

Key points to remember

- Land use planning is the systematic assessment of land and its resources and the choice and design of land uses and management practices that best meet people's needs while conserving the resource.
- Its concepts: land is limited; land has many possible uses; the best use is the one that fits the land; use must match capability; planning is for people, for the long term, on the watershed as the unit, with alternatives, participation and iteration.
- Its objectives: best use for each area, sustained production, conservation, environmental protection, higher living standards, reduced risk, resolution of conflicting demands, a basis for investment, equity and a framework for monitoring.
- The process has seven steps: goals, inventory, analysis, alternatives, evaluation and choice, plan and implementation, and monitoring and revision.
- Tropics (between 23.5 N and 23.5 S): high temperature, intense rainfall, weathered soils, erosion; response is soil cover, contour bunding, organic matter and watershed treatment.
- Subtropics (23.5 to 35 degrees): distinct cool season, monsoon seasonality, salinity and alkalinity; response is a season-based crop calendar, irrigation with drainage and reclamation.
- Temperate (35 to 60-65 degrees): cold winter, short season, acid soils, steep land; response is short-duration crops, terracing, liming and forest protection.

Short-answer questions (with model answers)

Q1. Define land use planning.

Model answer. Land use planning is the systematic assessment of land and its resources and the choice and design of land uses and management practices that will best meet the needs of the people while conserving the land resource for the future.

Q2. State any four objectives of land use planning.

Model answer. Any four of: to select the best use for each area of land; to increase and sustain production; to conserve soil and water; to protect the environment; to raise the standard of living; to reduce risk; to resolve conflicting demands for land; to provide a basis for investment.

Q3. Why is the watershed the unit of land use planning?

Model answer. Because land use is a system. Crops, livestock, forestry, water and settlement within a watershed are linked, and a change in one affects the others. Planning on the watershed treats the whole system together, and it is the natural unit for soil and water conservation.

Q4. Name the three main tropical problems that land use planning must address.

Model answer. Erosion by intense rainfall, loss of organic matter and nutrients under high temperature, and cultivation of marginal land under population pressure. Salinity and waterlogging follow where irrigation is introduced without drainage.

Q5. How does the subtropics differ from the tropics for planning?

Model answer. The subtropics have a distinct cool season, so a crop calendar can include both temperate crops in the cool season and tropical crops in the warm season. Rainfall is more seasonal, and salinity and alkalinity are major problems, so irrigation planning must be accompanied by drainage.

Q6. What is the chief constraint on land use planning in temperate regions?

Model answer. The short growing season and the cold winter, with frost setting a limit on crops. Steep slopes and acid soils add to the constraint, so forest protection, terracing, liming and horticulture dominate the plan.

Long-answer questions (with model answer outlines)

Q1. Explain the concepts, objectives and steps of land use planning.

Outline.

- Define land use planning.
- State the concepts: land is limited; land has many possible uses; the best use fits the land; use must match capability; planning is for people, for the long term, on the watershed unit, with alternatives, participation and iteration.
- State the objectives in full.
- Give the seven steps: goals, inventory, analysis, alternatives, evaluation and choice, plan and implementation, monitoring and revision.
- Explain the role of soil survey, capability and suitability as the evidence base.
- Give the Tamil Nadu delta illustration.
- Conclude with the four-part test of a good plan: production, conservation, acceptability and affordability.

Q2. Compare land use planning in tropical, subtropical and temperate regions.

Outline.

- Define each region by latitude.
- Tropics: temperature, rainfall, soils, problems, planning response.
- Subtropics: temperature, rainfall, soils, problems, planning response.
- Temperate: temperature, rainfall, soils, problems, planning response.
- Draw or reproduce the comparison table.
- Give an Indian example for each region.
- Explain the Tamil Nadu delta plan in detail as a tropical example.
- Conclude that the same planning process is used everywhere but the problems and responses differ with climate and soil.

Lecture 11 Remote Sensing - Definition, Components, Platforms and Sensors

A soil surveyor can examine only a small part of the ground directly. Remote sensing lets the surveyor examine the whole area at once, from a distance, by recording the radiation that the land surface reflects or emits. This lecture defines remote sensing, lists its components, and describes the platforms and sensors that carry it out.

11.1 Remote sensing - definition

Definition. Remote sensing is the science and art of obtaining information about an object, area or phenomenon without coming into physical contact with it, by recording and analysing the radiation reflected or emitted from it.

A more complete exam definition: remote sensing is the technique of acquiring data about the earth's surface from a distance, using sensors carried on aircraft or satellites that measure the electromagnetic radiation reflected or emitted by the surface, and then interpreting that data to derive information about the surface.

The essential idea

- Every object on the earth's surface reflects, absorbs and emits electromagnetic radiation in a way that depends on its properties.
- Two different objects - for example dry soil and green vegetation - reflect different amounts of radiation in different wavelengths.
- This difference is the **spectral signature** of the object.
- A sensor records the radiation and produces a numerical value for each wavelength band at each point.
- The interpreter converts these numbers into information about the object - its type, condition and extent.

11.2 Components of remote sensing

A complete remote sensing system has seven components. They are usually written in the order in which the signal travels, from the source to the user.

1. **Energy source or illumination.** The source of the radiation. It may be the sun (passive remote sensing) or an instrument on the platform that emits its own radiation (active remote sensing, such as radar and LiDAR).
2. **Atmosphere.** The radiation passes through the atmosphere on the way down to the surface and on the way back up. The atmosphere absorbs and scatters part of it, which affects what the sensor can measure.
3. **Target or object.** The feature being observed - soil, water, vegetation, a crop, a built-up area. Its properties determine how it reflects or emits radiation.
4. **Sensor.** The instrument that detects and records the radiation. It may be a photographic camera or an electronic scanner.
5. **Platform.** The vehicle that carries the sensor - a ground-based mast, a drone, an aircraft, a balloon or a satellite.
6. **Data transmission and storage.** The recorded signal is transmitted to a ground station and stored, usually as digital numbers on magnetic media or sent over a downlink.
7. **Data processing, analysis and interpretation.** The raw data are corrected, processed and analysed, and the results are interpreted and presented as a map, an image or a report to the user.

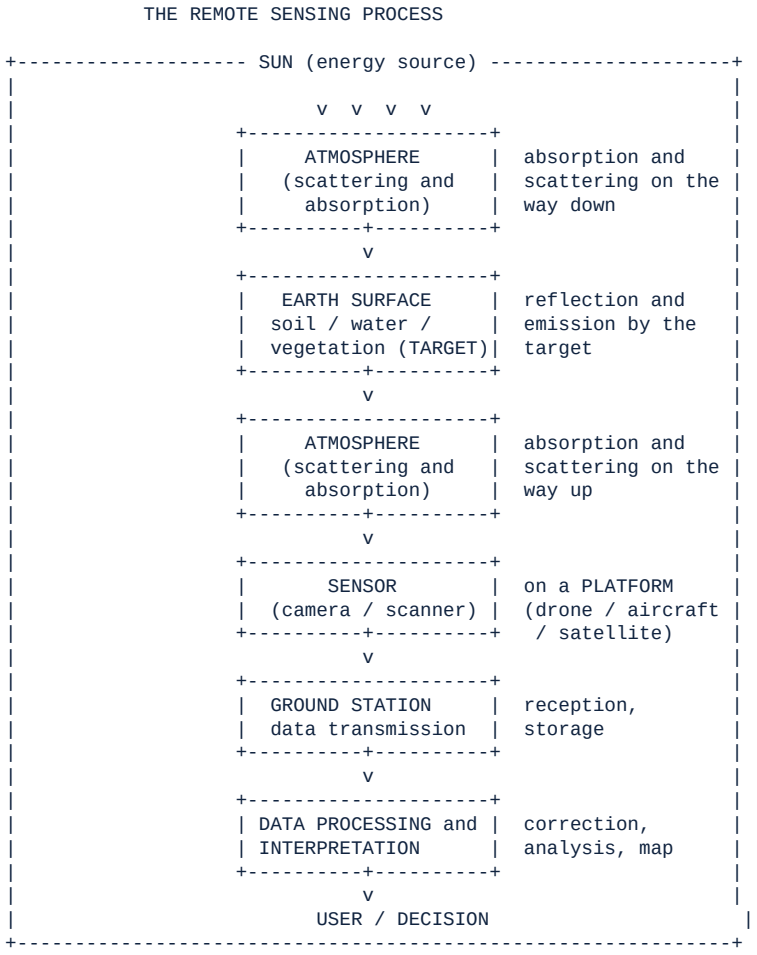


Fig. 11.1 The remote sensing process, from energy source to the user.

Component	Role	Indian or common example
Energy source	Supplies the radiation to be measured	The sun; or the radar antenna of RISAT
Atmosphere	Transmits, absorbs and scatters the radiation	A clear window is chosen for each band
Target	Reflects or emits the radiation	Soil, water, crop canopy, forest
Sensor	Detects and records the radiation	LISS-III, LISS-IV, AWiFS cameras on IRS satellites
Platform	Carries the sensor	IRS satellites, drones, aircraft
Data transmission and storage	Sends data to the ground and stores it	NRSC ground station at Shadnagar
Processing and interpretation	Turns data into information	Bhuvan, GIS and image processing software

Table 11.1 The seven components of a remote sensing system.

11.3 Types of remote sensing

By source of energy

- **Passive remote sensing.** The sensor measures radiation that comes from an external source, usually the sun, or radiation emitted by the object itself. Optical and thermal sensors are passive.
- **Active remote sensing.** The sensor carries its own source, sends out a pulse of radiation and measures the returning echo. Radar and LiDAR are active. They can work at night and through cloud.

By spectral region used

- **Optical remote sensing** - visible and near-infrared and short-wave infrared, using reflected sunlight.
- **Thermal remote sensing** - measures emitted thermal infrared radiation, giving surface temperature.
- **Microwave remote sensing** - radar, which penetrates cloud and can measure surface roughness and moisture.

By number of bands

- **Panchromatic** - one broad band, giving a black-and-white image of high spatial detail.
- **Multispectral** - several separate bands (for example green, red, near-infrared), giving colour information.
- **Hyperspectral** - many narrow contiguous bands (often 100 or more), giving a near-continuous spectrum for each pixel.

11.4 Remote sensing platforms

Definition. A platform is the vehicle or structure that carries the remote sensing sensor and supports it in a position from which it can view the target.

Ground-based platforms

- Handheld cameras, tripods, masts and towers.
- Used for close-range work: measuring the spectral reflectance of a soil sample, a leaf or a crop canopy.
- They provide the reference or ground-truth data used to check satellite measurements.

Airborne platforms

- **Balloons and kites** - low-altitude, low-cost, for small areas.
- **Drones or unmanned aerial vehicles (UAVs)** - very high resolution, flexible, low cost, now widely used for crop and soil monitoring over small areas.
- **Aircraft** - carry photographic and scanner sensors at altitudes from a few hundred metres to about 20 kilometres; used for aerial photography and for dedicated survey missions.

Spaceborne platforms

- **Satellites** - orbit the earth and carry sensors that cover large areas repeatedly.
- **Geostationary orbit** - about 36,000 km above the equator; the satellite stays over one point and views the same hemisphere continuously. Used for weather satellites such as INSAT.
- **Sun-synchronous polar orbit** - about 700 to 900 km altitude; the satellite crosses each latitude at about the same local solar time, so illumination is consistent between passes. Most land resources satellites, including the Indian Remote Sensing (IRS) series, use this orbit.

The Indian Remote Sensing (IRS) satellite series

India's IRS programme, operated by ISRO, provides the main satellite data used for soil and crop studies in the country. Well-known satellites in the series include IRS-1A, IRS-1B, IRS-1C, IRS-1D, IRS-P6 (Resourcesat-1), Resourcesat-2 and Resourcesat-2A, Cartosat series for high-resolution mapping, RISAT series for radar, and the Bhaskara satellites, which were India's first experimental remote sensing satellites. The data are distributed through the National Remote Sensing Centre, and are viewed and analysed through **Bhuvan**, the Indian geo-platform.

11.5 Remote sensing sensors

Definition. A sensor is the instrument that detects electromagnetic radiation and records it, usually as a digital number for each wavelength band and each point on the ground.

Two families of sensor

- **Framing or photographic sensors.** A camera records the whole scene at one instant on film or on a digital array, giving a photograph. The geometry is central projection, so it has relief displacement.
- **Scanning sensors.** The sensor builds the image line by line as the platform moves. The main kinds are:
- **Whiskbroom or across-track scanner** - a rotating mirror sweeps across the ground line by line. LISS and AWiFS sensors work this way.
- **Pushbroom or along-track scanner** - a linear array of detectors records a whole line at once, as in many modern sensors.
- **Radar** - an active sensor that sends a microwave pulse and measures the strength and time of the echo, giving surface roughness and moisture.
- **LiDAR** - an active sensor that sends laser pulses and measures the return time, giving precise elevation and, with multiple returns, vegetation height.

The four resolutions of a sensor

Resolution	Meaning	Example
Spatial resolution	The smallest ground area that one pixel represents, or the smallest object that can be distinguished	1 metre (Cartosat), 5.8 m (LISS-IV), 23.5 m (LISS-III), 56 m (AWiFS)
Spectral resolution	The number, width and position of the wavelength bands the sensor records	Multispectral with 4 bands; hyperspectral with 100 or more bands
Temporal resolution	The time between two successive images of the same area (the revisit period)	About 5 days for Resourcesat-2 with wide swath; 16 days for Landsat
Radiometric resolution	The number of distinct intensity levels the sensor can record, expressed in bits	7-bit (128 levels), 8-bit (256 levels), 10-bit or more

Table 11.2 The four resolutions of a remote sensing sensor. Spatial resolution values for the IRS sensors are the standard published figures.

Common sensors on Indian and other satellites

Sensor	Satellite	Spatial resolution	Bands	Typical use
LISS-IV	Resourcesat-2 / 2A	5.8 m	3 (green, red, NIR)	Detailed crop and soil mapping, land use
LISS-III	Resourcesat-2 / 2A	23.5 m	4 (green, red, NIR, SWIR)	District-level crop and soil mapping
AWiFS	Resourcesat-2 / 2A	56 m	4	State and national level crop inventory
PAN (Cartosat)	Cartosat series	About 1 to 2.5 m	1 (panchromatic)	Large-scale mapping, cadastral and urban
Cartosat-1 stereo	Cartosat-1	2.5 m	2 (fore and aft)	Digital elevation models, terrain mapping
RISAT-1 SAR	RISAT-1	About 3 to 50 m depending on mode	Microwave (C-band)	Soil moisture, crop monitoring, flood mapping
TM and OLI	Landsat series	30 m	Multispectral (6-11)	Long-term land cover change
MSI	Sentinel-2	10 to 60 m	13	Crop and vegetation monitoring

Table 11.3 Common sensors used for soil and crop studies. Resolutions are the standard published figures and vary with the satellite and the mode of operation.

11.6 Advantages and limitations of remote sensing

Advantages	Limitations
Covers large areas in a single image, so regional study is possible	Cloud cover blocks optical sensors in the monsoon, so a clear-season image is needed
Gives a synoptic view, so the pattern of the whole landscape is seen	The image shows only the surface; soil properties below the surface must still be examined in the field
Repeats at regular intervals, so change over time can be monitored	The pixel is an average of a ground area, so small features are lost at coarse resolution
Reaches remote and inaccessible areas	Interpretation needs skill and field checking (ground truth)
Is faster and cheaper than ground survey for large areas	Sensor and atmospheric effects must be corrected before use
Digital data can be processed and combined with GIS layers	Purchase and processing cost, and the need for computing facilities

Table 11.4 Advantages and limitations of remote sensing.

Remote sensing does not replace the soil survey. It reduces the number of field observations needed and shows the pattern of the landscape, but the soil profile must still be examined on the ground to identify the soil and to check the interpretation. The satellite image shows the pattern; the soil pit explains it.

Exam focus

Key points to remember

- Remote sensing is the science and art of obtaining information about an object without physical contact, by recording the radiation reflected or emitted from it.
- Its seven components are the energy source, the atmosphere, the target, the sensor, the platform, data transmission and storage, and data processing and interpretation.
- Passive remote sensing uses an external source such as the sun; active remote sensing carries its own source, as radar and LiDAR do.
- Platforms are ground-based, airborne (balloons, drones, aircraft) and spaceborne (satellites).
- Satellites use geostationary or sun-synchronous polar orbits; the IRS series uses sun-synchronous polar orbits.
- Sensors are framing (photographic) or scanning; scanners are whiskbroom, pushbroom, radar or LiDAR.
- The four resolutions are spatial, spectral, temporal and radiometric.
- The IRS series includes IRS-1A to 1D, Resourcesat, Cartosat, RISAT and the Bhaskara satellites; data are viewed through Bhuvan.

Short-answer questions (with model answers)

Q1. Define remote sensing.

Model answer. Remote sensing is the science and art of obtaining information about an object, area or phenomenon without coming into physical contact with it, by recording and analysing the radiation reflected or emitted from it.

Q2. Name the seven components of a remote sensing system.

Model answer. Energy source or illumination, atmosphere, target or object, sensor, platform, data transmission and storage, and data processing and interpretation.

Q3. Distinguish passive from active remote sensing.

Model answer. Passive remote sensing measures radiation from an external source, usually the sun, or radiation emitted by the object itself, as optical and thermal sensors do. Active remote sensing carries its own energy source, sends out a pulse and measures the returning echo, as radar and LiDAR do, so it can work at night and through cloud.

Q4. Name the four resolutions of a remote sensing sensor.

Model answer. Spatial resolution, the smallest ground area represented by one pixel; spectral resolution, the number and width of the wavelength bands; temporal resolution, the time between two images of the same area; and radiometric resolution, the number of intensity levels recorded.

Q5. Name any four satellites of the Indian Remote Sensing series.

Model answer. Any four of: IRS-1A, IRS-1B, IRS-1C, IRS-1D, IRS-P6 or Resourcesat-1, Resourcesat-2, Resourcesat-2A, the Cartosat series, the RISAT series, and the Bhaskara satellites.

Long-answer questions (with model answer outlines)

Q1. Define remote sensing, explain its components and describe the platforms and sensors used.

Outline.

- Define remote sensing and explain the idea of a spectral signature.
- List and explain the seven components in order.
- Draw the remote sensing process diagram.
- Describe the types of remote sensing by energy source, spectral region and number of bands.
- Describe the platforms: ground-based, airborne and spaceborne.
- Explain geostationary and sun-synchronous polar orbits.
- Describe the IRS satellite series and Bhuvan.
- Describe the sensors: framing and scanning, whiskbroom, pushbroom, radar and LiDAR.
- Explain the four resolutions with examples from IRS sensors.
- State the advantages and limitations, and conclude that remote sensing supplements but does not replace the field soil survey.

Q2. Describe the remote sensing platforms and sensors used in soil and crop studies, with their resolutions.

Outline.

- Define platform and sensor.
- Platforms: ground-based, balloons and kites, drones, aircraft, satellites.
- Satellite orbits: geostationary and sun-synchronous polar.
- IRS series: satellites and their purpose.
- Sensors: photographic and scanning families, with examples.
- The four resolutions and what each means.
- Give the sensor table with LISS-IV, LISS-III, AWiFS, Cartosat PAN, RISAT SAR and Landsat.
- Explain which sensor is chosen for which task - detailed mapping, district mapping, state inventory, terrain and soil moisture.
- Conclude with the advantage of matching resolution to the purpose.

Lecture 12 Electromagnetic Radiation - Principle and Atmospheric Interaction

Remote sensing works because energy travels from the target to the sensor. That energy is **electromagnetic radiation**. To interpret a satellite image correctly, the student must know what EMR is, how it is described, what happens to it on the way through the atmosphere, and which wavelength bands are useful. This lecture covers all four.

12.1 Electromagnetic radiation - definition

Definition. Electromagnetic radiation is a form of energy that travels through space as a wave made of oscillating electric and magnetic fields, at the speed of light, and that carries energy without the need for a material medium.

Electromagnetic radiation has a **dual nature**: it behaves as a wave when it travels and interacts with matter, and as a stream of discrete packets of energy called **photons** or quanta when it is emitted or absorbed. Both descriptions are needed - the wave theory explains reflection, refraction and interference, and the particle theory explains the photoelectric effect and the energy of the radiation.

12.2 The wave model and its parameters

- **Wavelength (lambda)** - the distance between two successive crests of the wave. It is measured in metres, commonly in micrometres (1 µm = one-millionth of a metre) for optical radiation, and in centimetres for microwave.
- **Frequency (nu)** - the number of wave crests passing a fixed point in one second, measured in hertz (Hz).
- **Amplitude** - the height of the wave crest, which is related to the intensity or brightness of the radiation.
- **Velocity (c)** - the speed at which the wave travels, which in a vacuum is about 3 x 10^8 metres per second (the speed of light).
- **Polarisation** - the plane in which the electric field of the wave oscillates. It is important in radar remote sensing.

The two basic relationships

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            THE TWO BASIC RELATIONSHIPS OF ELECTROMAGNETIC RADIATION

1.  c  =  lambda  x  nu          (wave relationship)

    c      = speed of light  ~  3 x 10^8 m/s
    lambda = wavelength (m)
    nu     = frequency (Hz)

    So: long wavelength  <->  low frequency
        short wavelength <->  high frequency

2.  E  =  h  x  nu               (Planck's relationship)

    E = energy of one photon (joule)
    h = Planck's constant ~ 6.626 x 10^-34 joule-second
    nu = frequency (Hz)

    So: short wavelength (high frequency)  ->  high energy
        long wavelength (low frequency)    ->  low energy

Consequence: X-rays and gamma rays are dangerous and are absorbed by
the atmosphere; microwaves are low energy and pass through cloud.
```

Fig. 12.1 The wave relationship and Planck's relationship, and their consequences.

12.3 The electromagnetic spectrum

Definition. The electromagnetic spectrum is the continuous range of all electromagnetic radiation, arranged in order of wavelength or frequency, from the shortest gamma rays to the longest radio waves.

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                THE ELECTROMAGNETIC SPECTRUM

Wavelength (m)   Region                Use in remote sensing
-------------------------------------------------------------------------
< 10^-11         Gamma rays            not used (absorbed by atmosphere)
10^-11 to 10^-8  X-rays                not used
10^-8 to 4x10^-7 Ultraviolet (UV)      limited use; absorbed by ozone
-------------------------------------------------------------------------
0.4 to 0.7 um    VISIBLE               main band for optical sensing
                  violet  0.40 - 0.45
                  blue    0.45 - 0.50
                  green   0.50 - 0.58
                  yellow  0.58 - 0.62
                  orange  0.62 - 0.65
                  red     0.65 - 0.70
-------------------------------------------------------------------------
0.7 to 1.3 um    Near infrared (NIR)   vegetation, water, soil moisture
1.3 to 3.0 um    Short-wave IR (SWIR)  soil moisture, minerals, vegetation
-------------------------------------------------------------------------
3.0 to 5.0 um    Mid-wave IR (MWIR)    high-temperature sources, fire
8.0 to 14 um     Thermal IR (TIR)      surface temperature, emitted energy
-------------------------------------------------------------------------
0.1 cm to 1 m    Microwave / radar     soil moisture, roughness, cloud-
                                       penetrating imaging
> 1 m            Radio waves           communication, not imaging
-------------------------------------------------------------------------
Visible + reflective IR (0.4-3.0 um) is called OPTICAL remote sensing.
```

Fig. 12.2 The electromagnetic spectrum with the regions used in remote sensing. Wavelength ranges are the standard published values.

The bands used in soil and crop remote sensing

Band	Wavelength (approx.)	What it shows	Typical use
Violet and blue	0.40 - 0.50 µm	Water, atmospheric haze, deep water; strongly scattered by the atmosphere	Water body mapping; least useful over land because of haze
Green	0.50 - 0.58 µm	Reflectance of healthy vegetation; greenness of the canopy	Vegetation vigour, crop discrimination
Red	0.62 - 0.70 µm	Chlorophyll absorption; soil colour	Vegetation and crop mapping; the red band enters NDVI
Near infrared (NIR)	0.70 - 1.30 µm	Leaf cell structure and biomass; low reflectance from water	Vegetation biomass, crop condition, water body delineation
Short-wave infrared (SWIR)	1.30 - 3.00 µm	Soil and plant moisture, clay and carbonate minerals	Soil moisture, soil mineralogy, crop residue
Thermal infrared (TIR)	8 - 14 µm	Surface temperature	Soil temperature, crop water stress, urban heat
Microwave (radar)	About 1 mm to 1 m	Surface roughness and dielectric constant, hence moisture	Soil moisture, crop structure, flood mapping, cloud-free imaging

Table 12.1 Electromagnetic bands used in soil and crop remote sensing, with their approximate wavelengths and uses.

The three bands worth memorising. Green (0.50-0.58 µm) for vegetation greenness, red (0.62-0.70 µm) where chlorophyll absorbs, and near infrared (0.70-1.30 µm) where healthy leaves reflect strongly. The contrast between the red and near-infrared reflectance of vegetation is the basis of the NDVI.

12.4 Emission of radiation and blackbody radiation

Every object with a temperature above absolute zero emits radiation. The amount and the peak wavelength of the emitted radiation depend on the temperature of the object. An ideal emitter is called a **blackbody**.

- **Stefan-Boltzmann law.** The total energy emitted by a blackbody is proportional to the fourth power of its absolute temperature. A hotter object emits far more energy.
- **Wien's displacement law.** The wavelength at which a blackbody emits most strongly is inversely proportional to its absolute temperature. A hotter object emits most strongly at a shorter wavelength.

- **Application.** The sun, with a surface temperature of about 6,000 K, emits most strongly in the visible region, at about 0.5 µm. The earth, with a surface temperature of about 300 K, emits most strongly in the thermal infrared, at about 10 µm. This is why solar radiation is used for reflective remote sensing and earth radiation for thermal remote sensing.

12.5 Interaction of EMR with the atmosphere

Radiation from the sun must pass through the atmosphere to reach the surface, and the reflected or emitted radiation must pass through the atmosphere again to reach the sensor. On both journeys the atmosphere changes the radiation in three ways: it **scatters** it, it **absorbs** it, and it **emits** its own radiation. Together these effects decide which wavelengths can be used for remote sensing.

Scattering

Definition. Scattering is the redirection of radiation by particles and gas molecules in the atmosphere, without absorption, so that the energy is spread in many directions.

- **Rayleigh scattering** - caused by gas molecules much smaller than the wavelength. It affects short wavelengths most strongly, which is why the sky appears blue and why the blue band of a satellite image is hazy. It is the dominant scattering above about 3 to 4 km altitude.
- **Mie scattering** - caused by particles about the same size as the wavelength, such as dust, smoke, pollen and water droplets. It affects longer wavelengths than Rayleigh scattering and is dominant in the lower atmosphere, near the surface, especially in dusty or polluted air.
- **Non-selective scattering** - caused by particles much larger than the wavelength, such as fog droplets, cloud droplets and large dust. It scatters all visible wavelengths about equally, which is why fog and cloud appear white.

Absorption

Definition. Absorption is the transfer of energy from the radiation to the molecules of the atmosphere, which converts it into internal energy of those molecules, so that the radiation is removed from the beam.

- **Ozone** absorbs strongly in the ultraviolet, which is why ultraviolet remote sensing from space is limited.
- **Carbon dioxide** absorbs in the mid and thermal infrared.
- **Water vapour** absorbs strongly in several infrared bands, including around 1.4 µm, 1.9 µm and 2.7 µm.
- **Oxygen** absorbs in narrow bands in the visible and near infrared.

Atmospheric windows

Definition. An atmospheric window is a wavelength region in which the atmosphere transmits radiation well, so that it can be used for remote sensing.

- The **visible window**, about 0.4 to 0.7 µm, transmits well and is the mainstay of optical remote sensing.
- The **reflective infrared window**, about 0.7 to 3.0 µm, transmits well except for the narrow water-vapour absorption bands.
- The **thermal infrared window**, about 8 to 14 µm, transmits well and is used for surface temperature measurement.
- The **microwave window**, roughly 1 mm to 1 m, transmits well and even penetrates cloud, which is why radar works in the monsoon.

Atmospheric process	Cause	Effect on the image	How it is handled
Rayleigh scattering	Gas molecules smaller than the wavelength	Blue haze, especially in the blue band and at high altitude	Avoid the blue band; apply atmospheric correction
Mie scattering	Dust, smoke, pollen and water droplets	General haze, worst near the surface and in polluted air	Choose a clear day; atmospheric correction

Atmospheric process	Cause	Effect on the image	How it is handled
Non-selective scattering	Fog and cloud droplets	A white, uniform veil that hides the ground	Use cloud-free images; use radar through cloud
Absorption	Ozone, carbon dioxide, water vapour, oxygen	Some bands carry no usable signal (absorption bands)	Use the atmospheric windows; avoid absorption bands
Path radiance	Scattered radiation that reaches the sensor without touching the ground	Adds a brightening haze to the whole image	Subtract path radiance in atmospheric correction

Table 12.2 Atmospheric effects on remote sensing and how each is handled.

Other atmospheric effects

- **Transmittance.** The fraction of the incident radiation that passes through the atmosphere. It is high in the windows and low in the absorption bands.
- **Path radiance.** The scattered radiation that reaches the sensor without ever touching the ground, which adds a haze to the image and reduces contrast.
- **Atmospheric emission.** The atmosphere also emits its own thermal radiation, which must be accounted for in thermal remote sensing.
- **Refraction and turbulence.** Bending and shimmer of the radiation caused by variation in air density, which affects high-resolution imagery.

Why remote sensing uses windows, not the whole spectrum. Radiation is useful only where the atmosphere lets it through. Absorption bands carry almost no signal from the ground, and heavy scattering destroys contrast. Sensors are therefore designed to work inside the atmospheric windows - visible, reflective infrared, thermal infrared and microwave.

Exam focus

Key points to remember

- Electromagnetic radiation is energy travelling as oscillating electric and magnetic fields at the speed of light, with a dual wave and particle nature.
- Its parameters are wavelength, frequency, amplitude, velocity and polarisation.
- c = lambda x nu, and E = h x nu: long wavelength means low frequency and low energy; short wavelength means high frequency and high energy.
- The electromagnetic spectrum runs from gamma rays at the shortest wavelength to radio waves at the longest; visible light is 0.4 to 0.7 µm.
- Stefan-Boltzmann law: emitted energy is proportional to the fourth power of absolute temperature. Wien's law: the peak wavelength is inversely proportional to absolute temperature.
- The atmosphere scatters radiation by Rayleigh, Mie and non-selective scattering, and absorbs it mainly by ozone, carbon dioxide, water vapour and oxygen.
- Atmospheric windows are wavelength regions of good transmission, and sensors work inside them.
- Rayleigh scattering causes blue haze; non-selective scattering by cloud and fog hides the ground; path radiance adds a brightening haze.

Short-answer questions (with model answers)

Q1. Define electromagnetic radiation.

Model answer. Electromagnetic radiation is a form of energy that travels through space as a wave of oscillating electric and magnetic fields at the speed of light, and that carries energy without needing a material medium. It has a dual wave and particle nature.

Q2. State the wave relationship and Planck's relationship.

Model answer. The wave relationship is c = lambda x nu, where c is the speed of light, lambda the wavelength and nu the frequency. Planck's relationship is E = h x nu, where E is the energy of one photon and h is Planck's constant. Together they show that short wavelength means high frequency and high energy.

Q3. Name the three types of atmospheric scattering.

Model answer. Rayleigh scattering, caused by gas molecules much smaller than the wavelength and affecting short wavelengths most, which makes the sky blue; Mie scattering, caused by particles about the size of the wavelength, such as dust and smoke; and non-selective scattering, caused by particles much larger than the wavelength, such as fog and cloud droplets, which scatters all visible wavelengths equally and appears white.

Q4. What is an atmospheric window? Give two examples.

Model answer. An atmospheric window is a wavelength region in which the atmosphere transmits radiation well, so that it can be used for remote sensing. Examples are the visible window of about 0.4 to 0.7 µm and the thermal infrared window of about 8 to 14 µm.

Q5. Which gases absorb radiation in the atmosphere?

Model answer. Ozone absorbs in the ultraviolet; carbon dioxide absorbs in the mid and thermal infrared; water vapour absorbs strongly in several infrared bands; and oxygen absorbs in narrow visible and near-infrared bands.

Long-answer questions (with model answer outlines)

Q1. Explain electromagnetic radiation, its properties and the electromagnetic spectrum.

Outline.

- Define EMR and its dual nature.
- Explain the wave parameters: wavelength, frequency, amplitude, velocity and polarisation.
- State c = lambda x nu and E = h x nu, and draw the conclusions from each.
- Describe the electromagnetic spectrum from gamma rays to radio waves.
- Give the visible wavelengths for violet, blue, green, yellow, orange and red.
- Explain the regions used in remote sensing: visible, NIR, SWIR, thermal IR and microwave.
- State the Stefan-Boltzmann law and Wien's displacement law, and apply them to the sun and the earth.
- Give the band table with wavelength and use.
- Conclude that the sun emits in the visible and the earth in the thermal infrared.

Q2. Explain the interaction of electromagnetic radiation with the atmosphere.

Outline.

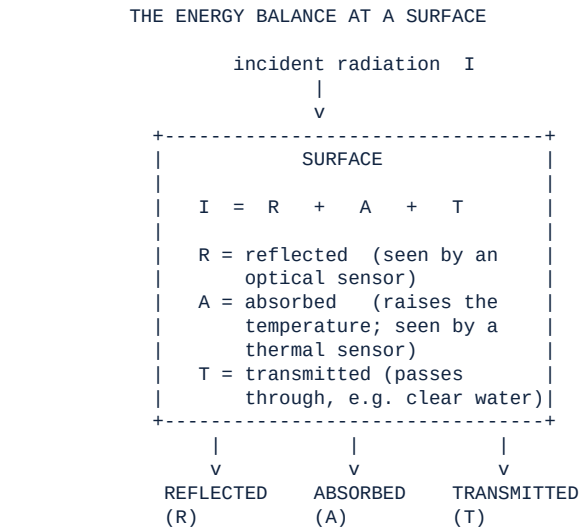
- State that radiation passes through the atmosphere twice.
- Define scattering and explain Rayleigh, Mie and non-selective scattering with their causes and effects.
- Explain the blue sky and the white cloud.
- Define absorption and name the absorbing gases - ozone, carbon dioxide, water vapour and oxygen.
- Define an atmospheric window and give the visible, reflective infrared, thermal infrared and microwave windows.
- Explain path radiance and its effect on the image.
- Give the atmospheric effects table with the remedy for each.
- Conclude that sensors are designed to work inside the windows, and that atmospheric correction is needed before quantitative use.

Lecture 13 Interaction of EMR with Earth Surface Features

The atmosphere decides which wavelengths reach the sensor. The earth's surface decides how much of that radiation comes back, and in what proportions. Because soil, water and vegetation return radiation differently, each has a characteristic **spectral signature**, and the signature is what lets the interpreter separate them on an image. This lecture explains the interaction for soil, water and vegetation, and shows how the difference is turned into an index such as the NDVI.

13.1 How radiation interacts with a surface

When radiation falls on a surface, it is divided into three parts. The energy that arrives is either reflected, absorbed or transmitted, so that the three fractions add to one.



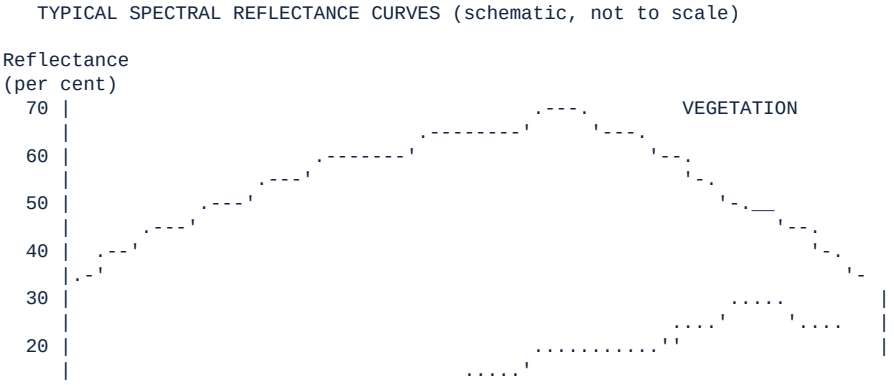
For most remote sensing we measure the REFLECTED part (optical) or the EMITTED part (thermal). Reflectance is usually expressed as a fraction (0 to 1) or a percentage, and is called the spectral reflectance.

reflectance = reflected energy / incident energy

Fig. 13.1 The energy balance at the earth's surface.

Reflectance and the spectral reflectance curve

- **Spectral reflectance** is the ratio of the energy reflected by a surface at a given wavelength to the energy incident on it at that wavelength. It is expressed as a fraction or a percentage.
- When reflectance is plotted against wavelength, the result is the **spectral reflectance curve** or spectral signature of the surface.
- The curve is the identity card of the surface. Soil, water and vegetation have curves that differ in level and in shape, and it is the shape of the curve, not the brightness at one wavelength, that allows the surface to be identified.



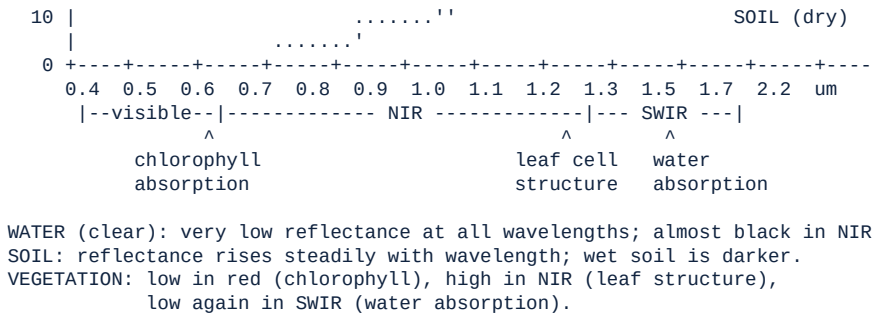


Fig. 13.2 Schematic spectral reflectance curves of vegetation, dry soil and clear water.

13.2 Interaction with vegetation

A green leaf is the most distinctive target in optical remote sensing. Its spectral behaviour is controlled by three components: leaf pigments, the internal cell structure, and the water in the leaf.

The three controls

- **Pigments.** Chlorophyll absorbs strongly in the blue (about 0.45 µm) and in the red (about 0.66 µm), which is why a healthy leaf looks green - green light is reflected. The two absorption bands are the reason for the **red dip** in the vegetation curve.
- **Cell structure.** In the near infrared, about 0.7 to 1.3 µm, the internal structure of the leaf - the spongy mesophyll and the air-water interfaces - scatters radiation strongly, so reflectance is high, often 40 to 60 per cent. This is the **near-infrared plateau**, and it is the single most useful feature for detecting vegetation and measuring biomass.
- **Water content.** In the short-wave infrared, beyond about 1.3 µm, water in the leaf absorbs strongly, so reflectance falls. The position and depth of these water-absorption bands indicate the moisture status of the crop.

The red edge

Between the red dip and the near-infrared plateau, reflectance rises steeply over a narrow wavelength range of about 0.68 to 0.75 µm. This steep rise is called the **red edge**. Its position and steepness are sensitive to chlorophyll content and to plant stress, so the red edge is widely used to detect the early onset of stress before it is visible to the eye.

What changes the vegetation signature

- **Leaf area index and canopy cover.** A denser canopy gives higher near-infrared reflectance, up to a saturation point.
- **Chlorophyll content and health.** A stressed or diseased crop has less chlorophyll, so reflectance in the red rises and the red edge shifts.
- **Water content.** A water-stressed crop has weaker water-absorption bands in the short-wave infrared.
- **Senescence and dry matter.** As a crop matures and dries, the near-infrared reflectance falls and the short-wave infrared rises, because dry plant material reflects more than green leaf.
- **Background.** Soil, shadow and weeds inside the pixel change the measured signature.

13.3 Interaction with water

Water has the lowest reflectance of the three features. In the visible region clear water reflects only a small part of the incident radiation, mostly from the surface, and in the near infrared it absorbs almost all of it, so that clear water appears almost black in a near-infrared image.

What controls the water signature

- **Reflectance falls with wavelength.** For clear water, reflectance is greatest in the blue-green and falls to nearly zero in the near infrared.

- **Suspended sediment.** Turbid water with silt and clay reflects more, especially in the red, and shifts the colour towards brown.
- **Chlorophyll and algae.** Water with algal blooms shows a reflectance peak in the green, which is why a bloom appears green.
- **Depth.** Shallow water over a bright bottom reflects more than deep water, because the bottom is partly visible.
- **Surface state.** A calm surface reflects the sky in one direction (sun glint) and is dark elsewhere; a rough, wind-blown surface scatters the reflection.
- **Turbulence and foam.** Breaking waves and foam reflect strongly and appear bright.

Why the near infrared is used to map water. Water absorbs near-infrared radiation almost completely, so water bodies appear very dark in the near infrared while land and vegetation appear bright. This strong contrast is why the near-infrared band, or the NDWI index built from it, is the standard way to delineate tanks, reservoirs, canals and flooded fields.

13.4 Interaction with soil

Soil reflectance rises steadily with wavelength from the visible into the short-wave infrared, without the sharp absorption features of vegetation. The level of the curve, and the slope, depend on the properties of the soil.

What controls the soil signature

- **Moisture.** Wet soil is darker than dry soil at all wavelengths, because water films reduce reflection and increase absorption. Soil moisture is therefore one of the strongest controls on soil brightness.
- **Organic matter.** Organic matter darkens the soil, so a soil with high organic matter reflects less, especially in the visible and near infrared.
- **Iron oxides.** Iron oxides and hydroxides, common in the red soils of Tamil Nadu, raise reflectance in the red and lower it in the blue, giving the soil its reddish colour.
- **Texture.** Coarse, sandy soils generally reflect more than fine, clayey soils when dry, because clay soils retain more moisture and have more internal shadow.
- **Surface roughness.** A smooth, clod-free surface reflects more directly than a rough, cloudy surface, which scatters the radiation.
- **Carbonates and salts.** Free lime and salt crusts raise reflectance strongly and give bright, white patches in the image, which is useful for mapping saline and calcareous soils.
- **Clay minerals.** Clay minerals show characteristic absorption features in the short-wave infrared, around 2.2 µm, which allows clay-rich soil to be distinguished from sandy soil.
- **Surface crust and tillage.** A sealed or crusted surface reflects more than a freshly tilled one.

Feature	Visible (red)	Near infrared	Short-wave infrared	Distinguishing behaviour
Healthy vegetation	Low (chlorophyll absorbs)	High (leaf structure scatters)	Low (water absorbs)	The red dip with a near-infrared peak - the classic vegetation signature
Stressed or sparse vegetation	Higher than healthy	Lower than healthy	Higher than healthy	The red-near-infrared contrast weakens as the crop declines
Dry plant residue	Moderate to high	Moderate	High	Bright in the short-wave infrared, unlike green vegetation
Clear water	Low	Very low	Very low	Almost black in the near infrared; low everywhere

Feature	Visible (red)	Near infrared	Short-wave infrared	Distinguishing behaviour
Turbid water	Moderate to high	Low	Very low	Bright in the red because of suspended sediment
Dry bare soil	Moderate	Moderate to high	High	Reflectance rises steadily with wavelength
Wet bare soil	Low	Low	Moderate	Darker than dry soil at all wavelengths
Saline or calcareous soil	High	High	High	Uniformly bright, especially in the visible
Iron-rich red soil	High in red, lower in blue	Moderate	Moderate	Reddish hue from iron oxide

Table 13.1 Spectral behaviour of soil, water and vegetation features. The descriptions are qualitative and give the standard direction of change, not fixed reflectance values.

13.5 From reflectance to an index - the NDVI

Because the red and near-infrared bands behave oppositely for vegetation, the contrast between them can be combined into a single number that measures green vegetation. The most widely used such index is the **Normalised Difference Vegetation Index (NDVI)**.

The formula

```
                THE NORMALISED DIFFERENCE VEGETATION INDEX

          NDVI  =  ( NIR  -  RED )  /  ( NIR  +  RED )

NIR = reflectance in the near-infrared band   (e.g. 0.7 - 1.1 um)
RED = reflectance in the red band             (e.g. 0.6 - 0.7 um)

Because the two reflectances are divided by their sum, NDVI is
normalised: it always lies between -1 and +1, whatever the overall
brightness of the scene. This makes images from different dates
and different sensors comparable.

Interpretation:
   NDVI  <  0        water, cloud, snow
   0  to  0.2        bare soil, rock, built-up land, senescent crop
   0.2 to 0.4        sparse vegetation, early crop growth, poor crop
   0.4 to 0.6        moderate to good vegetation cover
   0.6 to  0.8       dense, healthy vegetation
   >  0.8            very dense, vigorous canopy (approaching saturation)
```

Fig. 13.3 The NDVI, its formula and its interpretation range.

13.6 Worked example - interpreting NDVI

A satellite image of a block in the Cauvery delta is examined. Four sample pixels are chosen, one from each of four fields, and the reflectance values in the red and near-infrared bands are read from the image.

Field	Red reflectance	NIR reflectance	Description of the field
A	0.08	0.42	Paddy at maximum tillering, fully green
B	0.18	0.28	Paddy just transplanted, sparse cover
C	0.26	0.30	Bare ploughed field, no crop
D	0.05	0.04	Water in a flooded field

Table 13.2 Reflectance values for the four sample pixels of worked example 13.1.

Worked example 13.1 - solution

1 **Field A.** NDVI = (0.42 - 0.08) / (0.42 + 0.08) = 0.34 / 0.50 = **0.68**. This is in the 0.6 to 0.8 range, so Field A carries dense, healthy vegetation - as expected for paddy at maximum tillering.

2 **Field B.** NDVI = (0.28 - 0.18) / (0.28 + 0.18) = 0.10 / 0.46 = **0.22**. This is in the 0.2 to 0.4 range, so Field B carries sparse vegetation - as expected for freshly transplanted paddy with little canopy cover.

3 **Field C.** NDVI = (0.30 - 0.26) / (0.30 + 0.26) = 0.04 / 0.56 = **0.07**. This is in the 0 to 0.2 range, so Field C is bare soil - as expected for a ploughed field with no crop.

4 **Field D.** NDVI = (0.04 - 0.05) / (0.04 + 0.05) = -0.01 / 0.09 = **-0.11**. This is negative, which indicates water - as expected for the flooded field.

5 **Interpretation for the block.** The four values separate the four conditions cleanly. A map of NDVI over the whole block would therefore show, field by field, which fields are cropped and vigorous (A), which are young or sparse (B), which are still bare (C), and which are under water (D). Repeating the same calculation on an image taken two weeks later would show the growth of the crop as a rise in NDVI, and would let the planner estimate the cropped area and the crop condition without visiting every field.

What the NDVI does and does not tell you. The NDVI measures the amount and vigour of green vegetation. It does not identify the crop, and it saturates in very dense canopies. It must be supported by field observation and, for crop identification, by multi-date or multi-band analysis.

Other common vegetation indices

- **NDWI** (Normalised Difference Water Index) - uses the near infrared and short-wave infrared, or the green and near infrared, to delineate water bodies and crop moisture.
- **SAVI** (Soil Adjusted Vegetation Index) - adjusts the NDVI for the influence of bare soil, useful in sparse canopies.
- **EVI** (Enhanced Vegetation Index) - reduces atmospheric and canopy background effects and is less prone to saturation than the NDVI.
- **NDMI** (Normalised Difference Moisture Index) - uses the near infrared and short-wave infrared to measure crop and soil moisture.

Exam focus

Key points to remember

- Incident radiation is divided into reflected, absorbed and transmitted parts, which add to one; reflectance = reflected energy / incident energy.
- The spectral reflectance curve, or spectral signature, is the identity card of a surface.
- Vegetation: low reflectance in the red because chlorophyll absorbs; high in the near infrared because leaf cell structure scatters; low in the short-wave infrared because water absorbs. The steep rise between red and near infrared is the red edge.
- Water: low reflectance at all wavelengths, almost black in the near infrared; turbidity, algae, depth and surface state change the signature.
- Soil: reflectance rises steadily with wavelength; moisture, organic matter, iron oxides, texture, roughness and salts control its level.
- NDVI = (NIR - RED) / (NIR + RED), always between -1 and +1; below 0 is water, 0 to 0.2 bare soil, 0.2 to 0.4 sparse vegetation, 0.4 to 0.6 moderate, 0.6 to 0.8 dense healthy vegetation.
- The near-infrared band is used to delineate water bodies because water absorbs near-infrared radiation almost completely.
- Other indices: NDWI, SAVI, EVI and NDMI.

Short-answer questions (with model answers)

Q1. What is a spectral reflectance curve?

Model answer. A spectral reflectance curve is a graph of the reflectance of a surface against wavelength. It is the spectral signature of the surface and is used to identify the surface, because soil, water and vegetation each have a curve of different level and shape.

Q2. Why does healthy vegetation appear bright in the near infrared?

Model answer. Because the internal structure of the leaf - the spongy mesophyll and the air-water interfaces - scatters near-infrared radiation strongly, so reflectance is high, often 40 to 60 per cent. This near-infrared plateau is the most useful feature for detecting vegetation and measuring biomass.

Q3. Why is water almost black in a near-infrared image?

Model answer. Because water absorbs near-infrared radiation almost completely, so very little is reflected back to the sensor. Land and vegetation, by contrast, reflect strongly in the near infrared, so the strong contrast makes water bodies easy to delineate.

Q4. Name four soil properties that control soil reflectance.

Model answer. Any four of: moisture, organic matter content, iron oxide content, texture, surface roughness, free lime or salt content, and clay mineral content.

Q5. Write the NDVI formula and state what an NDVI of -0.1 indicates.

Model answer. NDVI = (NIR - RED) / (NIR + RED). An NDVI of -0.1 is negative, which indicates water, cloud or snow, because water absorbs the near infrared and reflects little.

Q6. What is the red edge?

Model answer. The red edge is the steep rise in reflectance between about 0.68 and 0.75 µm, between the red chlorophyll absorption dip and the near-infrared plateau. Its position and steepness are sensitive to chlorophyll content and plant stress, so it is used to detect stress early.

Long-answer questions (with model answer outlines)

Q1. Explain the interaction of electromagnetic radiation with soil, water and vegetation.

Outline.

- State the energy balance: incident equals reflected plus absorbed plus transmitted.
- Define reflectance and the spectral reflectance curve.
- Vegetation: explain the three controls - pigments, cell structure and water content.
- Explain the red dip, the near-infrared plateau and the red edge.
- List the factors that change the vegetation signature.
- Water: explain the low reflectance, the fall with wavelength, and the effect of turbidity, algae, depth and surface state.
- Soil: explain the steady rise with wavelength and the control by moisture, organic matter, iron oxides, texture, roughness and salts.
- Draw the schematic spectral curves.
- Conclude that the differences between the three signatures are the basis of image interpretation.

Q2. Explain the NDVI, its formula, its interpretation, and work an example.

Outline.

- Explain why a vegetation index is needed - the red and near-infrared bands behave oppositely for vegetation.
- Write the formula NDVI = (NIR - RED) / (NIR + RED) and explain each term.
- Explain why dividing by the sum normalises the index between -1 and +1.
- Give the interpretation range from negative values for water to above 0.8 for very dense canopy.
- Work the four-field example: A 0.68 dense crop, B 0.22 sparse crop, C 0.07 bare soil, D -0.11 water.

- Explain how an NDVI map is used to estimate cropped area and crop condition.
- State the limitations of the NDVI - it does not identify the crop and it saturates in dense canopies.
- Name other indices - NDWI, SAVI, EVI and NDMI.
- Give a Tamil Nadu illustration from the Cauvery delta.

Lecture 14 Basemaps, Satellite Data Products and Image Interpretation Elements

The maps and images that a soil surveyor or a land use planner works with are of four main kinds: the cadastral map, the topographic sheet, the aerial photograph and the satellite image. This lecture introduces each, shows how scale and area are calculated from them, and then sets out the elements by which a feature is recognised on an image.

14.1 What a basemap is

Definition. A basemap is the map or image that provides the spatial framework and the reference detail on which other information is plotted. It fixes positions, so that any observation can be located and any boundary drawn.

- A basemap must be at a known scale, in a known projection and datum, and must carry enough reference detail to fix a position.
- The four kinds used in Indian soil and land work are the cadastral map, the topographic sheet, the aerial photograph and the satellite image.
- All four can be brought into a GIS, where they are georeferenced to a common coordinate system.

14.2 The cadastral map

Definition. A cadastral map is a large-scale map of a village or a group of survey numbers, showing the boundaries of individual land parcels and their survey numbers, prepared for land revenue and land record purposes.

- **Scale.** Usually about 1:4,000 to 1:10,000, so each field is large enough on the map to be identified.
- **Content.** Field boundaries, survey numbers, sub-divisions, village boundaries, tanks, roads, canals and habitation.
- **Use in soil survey.** It is the standard basemap for detailed farm-level soil surveys, because every field can be identified on the ground by its survey number.
- **Limitation.** It carries no contours and no elevation information, so it cannot be used to judge slope. Slope must be measured separately.
- **In Tamil Nadu.** Village cadastral maps, maintained by the Department of Survey and Settlement, are used in detailed soil surveys, watershed planning and the digitisation of land records.

14.3 The topographic sheet (toposheet)

Definition. A topographic sheet is a map prepared by the Survey of India showing the surface features of an area - relief, drainage, roads, settlements, land use and administrative boundaries - at a stated scale and with contours.

- **Scales.** The main series are 1:1,000,000, 1:250,000, 1:50,000 and 1:25,000. The 1:50,000 series is the standard base for medium-scale soil and land resource surveys.
- **Contours.** Contour lines join points of equal elevation. The contour interval is stated on the sheet, commonly 20 m on a 1:50,000 sheet and 10 m on a 1:25,000 sheet. Contours are the source of slope information.
- **Sheet numbering.** Sheets are numbered in a systematic way based on the international map of the world, so that any sheet can be located from its number.
- **Use in soil survey.** Toposheets provide relief and drainage, which explain the soil pattern, and they cover the whole country on a uniform standard.

- **Limitation.** Toposheets are revised only at long intervals, so they may not show recent changes in land use or new roads.

Reading slope from contours

Slope is the angle or the percentage rise of the ground. From a toposheet it is obtained from the contour interval and the horizontal distance between two contours.

```
          SLOPE FROM CONTOURS ON A TOPOSHEET

Contour interval (CI) = vertical rise between two successive contours
                        (given on the sheet, e.g. 20 m)

Horizontal distance (HD) = the map distance between two successive
                           contours, converted to ground distance

                     CI
   Slope (per cent) = ----- x 100
                     HD

   Slope (degrees)  = arctan ( CI / HD )

Example: CI = 20 m; two contours are 0.5 cm apart on a 1:50,000 sheet.
         HD = 0.5 cm x 50,000 = 25,000 cm = 250 m.
         Slope = (20 / 250) x 100 = 8 per cent.

Close contours = steep slope. Wide contours = gentle slope.
Evenly spaced contours = uniform slope. Contours crossing a stream form a V
pointing upstream.
```

Fig. 14.1 Obtaining slope from the contour interval and the horizontal distance between contours.

14.4 The aerial photograph

Definition. An aerial photograph is a photograph of the ground taken from an aircraft, using a camera pointing downwards, and used as a basemap and as a source of information on land features.

Types of aerial photograph

- **Vertical** - the camera axis is vertical, so the ground is seen from directly above. Vertical photographs are the standard kind used for mapping.
- **Oblique** - the camera axis is tilted. A **high oblique** shows the horizon; a **low oblique** does not. Oblique photographs give a familiar perspective view but are not used for accurate mapping.

Scale of a vertical aerial photograph

- The scale is the ratio of the focal length of the camera to the flying height above the ground: **scale = f / H**, where f is the focal length and H the height of the camera above the ground.
- The scale is uniform only if the ground is level. Over uneven ground, the scale varies with elevation, which causes **relief displacement** - tall objects lean away from the centre of the photograph.
- **Stereoscopy.** Two overlapping vertical photographs of the same strip, taken from two positions, can be viewed as a pair in a stereoscope to give a three-dimensional view. This is how elevation, slope and tree height are measured from photographs.

14.5 Satellite images and data products

Definition. A satellite image is a digital picture of the earth's surface produced by a sensor on a satellite, in which each picture element (pixel) carries a numerical value for the radiation recorded in each spectral band.

Standard satellite data products

- **Raw or level-0 data** - the unprocessed signal as received, with no corrections.
- **Radiometrically corrected data** - sensor and detector variations removed, so that equal targets give equal values.

- **Geometrically corrected data** - distortions removed and the image fitted to a map projection, so that it can be overlaid on a toposheet or a cadastral map.
- **Geocoded or orthorectified data** - corrected for terrain displacement and supplied with map coordinates, so that positions are accurate on hilly ground.
- **False colour composite (FCC)** - a colour image made by displaying three bands through the red, green and blue channels of a display. The standard FCC for vegetation and soil work displays near infrared as red, red as green and green as blue, so that vegetation appears bright red.
- **Panchromatic and merged products** - a high-resolution panchromatic image merged with a lower-resolution multispectral image to give a sharp colour image.
- **Digital elevation models and derived products** - elevation, slope and aspect maps, from stereo or radar data.
- **Value-added products** - land use and land cover maps, crop maps, soil maps and wasteland maps, prepared by national agencies such as the National Remote Sensing Centre and the National Bureau of Soil Survey and Land Use Planning.

Common Indian data products and portals

- **Bhuvan** - the Indian geo-platform of ISRO, giving access to satellite imagery, thematic maps and tools for visualisation and analysis.
- **Bhaskara** - India's first experimental remote sensing satellites, which demonstrated the use of satellite data for earth resources.
- **IRS Resourcesat data** - LISS-III and LISS-IV imagery widely used for land use, crop and soil mapping.
- **National land use and land cover maps** - periodic maps at 1:250,000 and 1:50,000 prepared from satellite data.
- **Soil resource maps** - prepared by the National Bureau of Soil Survey and Land Use Planning from satellite data supported by field survey.

14.6 Worked example - scale and area from a toposheet

Part 1 - scale

A toposheet carries the representative fraction 1:50,000. Express the scale as a statement of scale in words, in metres and kilometres per centimetre.

1. The representative fraction 1:50,000 means that 1 unit on the map represents 50,000 of the same units on the ground.
2. So 1 cm on the map represents 50,000 cm on the ground.
3. Convert to metres: 50,000 cm / 100 = 500 m.
4. Convert to kilometres: 500 m / 1,000 = 0.5 km.
5. **Statement of scale:** 1 cm on the map represents 500 m, or 0.5 km, on the ground.

Part 2 - area of a field

A field on the same 1:50,000 toposheet measures 4.0 cm by 3.0 cm on the map. Find its area on the ground in hectares and in acres.

1. **Ground dimensions.** 4.0 cm on the map represents 4.0 x 500 m = 2,000 m. 3.0 cm represents 3.0 x 500 m = 1,500 m.
2. **Ground area.** Area = 2,000 m x 1,500 m = 3,000,000 m2.
3. **Convert to hectares.** 1 hectare = 10,000 m2. So area = 3,000,000 / 10,000 = **300 hectares**.
4. **Convert to acres.** 1 hectare = 2.471 acres. So area = 300 x 2.471 = **about 741 acres**.

5 **Short cut.** At 1:50,000, one square centimetre on the map represents 500 m x 500 m = 250,000 m2 = 25 hectares. The map area is 4.0 x 3.0 = 12.0 cm2, so the ground area is 12.0 x 25 = **300 hectares**. The two methods agree.

The three conversions worth memorising. At 1:10,000 one square centimetre on the map is 1 hectare on the ground. At 1:25,000 it is 6.25 hectares. At 1:50,000 it is 25 hectares. These three cover almost every calculation asked in an examination.

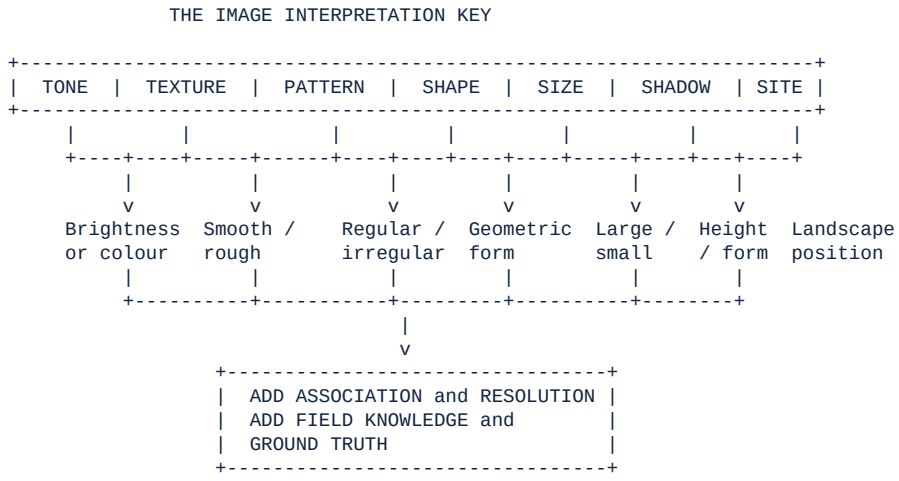
14.7 Elements of image interpretation

Definition. Image interpretation is the act of identifying features on an image and judging their condition, by examining a set of visual elements that together make up the appearance of the feature.

The elements below are the standard set. A single element rarely identifies a feature; the interpreter uses several together, and adds field knowledge.

Element	Meaning	Example on an image
Tone or hue	The brightness or colour of the feature, from black through grey to white, or its colour in a false colour composite	Water is dark in the near infrared; dry bare soil is light; healthy vegetation is bright red in the standard FCC
Texture	The frequency of tonal change in a small area, giving a smooth or rough appearance	A smooth texture for a water surface; a mottled texture for a mixed crop area
Pattern	The spatial arrangement of features in a recognisable design	The rectangular grid of an orchard; the parallel strips of a terraced hillside
Shape	The geometric form of the feature	The rectangle of a field; the curve of a canal; the circle of a tank
Size	The area or dimension of the feature, judged against the scale or against known objects	A large tank against a small pond; a wide canal against a field channel
Shadow	The dark area cast by an object, which indicates its height and shape	The shadow of a tall building or a tree reveals its height and profile
Site	The position of the feature in the landscape	A tank is at the lowest point; a settlement is near a road or a river
Association	The presence of other features that normally occur with the feature	A canal with distributaries and field channels; a settlement with an approach road
Resolution	The smallest object that can be seen, fixed by the sensor	Field boundaries are visible at 5.8 m resolution but not at 56 m resolution
Season and date	The time of year and time of day of the image, which changes the appearance of crops	Paddy fields are flooded and dark at transplanting but bright green later

Table 14.1 The elements of image interpretation.



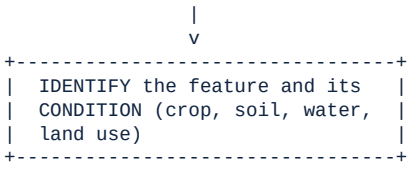


Fig. 14.2 The image interpretation key - the elements used together to identify a feature.

Interpreting the three main land features

- **Water.** Dark tone, especially in the near infrared; smooth texture; sharply defined boundary; shape follows the low ground; association with canals, bunds and field channels.
- **Vegetation.** Bright red in the standard false colour composite; mottled texture; a pattern that follows field boundaries or slope contours; the tone changes with the season.
- **Soil and bare land.** Light to moderate tone; smooth or cloudy texture depending on tillage; a geometric pattern of fields; the tone varies with moisture, so the same field is darker after rain.

Exam focus

Key points to remember

- A basemap provides the spatial framework and reference detail on which other information is plotted.
- The four kinds of basemap are the cadastral map, the topographic sheet, the aerial photograph and the satellite image.
- A cadastral map, at about 1:4,000 to 1:10,000, shows field boundaries and survey numbers but carries no contours.
- A toposheet, mainly at 1:50,000 and 1:25,000, shows relief with contours, drainage, roads and settlements, and is the standard base for medium-scale soil surveys.
- An aerial photograph may be vertical or oblique; for a vertical photograph, scale = f / H, and relief displacement causes tall objects to lean away from the centre. Two overlapping photographs give a stereoscopic three-dimensional view.
- Satellite data products include raw, radiometrically corrected, geometrically corrected, geocoded or orthorectified data, false colour composites and value-added maps.
- The elements of image interpretation are tone, texture, pattern, shape, size, shadow, site, association, resolution and season or date.

Short-answer questions (with model answers)

Q1. What is a basemap and name its four common kinds.

Model answer. A basemap is the map or image that provides the spatial framework and the reference detail on which other information is plotted. The four common kinds are the cadastral map, the topographic sheet, the aerial photograph and the satellite image.

Q2. What is a cadastral map and what is its main limitation?

Model answer. A cadastral map is a large-scale map of a village or group of survey numbers showing field boundaries and survey numbers, prepared for land revenue purposes. Its main limitation is that it carries no contours or elevation information, so slope cannot be judged from it.

Q3. What is a false colour composite and why is it used?

Model answer. A false colour composite is a colour image made by displaying three spectral bands through the red, green and blue channels of a display. In the standard composite for vegetation and soil work, near infrared is shown as red, red as green and green as blue, so that healthy vegetation appears bright red and is easy to distinguish from soil and water.

Q4. Name the elements of image interpretation.

Model answer. Tone or hue, texture, pattern, shape, size, shadow, site, association, resolution, and season or date.

Q5. Define relief displacement.

Model answer. Relief displacement is the leaning of objects away from the centre of a vertical aerial photograph because of their height above the ground. It is caused by the variation of scale with elevation and is why tall objects such as buildings and trees appear to lean outwards.

Long-answer questions (with model answer outlines)

Q1. Describe the four kinds of basemap used in soil and land resource work, and explain the elements of image interpretation.

Outline.

- Define a basemap and state its requirements.
- Cadastral map: scale, content, use and limitation.
- Toposheet: scales, contours, sheet numbering, use and limitation.
- Aerial photograph: vertical and oblique, scale = f / H, relief displacement and stereoscopy.
- Satellite image: pixel, bands, and the standard data products.
- Explain the false colour composite and Bhuvan and the IRS series.
- List and explain the elements of image interpretation with examples.
- Apply the elements to water, vegetation and soil.
- Conclude that interpretation needs several elements together plus field knowledge and ground truth.

Q2. Explain how scale and area are calculated from a toposheet, and work an example.

Outline.

- Explain the representative fraction and the statement of scale.
- Show the conversion of 1:50,000 into metres and kilometres per centimetre.
- Explain the method of finding ground area from map dimensions.
- Work the example: a field 4.0 cm by 3.0 cm at 1:50,000, giving 2,000 m by 1,500 m and 3,000,000 m2.
- Convert to hectares (300 ha) and to acres (about 741 acres).
- Show the short cut using one square centimetre = 25 hectares at 1:50,000.
- State the three useful conversions at 1:10,000, 1:25,000 and 1:50,000.
- Explain how slope is obtained from the contour interval and the horizontal distance between contours.
- Conclude with a Tamil Nadu example using a village cadastral map and a 1:50,000 toposheet.

Lecture 15 Geographic Information System - Definition, Components and Data Models

Soil survey, remote sensing and GPS all produce spatial data - information tied to a location on the earth. A **Geographic Information System** is the tool that stores that data, keeps it linked to its location, and analyses it to answer questions. This lecture defines GIS, lists its components and functions, and explains the two data models on which all GIS data are built: raster and vector.

15.1 GIS - definition

Definition. A Geographic Information System is a computer-based system for capturing, storing, checking, integrating, manipulating, analysing and displaying data that are spatially referenced to the earth.

A shorter exam definition: a GIS is a computer system that combines spatial data with attribute data and provides tools to store, manipulate, analyse and display them for decision-making.

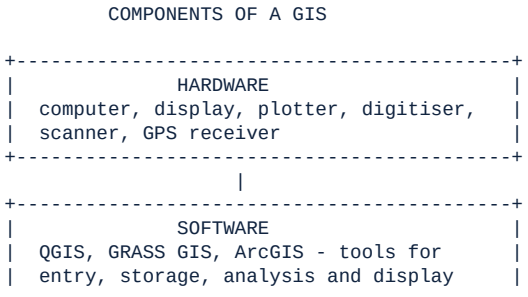
What makes a GIS different from a map

- A map shows the data. A GIS **stores** the data in a database, so it can be queried and updated.
- A GIS keeps **attribute data** - the properties of each feature - linked to its geometry, so a question can be asked of the properties.
- A GIS can **combine layers** - soil, slope, rainfall, land use - and analyse the combination, which a paper map cannot.
- A GIS can **answer questions** such as which fields are both saline and poorly drained, and can produce a new map as the answer.

15.2 Components of a GIS

A working GIS has five components. All five must be present; the absence of any one of them stops the system from working.

1. **Hardware.** The physical equipment - the computer, the display, the printer or plotter, and the devices that bring data in, such as a digitising table or a scanner, and the GPS receiver that supplies positions.
2. **Software.** The programs that carry out the GIS functions - data entry, storage, analysis and display. Examples are open-source packages such as QGIS and GRASS GIS, and commercial packages such as ArcGIS.
3. **Data.** The most important and usually the most expensive component - the spatial data (the geometry of the features) and the attribute data (the properties of the features). Data may come from a soil survey, a toposheet, a satellite image, a GPS survey or a government database.
4. **People.** The users - the surveyor, the analyst, the planner and the decision-maker. A GIS without trained people is only a set of programs.
5. **Methods or procedures.** The rules by which the work is done - the standard operating procedures, the data standards, the accuracy requirements and the analysis workflow. Good methods are what make a GIS produce consistent and defensible results.



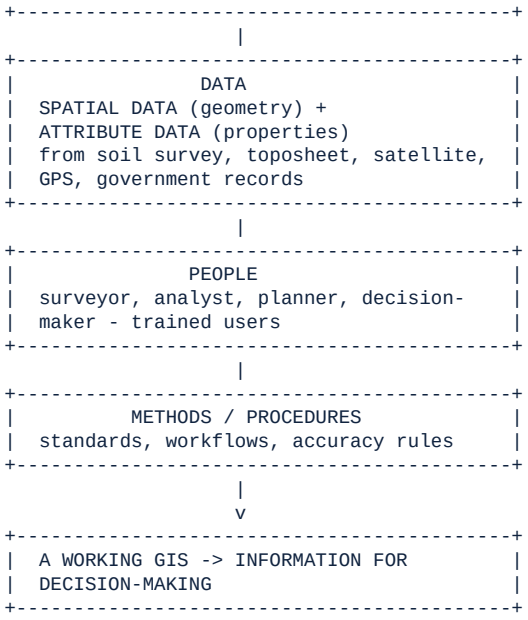


Fig. 15.1 The five components of a Geographic Information System.

15.3 Functions of a GIS

Function	What it does	Example in soil and land work
Data capture or input	Brings spatial data into the system	Digitising a village cadastral map; importing a satellite image; loading GPS points of soil sampling sites
Data storage and management	Stores the data in a database and keeps geometry linked to attributes	A soil map with a table of each mapping unit's depth, texture and pH
Data manipulation and analysis	Combines, converts and analyses layers	Overlaying soil, slope and rainfall to find suitable areas; buffering a canal; reclassifying a land use map
Data query	Answers questions about the data	Which mapping units have a slope greater than 8 per cent and a depth below 50 cm?
Data display and output	Presents the results as maps, tables and reports	Printing a soil suitability map; exporting an area statement by village

Table 15.1 The functions of a GIS with examples from soil and land resource work.

Common spatial analysis operations

- **Overlay.** Two or more layers are combined so that the result carries the properties of all of them - for example soil plus slope plus rainfall to give a suitability map.
- **Buffer.** A zone of a stated width is drawn around a feature - for example a 50 m buffer along a canal to mark the area to be kept free of cultivation.
- **Reclassify.** The values in a layer are regrouped - for example a slope map in degrees is reclassified into slope classes.
- **Interpolation.** Point data are converted into a continuous surface - for example soil pH at sampling points is interpolated into a pH map.
- **Query and select.** Features are selected by their attributes or their location.
- **Area calculation.** The area of each class is computed, giving the area statement that planners need.

15.4 Spatial data and attribute data

- **Spatial data** (also called geometric or locational data) record **where** a feature is - its position, shape and extent, given as coordinates in a known coordinate system.
- **Attribute data** (also called non-spatial or thematic data) record **what** the feature is - its properties, held in a table linked to the feature.
- **Linkage.** The two are joined by a common identifier, so that selecting a polygon on the map selects its row in the attribute table, and the reverse. This linkage is what makes a GIS more than a drawing tool.

15.5 The two data models

15.5.1 The vector data model

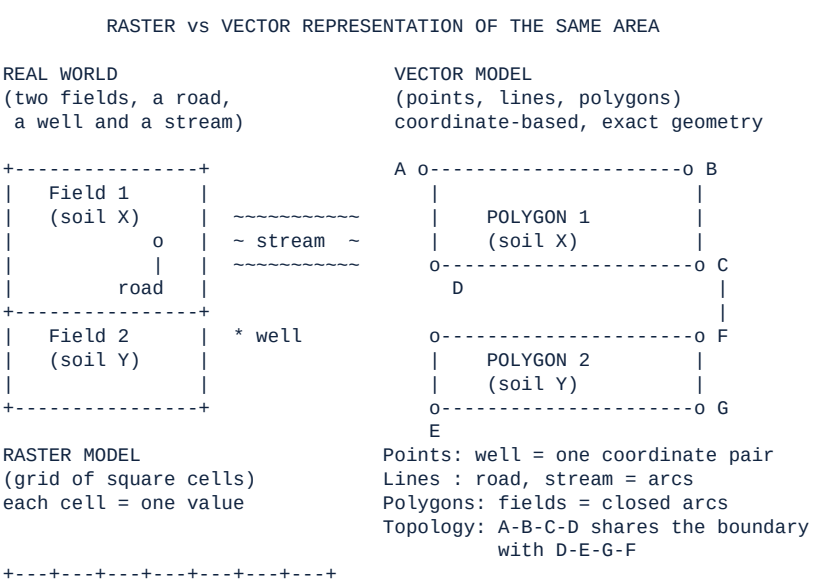
Definition. In the vector data model, spatial features are represented as discrete geometric objects defined by points, lines and polygons, whose positions are given by pairs of coordinates.

- **Point.** A single coordinate pair, used for features too small to have area at the map scale - a well, a soil sampling site, a tree.
- **Line or arc.** A string of connected coordinates, used for linear features - a road, a canal, a stream, a contour.
- **Polygon or area.** A closed line that encloses an area, used for features with extent - a field, a soil mapping unit, a village, a lake.
- **Topology.** The stored record of how the objects are related - which polygons share a boundary, which lines connect at a node, which point lies inside which polygon. Topology allows the GIS to detect errors and to carry out analyses such as finding the neighbours of a polygon.

15.5.2 The raster data model

Definition. In the raster data model, the area is divided into a regular grid of square cells or pixels, and each cell carries a single value that represents the condition of the ground within that cell.

- **Cell or pixel.** The unit of the grid. Its size is the spatial resolution of the raster.
- **Cell value.** A single number per cell, which may be a spectral value, an elevation, a soil code or a class number.
- **Origin and orientation.** The raster is defined by the coordinate of its top-left corner, the cell size and the number of rows and columns.
- **Uses.** Rasters are the natural form for satellite images, elevation models, scanned maps, rainfall surfaces and any continuous variable.



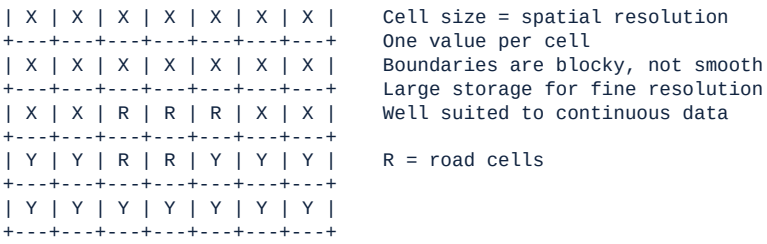


Fig. 15.2 The same area represented in the raster and the vector data models.

15.6 Raster versus vector - comparison

Point of comparison	Raster model	Vector model
Unit of representation	A grid cell or pixel	A point, a line or a polygon
Geometry	Approximate; boundaries are blocky and follow the cell grid	Exact; boundaries follow the true shape of the feature
Data storage	One value per cell; storage grows rapidly as resolution is refined	Coordinates of the vertices; compact for simple features
Data volume	Usually large	Usually smaller for the same feature
Topology	Not stored explicitly; derived from cell position	Stored explicitly; adjacency and connectivity are known
Best suited to	Continuous data - elevation, temperature, rainfall, reflectance, soil depth surface	Discrete data - boundaries, parcels, roads, streams, soil polygons
Source data	Satellite images, scanned maps, interpolated surfaces	Digitised maps, GPS surveys, cadastral records
Analysis strength	Map algebra, overlay of continuous surfaces, filtering, image classification	Network analysis, proximity and buffer, accurate area and length measurement
Accuracy of area and length	Approximate; depends on cell size	High, if the map was accurately digitised
Display quality	Blocky at coarse resolution; can look pixelated	Smooth, cartographic quality lines and boundaries
Typical soil use	A soil depth or slope surface from interpolation; a classified satellite image	A soil map with mapping units as polygons and a linked attribute table

Table 15.2 Comparison of the raster and vector data models.

Choosing a model. Choose raster for continuous data that varies everywhere, such as elevation, rainfall, reflectance or interpolated soil properties. Choose vector for discrete data with definite boundaries, such as fields, soil mapping units, roads and canals. Most real GIS projects use both, and a GIS can convert between them.

15.7 Coordinate systems, projection and scale

- **Geographic coordinates** - latitude and longitude in degrees, measured on the curved surface of the earth.
- **Projected coordinates** - positions on a flat map, in metres or feet, obtained by projecting the curved surface onto a plane. In India, the common systems are the Universal Transverse Mercator and the Survey of India polyconic and Lambert systems.
- **Datum** - the reference model of the shape of the earth used to define coordinates. In India the historical datum is Everest, and the modern global datum is WGS 84.
- **Georeferencing** - assigning real-world coordinates to a map or image so that it can be overlaid on other layers.

- **Why it matters.** Layers can be combined only if they share a coordinate system, datum and units. A mismatch produces layers that do not line up, and therefore wrong results.

15.8 Tamil Nadu illustration

In Tamil Nadu, a typical land resource GIS for a block would hold several layers: a village cadastral boundary layer digitised from cadastral maps; a soil layer with mapping units digitised from the soil survey map and linked to an attribute table of depth, texture, pH and drainage; a slope layer derived from a topographic sheet or a digital elevation model; a land use layer from a Resourcesat LISS-III or LISS-IV image; a rainfall layer interpolated from rain-gauge data; and a groundwater layer from state records. An overlay of soil, slope and rainfall produces a land suitability map; a query on the soil attribute table lists the fields that are saline and poorly drained. The same system supports watershed planning, crop area estimation and the monitoring of land use change.

Exam focus

Key points to remember

- A GIS is a computer-based system for capturing, storing, checking, integrating, manipulating, analysing and displaying spatially referenced data.
- Its five components are hardware, software, data, people, and methods or procedures.
- Its functions are data input, storage and management, manipulation and analysis, query, and display and output.
- Spatial data record where a feature is; attribute data record what the feature is; the two are linked by a common identifier.
- The vector model represents features as points, lines and polygons defined by coordinates, with topology stored explicitly.
- The raster model divides the area into a grid of cells, each carrying one value; the cell size is the spatial resolution.
- Vector suits discrete features with definite boundaries; raster suits continuous data such as elevation, rainfall and reflectance.
- Layers can be combined only if they share a coordinate system, datum and units.

Short-answer questions (with model answers)

Q1. Define a Geographic Information System.

Model answer. A Geographic Information System is a computer-based system for capturing, storing, checking, integrating, manipulating, analysing and displaying data that are spatially referenced to the earth.

Q2. Name the five components of a GIS.

Model answer. Hardware, software, data, people, and methods or procedures.

Q3. Distinguish spatial data from attribute data.

Model answer. Spatial data record where a feature is - its position, shape and extent in coordinates. Attribute data record what the feature is - its properties, held in a table. The two are linked by a common identifier so that selecting a feature selects its attributes.

Q4. What is the vector data model? Name its three basic objects.

Model answer. In the vector model, spatial features are represented as discrete geometric objects defined by coordinates. Its three basic objects are the point, used for features too small to have area; the line or arc, used for linear features; and the polygon, used for features with extent.

Q5. What is a raster and what is its cell size?

Model answer. A raster is a representation in which the area is divided into a regular grid of square cells, each carrying a single value. The cell size is the spatial resolution of the raster - the ground area that one cell

represents.

Q6. What is topology in a GIS?

Model answer. Topology is the stored record of how spatial objects are related - which polygons share a boundary, which lines connect at a node, and which point lies inside which polygon. It allows the GIS to detect errors and to carry out analyses such as finding neighbours.

Long-answer questions (with model answer outlines)

Q1. Define GIS, explain its components and functions, and describe the raster and vector data models.

Outline.

- Define GIS and state how it differs from a map.
- Explain the five components: hardware, software, data, people and methods.
- Draw the components diagram.
- Explain the functions: input, storage, analysis, query and output.
- Describe common analysis operations - overlay, buffer, reclassify, interpolation, query and area calculation.
- Explain spatial data and attribute data and their linkage.
- Describe the vector model with its points, lines, polygons and topology.
- Describe the raster model with its cells and cell values.
- Draw the raster versus vector diagram.
- Give a Tamil Nadu illustration of a block-level land resource GIS.

Q2. Compare the raster and vector data models and state when each is used.

Outline.

- Define both models.
- Compare the unit of representation.
- Compare the geometry and the accuracy of boundaries.
- Compare storage and data volume.
- Compare topology.
- Compare the kind of data each suits - continuous versus discrete.
- Compare the source data and the analysis strengths.
- Compare display quality.
- State the rule for choosing a model.
- Give a soil-survey example of each, and note that a GIS converts between them.

Lecture 16 GPS and GNSS - Functions, Segments, Working Principle and Errors

A GIS needs positions, and a soil survey needs to record exactly where each observation was made. The **Global Positioning System** and the wider family of **Global Navigation Satellite Systems** provide those positions anywhere on the earth, at any time, in all weather. This lecture explains what they do, how they are built, how they work, and what makes them inaccurate.

16.1 Definitions

GNSS - Global Navigation Satellite System. The general term for any satellite system that provides global, autonomous position and time information to users with a suitable receiver. The major systems are the American GPS, the Russian GLONASS, the European Galileo, the Chinese BeiDou and the Indian NavIC.

GPS - Global Positioning System. The satellite navigation system developed and operated by the United States, which is the most widely used GNSS. It provides three-dimensional position (latitude, longitude and altitude), velocity and precise time, continuously, anywhere on or near the earth.

What a GPS provides

- **Position** - latitude, longitude and, with a good fix, altitude.
- **Velocity** - the speed and direction of movement of the receiver.
- **Time** - precise time, from the atomic clocks carried on the satellites.
- **Navigation** - guidance from one point to another, with distance and bearing.

16.2 Functions of GPS and GNSS

- **Positioning.** To determine the position of a point on the earth's surface to a known accuracy.
- **Navigation.** To guide a person or a vehicle from one place to another along a route.
- **Surveying.** To fix the coordinates of ground control points, soil sampling sites, profile pits and boundaries.
- **Mapping and GIS data capture.** To record the position of features such as roads, canals, field boundaries and problem areas, and to bring them into a GIS.
- **Timing.** To supply precise time for telecommunications, power grids and scientific work.
- **Machine guidance.** To guide tractors and harvesters for precision agriculture, and to guide aircraft and ships.
- **Monitoring.** To track the movement of vehicles, animals, tectonic plates and glaciers, and to monitor crop and soil operations.

16.3 Segments of GPS

GPS has three segments. Together they produce the signal that reaches the receiver.

Space segment

- The constellation of satellites in orbit, each carrying an atomic clock and a radio transmitter.
- The nominal GPS constellation has 24 operational satellites, arranged in six orbital planes at an inclination of about 55 degrees, at an altitude of about 20,200 km, with an orbital period of about 12 hours.
- The arrangement guarantees that at least four satellites are in view from any point on the earth at any time.
- Each satellite broadcasts a navigation message giving its position (ephemeris), the approximate positions of all satellites (almanac), clock correction data and system status.

Control segment

- The ground-based network that monitors and controls the satellites.
- It consists of a **master control station**, a number of **monitor stations** distributed around the world, and **ground antennas** that uplink commands.
- The monitor stations track the satellites and collect ranging data; the master control station computes the orbit and clock corrections; the ground antennas upload the corrected navigation message to each satellite.
- The control segment is what keeps the satellite clocks and orbits accurate.

User segment

- The receivers owned and operated by users - handheld units, mobile phone chips, survey-grade receivers, vehicle trackers and machine guidance systems.
- The receiver receives the satellite signals, measures the travel time of each signal, computes the distance to each satellite and solves for its own position.
- The user segment is passive - the receiver only listens; it does not transmit.

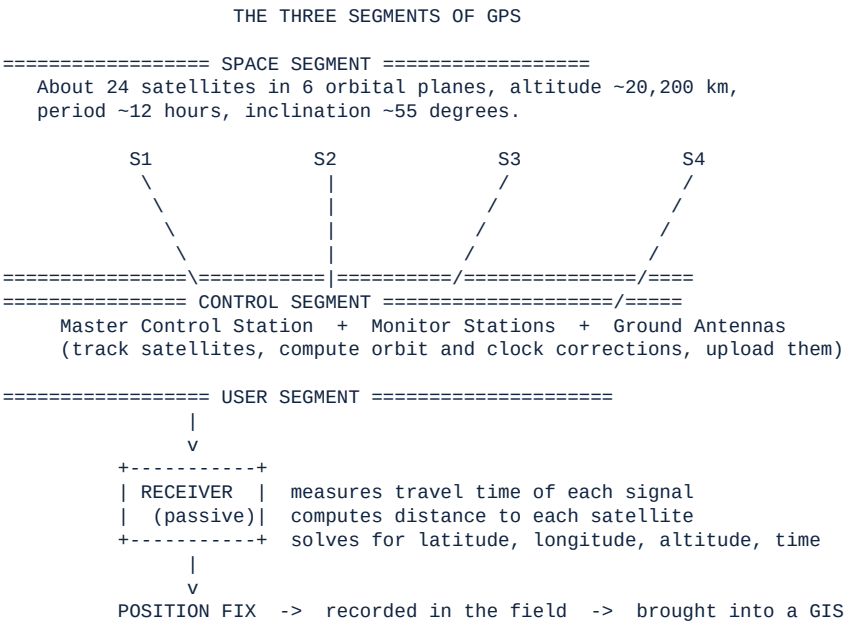


Fig. 16.1 The space, control and user segments of GPS.

Segment	What it contains	Function
Space segment	About 24 satellites in six orbital planes at about 20,200 km altitude, each with an atomic clock and a transmitter	Broadcasts the navigation signal and the navigation message giving position, clock and system data
Control segment	Master control station, worldwide monitor stations and ground uplink antennas	Tracks the satellites, computes orbit and clock corrections, and uploads the corrected navigation message
User segment	Receivers in handheld units, phones, survey instruments and vehicles	Receives the signals, measures travel time, computes position, velocity and time, and stores the data

Table 16.1 The three segments of GPS and their functions.

16.4 Working principle

GPS works on the principle of **trilateration**, also called trilateration by ranging: if the distance to a known point is known, the receiver must lie somewhere on a sphere around that point; the distance to a second known point narrows it to a circle; and the distance to a third narrows it to two points, one of which is the receiver's position.

How the distance is measured

1. Each satellite broadcasts a coded signal, and the moment of transmission is stamped on the signal using the satellite's atomic clock.
2. The receiver generates the same code internally and compares the incoming code with its own copy, to find how much the incoming code is delayed.
3. The delay is the **travel time** of the signal from the satellite to the receiver.
4. The distance is the travel time multiplied by the speed of light: **distance = travel time x speed of light**. Because the receiver clock is not perfectly synchronised with the satellite clock, this computed distance is called a **pseudorange**.
5. The receiver repeats this for several satellites and solves the simultaneous equations for its three position coordinates and the clock error.

Why at least four satellites are needed

- Three satellites give three distances, which would fix the position in three dimensions if the receiver clock were perfect.
- The receiver clock is a cheap quartz clock, not an atomic clock, so it carries an unknown error, which is a fourth unknown.
- Four unknowns - latitude, longitude, altitude and clock error - require four equations, hence four satellites.
- A fix with four or more satellites is a **three-dimensional fix**; with only three satellites the receiver assumes a known altitude and gives a **two-dimensional fix**.
- More satellites in view improve accuracy, because the solution is over-determined and the errors average out.

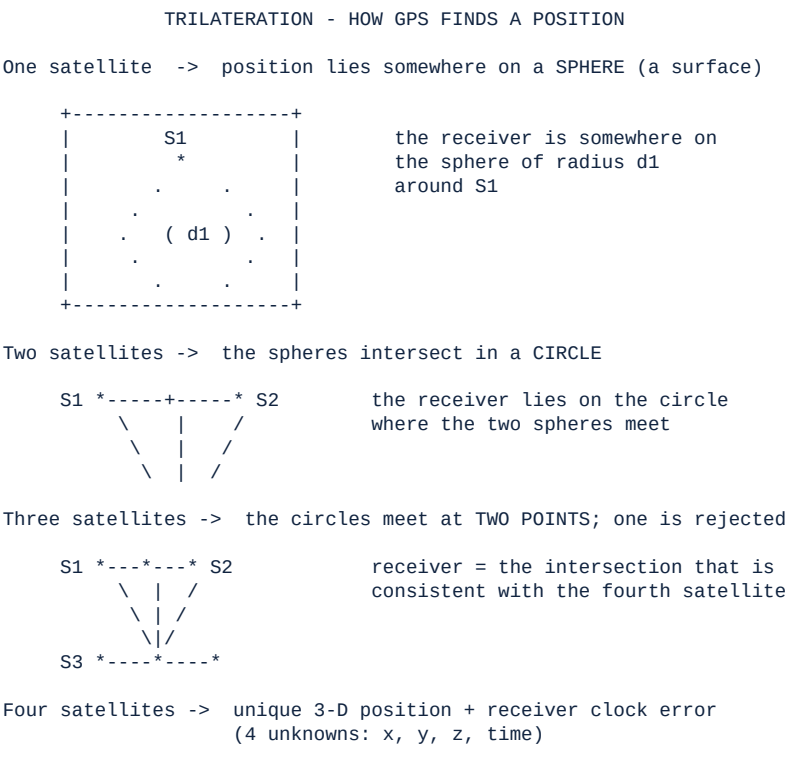


Fig. 16.2 The principle of trilateration and why four satellites are needed.

Signals and observables

- **Carrier frequencies.** GPS broadcasts on the L1 band at 1575.42 MHz, the L2 band at 1227.60 MHz and, on modern satellites, the L5 band at 1176.45 MHz.
- **Codes.** The **C/A code** (coarse acquisition) is the standard civilian code, giving the pseudorange used by ordinary receivers. The **P code** (precise) is used for higher-precision work.
- **Carrier phase.** The receiver can also measure the phase of the carrier wave itself, which is far more precise than the code. Carrier phase measurement is the basis of survey-grade GPS.
- **Navigation message.** The data stream that carries the satellite ephemeris, the almanac, the clock corrections and the satellite health status.

16.5 GPS errors

The accuracy of a GPS position depends on the errors present and on the geometry of the satellites used. The main error sources are listed below.

Error source	Cause	Approximate effect	How it is reduced
Ionospheric delay	The ionosphere slows and bends the radio signal, the delay varying with the free-electron content and the frequency	Several metres, the largest single source	Dual-frequency receivers; ionospheric models; differential correction
Tropospheric delay	Water vapour and dry air in the lower atmosphere slow the signal	About 0.5 to 2 metres	Tropospheric models; differential correction
Satellite clock error	The satellite atomic clock drifts slightly	About 1 to 2 metres	Clock corrections broadcast in the navigation message; differential correction
Ephemeris or orbit error	The broadcast satellite position differs from its true position	About 1 to 2 metres	Precise ephemeris; differential correction
Multipath	The signal reaches the receiver by reflection from buildings, walls, water or the ground, as well as directly	From a few metres to tens of metres	Choose an open site; avoid reflective surfaces; use a choke-ring antenna; average over time
Receiver noise	Electronic noise and code-tracking error in the receiver	About 0.3 to 1 metre	Better receiver; carrier-phase measurement; longer observation
Satellite geometry (DOP)	Poor distribution of satellites in the sky magnifies every other error	A multiplier on the other errors - GDOP, PDOP, HDOP, VDOP	Plan observation for a time of good geometry; observe when DOP is low
Selective availability	A deliberate degradation of the signal by the system operator, applied historically	Up to about 100 metres when it was in force	Discontinued since 2000; differential correction if it returned
Atmospheric and user errors	Incorrect antenna height, wrong datum or coordinate settings, poor field procedure	Variable, often several metres	Careful field procedure; correct datum and coordinate system

Table 16.2 Sources of GPS error and how each is reduced. The effects are approximate and depend on the receiver, the satellite geometry and the correction method used.

The one error the user controls. Satellite geometry and multipath are the errors most under the user's control. A receiver in an open field, away from walls and water, observing when the satellites are well spread out, gives a much better position than the same receiver in a courtyard between buildings. Good field procedure beats an expensive receiver.

Dilution of precision

- **DOP** is a multiplier that describes how much the satellite geometry magnifies the ranging errors. A low DOP means good geometry and a small error; a high DOP means poor geometry and a large error.
- **GDOP** - geometric dilution of precision, covering position and time.
- **PDOP** - position dilution of precision, covering the three coordinates.
- **HDOP** - horizontal dilution of precision; **VDOP** - vertical dilution of precision. Vertical accuracy is always poorer than horizontal, because all satellites are above the receiver.
- As a rough guide, PDOP below about 4 is good, 4 to 8 is acceptable, and above 8 is poor.

16.6 Improving accuracy - differential and augmented systems

Method	How it works	Typical accuracy
Autonomous GPS	A single receiver, using the broadcast signals alone	About 5 to 15 m
Differential GPS (DGPS)	A base receiver at a known point computes the error and broadcasts a correction to the roving receiver	About 0.5 to 3 m
WAAS and GAGAN	Space-based augmentation: a correction is broadcast from a geostationary satellite, as the Indian GAGAN system does	About 1 to 3 m
Real-Time Kinematic (RTK)	Carrier-phase measurement with a base station and a radio or internet link, giving corrections in real time	About 1 to 5 cm
Post-processed kinematic (PPK)	Carrier-phase data corrected after the survey, in the office	About 1 to 5 cm
Static survey-grade GPS	The receiver occupies the point for a long period and the carrier phase is post-processed	Sub-centimetre to a few centimetres

Table 16.3 GPS accuracy by method. The figures are the usual published ranges and depend on the receiver, the observation time and the satellite geometry.

Indian systems - GAGAN and NavIC

- **GAGAN** (GPS Aided GEO Augmented Navigation) is India's space-based augmentation system, developed by ISRO and the Airports Authority of India. It broadcasts corrections from geostationary satellites, improving GPS accuracy to the metre level over the Indian region, and supports aviation.
- **NavIC** (Navigation with Indian Constellation), formerly IRNSS, is India's own regional satellite navigation system. It provides position service over India and a surrounding region, and is used for terrestrial, aerial and marine navigation, disaster management and vehicle tracking.

16.7 GPS in soil survey - practical use

- Recording the exact position of every profile pit, auger bore and soil sample, so that the point can be revisited.
- Tracing field boundaries, canals, roads and problem areas as lines or polygons, for import into a GIS.
- Navigating back to a previously recorded sampling point, which is essential for monitoring change over time.
- Providing ground control points to georeference a toposheet, an aerial photograph or a satellite image.
- Supporting precision agriculture, where the tractor, the fertiliser spreader and the yield monitor all need the same accurate position.

Exam focus

Key points to remember

- GNSS is the general term for any global satellite navigation system; GPS is the American system and the most widely used GNSS.
- GPS provides position, velocity, precise time and navigation.
- Its three segments are the space segment (about 24 satellites in six planes at about 20,200 km), the control segment (master control station, monitor stations, uplink antennas) and the user segment (receivers).
- GPS works by trilateration: distance = travel time x speed of light, giving a pseudorange to each satellite.
- At least four satellites are needed, because there are four unknowns - latitude, longitude, altitude and receiver clock error.
- Main error sources are ionospheric delay, tropospheric delay, satellite clock error, ephemeris error, multipath, receiver noise and poor satellite geometry (DOP).
- Ionospheric delay is the largest single error source; multipath is the one most under the user's control.
- Differential GPS, GAGAN and RTK improve accuracy; GAGAN and NavIC are the Indian systems.

Short-answer questions (with model answers)

Q1. Define GNSS and GPS.

Model answer. GNSS, Global Navigation Satellite System, is the general term for any satellite system that provides global position and time information. GPS, Global Positioning System, is the American satellite navigation system, and is the most widely used GNSS. It provides three-dimensional position, velocity and precise time anywhere on the earth.

Q2. Name the three segments of GPS.

Model answer. The space segment, the constellation of about 24 satellites carrying atomic clocks and transmitters; the control segment, the master control station, monitor stations and ground antennas that track and correct the satellites; and the user segment, the receivers that receive the signals and compute position.

Q3. Why are at least four satellites needed for a position fix?

Model answer. Because there are four unknowns - latitude, longitude, altitude and the receiver clock error, the last arising because the receiver has a cheap quartz clock rather than an atomic clock. Four unknowns require four equations, hence four satellites. With only three satellites the receiver must assume a known altitude.

Q4. What is multipath error?

Model answer. Multipath error occurs when the satellite signal reaches the receiver by reflection from buildings, walls, water or the ground as well as by the direct path, so the measured travel time is too long and the position is wrong. It is reduced by choosing an open site, avoiding reflective surfaces and averaging over time.

Q5. What is DOP?

Model answer. DOP, dilution of precision, is a multiplier that describes how much the distribution of satellites in the sky magnifies the ranging errors. A low DOP means good geometry and a small error; a high DOP means poor geometry and a large error. Its forms include GDOP, PDOP, HDOP and VDOP.

Q6. Name two Indian satellite navigation systems.

Model answer. GAGAN, the GPS Aided GEO Augmented Navigation system, which broadcasts corrections from geostationary satellites and improves GPS accuracy over India; and NavIC, formerly IRNSS, India's own regional satellite navigation system.

Long-answer questions (with model answer outlines)

Q1. Explain the segments and working principle of GPS.

Outline.

- Define GNSS and GPS and list what a GPS provides.
- Describe the space segment - number of satellites, orbits, altitude, period, signals broadcast.
- Describe the control segment - master control station, monitor stations, uplink antennas, and their function.
- Describe the user segment - receivers, passive listening.
- Draw the three-segment diagram.
- Explain the principle of trilateration with one, two, three and four satellites.
- Explain how the pseudorange is measured - the code, the travel time and the speed of light.
- Explain why four satellites are needed.
- Describe the signals and observables - L1, L2, L5, C/A code, P code and carrier phase.
- Conclude with the practical uses in soil survey.

Q2. Explain the sources of GPS error and the methods used to reduce them.

Outline.

- State that accuracy depends on the errors present and on satellite geometry.
- Ionospheric delay: cause, effect, reduction.
- Tropospheric delay: cause, effect, reduction.
- Satellite clock error and ephemeris error: cause and reduction.
- Multipath: cause, effect, reduction.
- Receiver noise: cause and reduction.
- Dilution of precision: define GDOP, PDOP, HDOP and VDOP and give roughly acceptable values.
- Historical selective availability and its discontinuation.
- Explain DGPS, GAGAN, RTK and static survey-grade GPS with their accuracies.
- Conclude that good field procedure controls multipath and geometry, and that differential correction controls the atmospheric errors.

Lecture 17 Geoinformatics in Agricultural Applications

The earlier lectures built the separate tools - soil survey, remote sensing, GIS and GPS. This lecture brings them together under the name **geoinformatics** and shows what the combination can do for soil and crop studies. The applications below are the ones actually in use in India and in Tamil Nadu.

17.1 Geoinformatics - definition

Definition. Geoinformatics is the science and technology of acquiring, storing, processing, analysing, modelling and presenting geographic information, by integrating remote sensing, Geographic Information Systems, the Global Navigation Satellite System and related disciplines.

In agriculture, geoinformatics provides the spatial intelligence needed to manage land, soil, water and crops, field by field and district by district.

The components of geoinformatics

- **Remote sensing** - provides the imagery from which land, soil and crop conditions are read.
- **Geographic Information System** - provides the database and the analysis tools that combine the layers.
- **Global Navigation Satellite System** - provides the accurate ground positions that tie field observations to the imagery and the GIS.
- **Field survey and soil testing** - provides the ground truth and the soil property data without which the imagery cannot be interpreted.
- **Database and information technology** - provides the storage, processing and delivery of the resulting information.
- **Models and decision tools** - convert the data into forecasts and recommendations, such as yield forecasts and fertiliser prescriptions.

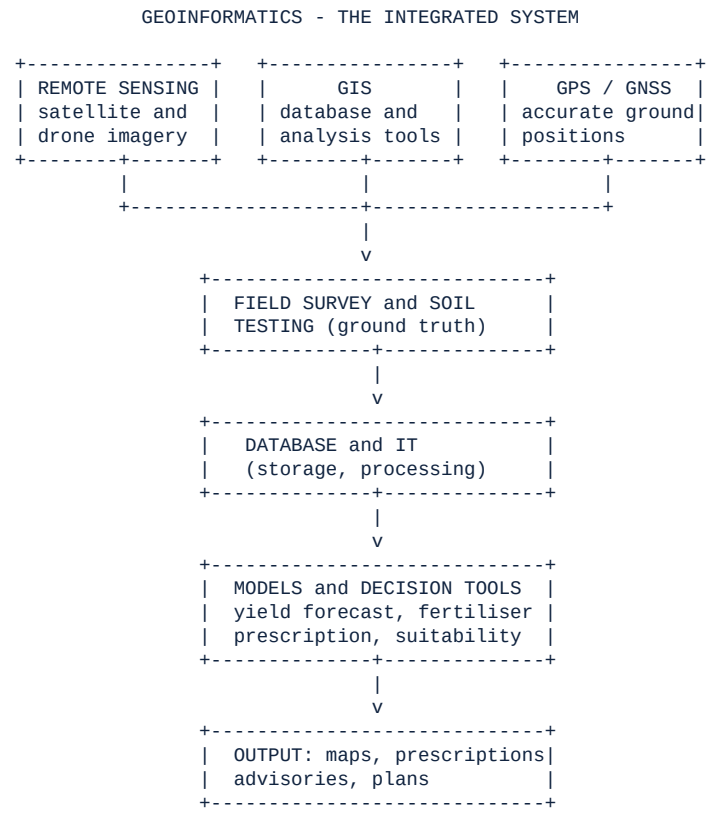


Fig. 17.1 Geoinformatics as the integration of remote sensing, GIS, GPS, field survey and decision tools.

17.2 Geoinformatics in soil studies

1. Soil survey and soil mapping

- Satellite imagery and aerial photographs are interpreted to identify the soil-landscape units, and the boundaries are drawn on the image before the field visit.
- Field observation is then targeted at the boundaries and at representative points, which greatly reduces the number of pits needed.
- The soil map is digitised, the mapping units become polygons in a GIS, and the soil properties of each unit are held in an attribute table.
- GPS records the exact position of every profile pit and sample.
- **Indian example.** The soil resource maps of the National Bureau of Soil Survey and Land Use Planning are prepared by this method, and the data are delivered through Bhuvan and the state soil survey organisations.

2. Soil moisture mapping

- Microwave and thermal infrared data, and optical indices such as the NDMI, are used to map surface soil moisture.
- Radar, such as the RISAT C-band sensor, is particularly useful because the dielectric constant of the soil, and hence the radar backscatter, depends strongly on moisture, and because radar works through cloud.
- The moisture map guides irrigation scheduling and the timing of field operations.

3. Soil erosion assessment and mapping

- The factors of the Universal Soil Loss Equation and its revised version - rainfall erosivity, soil erodibility, slope length and steepness, crop cover and conservation practice - are each prepared as a GIS layer.
- Rainfall erosivity comes from rainfall records; soil erodibility from soil texture and organic matter; the slope factor from a digital elevation model; and the cover factor from a land use map prepared from satellite data.
- Overlaying the layers in the GIS gives an erosion risk map, which directs conservation work to the fields that lose the most soil.

4. Soil salinity and alkalinity mapping

- Salt crusts and bare, bright patches are visible on satellite images, and the brightness rises with salinity in the visible and near-infrared bands.
- Salinity indices built from the visible and near-infrared bands separate saline from normal soil.
- Field measurement of electrical conductivity confirms and calibrates the map.
- **Indian and Tamil Nadu example.** Salinity mapping of the coastal and tail-end areas of the Cauvery delta and of the coastal tracts of Nagapattinam and Ramanathapuram districts supports reclamation planning.

5. Soil fertility and nutrient mapping

- Soil samples are collected on a grid or on a soil map unit basis, and the positions are recorded by GPS.
- The laboratory results - pH, electrical conductivity, organic carbon, available nitrogen, phosphorus and potassium, and micronutrients - are entered as attribute data.
- The values at the sampling points are interpolated into continuous surfaces, giving fertility maps.
- These maps feed the fertiliser recommendation, allowing variable-rate application instead of a uniform dose.

6. Land degradation and wasteland mapping

- Satellite data are used to map gullied and ravine land, waterlogged and marshy areas, saline and alkaline patches, shifting cultivation and barren rocky land.
- The maps prepared in this way are the basis of state and national wasteland atlases.

7. Digital soil mapping

- Digital soil mapping predicts soil properties from environmental covariates - elevation, slope, climate, parent material and vegetation - using statistical models, with the field observations as the training data.
- The output is a continuous surface of the predicted property, with an estimate of the uncertainty, rather than a polygon map.
- It is useful where field observations are few and the soil pattern is closely related to terrain.

17.3 Geoinformatics in crop studies

1. Crop identification and area estimation

- Different crops have different spectral signatures, and the same crop has a different signature at different growth stages.
- Multi-date satellite images are therefore used: a crop is identified from the way its spectral signature changes over the season.
- The classified image gives the area under each crop, which is the basis of the crop production forecast.
- **Indian example.** National crop acreage estimation uses AWiFS, LISS-III and LISS-IV data, and in Tamil Nadu the technique is used for paddy in the Cauvery delta, sugarcane and groundnut.

2. Crop yield and production forecasting

- The crop area from the satellite image is combined with a yield estimate to give the production forecast.
- The yield estimate is obtained from weather models, from the vegetation index and its peak, and from field crop-cutting experiments.
- The forecast supports procurement planning, price policy and food security management.

3. Crop condition and drought monitoring

- The NDVI and similar indices are computed for each period and compared with the long-term normal for the same period.
- A negative departure indicates a stressed or failing crop; a positive departure indicates a good crop.
- The comparison is used for drought declaration, relief planning and advisory services.

4. Crop water stress and irrigation scheduling

- Thermal infrared data give the canopy temperature; a crop that is short of water closes its stomata and warms up, so a rise in canopy temperature relative to the air indicates water stress.
- Microwave and optical data give soil moisture and crop moisture.
- Together they guide the timing and quantity of irrigation, and identify the fields that need water first.

5. Precision agriculture

- GPS-guided tractors, variable-rate fertiliser spreaders and yield monitors allow management to be varied within a field rather than applied uniformly.
- The prescription comes from the fertility, moisture and yield maps held in the GIS.
- **Indian and Tamil Nadu example.** Precision farming demonstrations in Tamil Nadu, including in the delta and in horticultural blocks, use soil test maps, GPS-guided planting and drip and fertigation control.

6. Pest and disease monitoring

- A pest or disease outbreak changes the reflectance of the crop, often before it is visible to the eye, and the change can be detected in the image.
- Early detection allows targeted spraying instead of blanket spraying, saving cost and reducing chemical load.

7. Horticulture and plantation inventory

- High-resolution data from LISS-IV, Cartosat or drones are used to identify and count trees, map orchards and estimate the area under horticultural crops.
- The inventory supports market planning, insurance and subsidy administration.

8. Land suitability and land use planning

- Soil, slope, rainfall, irrigation and market layers are combined in the GIS to produce crop suitability maps.
- The maps guide crop diversification, the location of processing units and the planning of irrigation and watershed development.

9. Watershed and natural resource management

- A watershed GIS holds the boundary, drainage, soil, slope, land use and rainfall layers.
- It is used to plan the location of check dams, percolation ponds, contour bunds, gully control structures and afforestation, and to estimate the runoff and the soil loss.

10. Crop insurance and disaster assessment

- Satellite data allow the extent of flood, drought, cyclone or hailstorm damage to be assessed objectively and quickly, over a large area.
- The assessment supports compensation and insurance settlement, which otherwise depend on slow and uneven field inspection.

17.4 Summary of applications

Area	Application	Main data used	Benefit
Soil	Soil survey and mapping	Satellite image, aerial photo, toposheet, GPS, field pits	Faster survey, fewer pits, a digital soil map in a GIS
Soil	Soil moisture mapping	Microwave radar, thermal and optical data	Better irrigation scheduling and field-operation timing
Soil	Erosion assessment and mapping	Rainfall, soil, digital elevation model and land use layers	Conservation directed to the fields that lose the most soil
Soil	Salinity and alkalinity mapping	Visible and near-infrared imagery, field electrical conductivity	Reclamation planning for the delta tail-ends and the coast
Soil	Fertility and nutrient mapping	GPS-located soil samples, laboratory results, interpolation	Fertility maps and variable-rate fertiliser prescription
Soil	Land degradation and wasteland mapping	Multi-season satellite data	Wasteland atlas and reclamation priority
Crop	Crop identification and area estimation	Multi-date multispectral imagery	Crop area and production forecast
Crop	Yield and production forecasting	Crop area, vegetation index, weather data, crop-cutting experiments	Procurement and price planning
Crop	Crop condition and drought monitoring	NDVI time series compared with the normal	Drought declaration and relief planning
Crop	Water stress and irrigation scheduling	Thermal infrared canopy temperature, soil moisture	Timely and precise irrigation

Area	Application	Main data used	Benefit
Crop	Precision agriculture	GPS, yield monitor, fertility and moisture maps	Variable-rate inputs, lower cost, higher efficiency
Crop	Pest and disease monitoring	High-resolution and multi-date imagery	Early detection and targeted spraying
Crop	Horticulture and plantation inventory	LISS-IV, Cartosat and drone imagery	Tree counts, orchard area, market and subsidy planning
Both	Land suitability and land use planning	Soil, slope, rainfall, irrigation and market layers	Crop diversification and development planning
Both	Watershed and natural resource management	Watershed boundary, drainage, soil, slope, land use layers	Structure layout and runoff and soil loss estimation
Both	Crop insurance and disaster assessment	Satellite imagery before and after the event	Objective, rapid damage assessment

Table 17.1 Applications of geoinformatics in soil and crop studies.

17.5 Tamil Nadu illustration

In Tamil Nadu, geoinformatics is used at three levels. At the state level, Resourcesat AWiFS and LISS-III data support the seasonal assessment of paddy area in the Cauvery delta and of sugarcane, groundnut and cotton in the districts, and the state wasteland and land use maps are prepared from satellite data. At the district and block level, soil survey maps, slope and land use layers are combined in a GIS to prepare land capability and land suitability maps, and to plan watershed development. At the field level, GPS-guided precision farming demonstrations use soil test maps and drip and fertigation control to vary inputs within the field. Across all three levels the same chain is followed: imagery and survey give the data, the GIS stores and analyses it, the GPS fixes the ground positions, and the model turns the result into advice.

17.6 Limitations to remember

- Geoinformatics supplements field observation; it does not replace the soil pit or the crop-cutting experiment.
- Optical imagery is blocked by cloud in the monsoon, so radar or multi-season data may be needed.
- Coarse spatial resolution hides small features, so the resolution must match the purpose.
- Interpretation needs trained people and reliable ground truth, or the maps will be plausible but wrong.
- Data purchase, processing and skilled staff carry a cost that must be justified by the benefit.

Exam focus

Key points to remember

- Geoinformatics is the science and technology of acquiring, storing, processing, analysing, modelling and presenting geographic information by integrating remote sensing, GIS and GNSS.
- Its components are remote sensing, GIS, GNSS, field survey and soil testing, database and information technology, and models and decision tools.
- In soil studies it supports soil survey and mapping, soil moisture mapping, erosion assessment, salinity and alkalinity mapping, fertility mapping, land degradation mapping and digital soil mapping.
- In crop studies it supports crop identification and area estimation, yield forecasting, crop condition and drought monitoring, water stress and irrigation scheduling, precision agriculture, pest and disease monitoring, horticulture inventory, suitability mapping, watershed management and disaster assessment.
- Multi-date imagery is used for crop identification because a crop's spectral signature changes over the season.

- Salinity raises reflectance, so salt-affected soil appears bright in the visible and near-infrared bands.
- Geoinformatics supplements but does not replace field observation, and its accuracy depends on resolution and ground truth.

Short-answer questions (with model answers)

Q1. Define geoinformatics.

Model answer. Geoinformatics is the science and technology of acquiring, storing, processing, analysing, modelling and presenting geographic information, by integrating remote sensing, Geographic Information Systems, the Global Navigation Satellite System and related disciplines.

Q2. Name four components of geoinformatics.

Model answer. Any four of: remote sensing, Geographic Information System, Global Navigation Satellite System, field survey and soil testing, database and information technology, and models and decision tools.

Q3. How is remote sensing used in soil survey?

Model answer. Satellite imagery and aerial photographs are interpreted to identify the soil-landscape units and to draw the boundaries before the field visit. Field observation is then targeted at the boundaries and at representative points, which greatly reduces the number of pits needed, and the soil map is digitised into a GIS.

Q4. Why are multi-date satellite images used for crop identification?

Model answer. Because a crop's spectral signature changes over the season as it germinates, grows, matures and dries. Different crops change at different times, so the pattern of change over several dates identifies the crop more reliably than a single image.

Q5. How is soil salinity mapped from satellite data?

Model answer. Salt crusts and bare, bright patches raise the reflectance of the soil in the visible and near-infrared bands, so saline areas appear bright. Salinity indices built from these bands separate saline from normal soil, and field measurement of electrical conductivity confirms and calibrates the map.

Q6. State one limitation of geoinformatics in agriculture.

Model answer. Optical satellite imagery is blocked by cloud during the monsoon, so radar or multi-season data may be needed; alternatively, coarse spatial resolution hides small features, so the resolution must match the purpose.

Long-answer questions (with model answer outlines)

Q1. Explain the applications of geoinformatics in soil studies.

Outline.

- Define geoinformatics and list its components.
- Soil survey and mapping: image interpretation before the field visit, targeted observation, digitisation into a GIS, GPS fixing of pits.
- Soil moisture mapping: microwave, thermal and optical data, and the role of radar.
- Erosion assessment: the factors of the soil loss equation as GIS layers and their overlay.
- Salinity and alkalinity mapping: brightness of salt crusts, salinity indices and field confirmation.
- Fertility and nutrient mapping: GPS-located samples, laboratory results, interpolation into surfaces.
- Land degradation and wasteland mapping.
- Digital soil mapping: prediction from covariates with uncertainty.
- Give Tamil Nadu examples from the delta and the coastal districts.
- Conclude that geoinformatics makes the soil survey faster and the result digital, but does not replace the soil pit.

Q2. Explain the applications of geoinformatics in crop studies.

Outline.

- Define geoinformatics briefly.
- Crop identification and area estimation using multi-date imagery.
- Yield and production forecasting from area, index and weather.
- Crop condition and drought monitoring by comparing the NDVI with the normal.
- Water stress and irrigation scheduling using thermal infrared and soil moisture.
- Precision agriculture with GPS, variable-rate application and yield monitoring.
- Pest and disease monitoring by early spectral change.
- Horticulture and plantation inventory from high-resolution data.
- Land suitability mapping and watershed management.
- Crop insurance and disaster assessment.
- Give the Tamil Nadu example of paddy area estimation in the Cauvery delta and precision farming demonstrations.
- Conclude with the limitations - cloud, resolution, cost and the need for ground truth.

Glossary

The terms below are those used in this book and in the SAC 201 syllabus. The definition given is the exam-oriented one.

A

- **Absorption (atmosphere).** The transfer of energy from radiation to atmospheric molecules, so that the radiation is removed from the beam.
- **Active remote sensing.** Remote sensing in which the sensor carries its own energy source, as radar and LiDAR do.
- **Aerial photograph.** A photograph of the ground taken from an aircraft, vertical or oblique, used as a basemap and for interpretation.
- **Alfisols.** A soil order of moderately weathered soils with an argillic horizon and a base saturation above 35 per cent; common in the red soil uplands of Tamil Nadu.
- **Attribute data.** The non-spatial data that record what a feature is, held in a table linked to the feature's geometry.
- **Auger bore.** A boring made with a hand auger to examine soil at successive depths quickly, without digging a pit.
- **Azimuth.** The horizontal angle of a direction measured clockwise from north.

B

- **Base map.** The map or image that provides the spatial framework and reference detail on which other information is plotted.
- **Bhaskara.** India's first experimental remote sensing satellites, which demonstrated satellite data for earth resources.
- **Bhuvan.** The Indian geo-platform of ISRO for visualising and analysing satellite imagery and thematic maps.
- **Blackbody.** An ideal emitter and absorber of radiation, used as the reference in radiation laws.

C

- **Cadastral map.** A large-scale village map showing field boundaries and survey numbers, prepared for land revenue purposes.
- **Capability class.** One of eight classes in Land Capability Classification, showing the degree of limitation on sustained agricultural use.
- **Capability subclass.** A subdivision of a capability class by the kind of limitation - e erosion, w wetness, s soil, c climate.
- **Carrier phase.** The phase of the GPS carrier wave, measured by a receiver; it is far more precise than the code and is the basis of survey-grade GPS.
- **Catena.** A sequence of related soils on a slope, from summit to valley, formed under the same climate and from similar parent material but differing because of relief and drainage.
- **Consociation.** A soil mapping unit that is dominantly one kind of soil with only minor inclusions.
- **Contour.** A line on a map joining points of equal elevation.
- **Control section.** The fixed depth interval of the soil profile used for a particular classification or evaluation decision; for the texture, mineralogy, CEC and reaction of a soil family it is 25-100 cm.

- **Correlation (soil).** The process of deciding whether soils in different places are the same or different, and giving each kind of soil one agreed name.
- **Crop acreage estimation.** Estimating the area under a crop from satellite imagery, usually multi-date.

D

- **DGPS.** Differential GPS, in which a base receiver at a known point computes the error and broadcasts a correction to the roving receiver.
- **Digital elevation model.** A raster of ground elevation values, from which slope and aspect maps are derived.
- **Digital soil mapping.** Predicting soil properties from environmental covariates using statistical models, with field observations as training data.
- **Dilution of precision (DOP).** A multiplier describing how much the distribution of satellites in the sky magnifies ranging errors; low DOP is good.
- **Detailed soil survey.** A large-scale survey, usually 1:5,000 to 1:10,000, mapping soil series and phases for farm and watershed planning.
- **Detailed-reconnaissance soil survey.** An intermediate survey, usually about 1:50,000 to 1:100,000, detailed along traverses and reconnaissance between them.

E

- **Electromagnetic radiation.** Energy travelling as oscillating electric and magnetic fields at the speed of light, with a dual wave and particle nature.
- **Electromagnetic spectrum.** The continuous range of all electromagnetic radiation from gamma rays to radio waves.
- **Entisols.** A soil order of weakly developed soils with little or no horizon development; common in recent alluvium and coastal sands.
- **Ephemeris.** The precise orbital position of a satellite at a given time, broadcast in the GPS navigation message.
- **False colour composite.** A colour image made by displaying three spectral bands through the red, green and blue channels of a display.
- **Fertility Capability Classification.** A technical classification of soils for fertility management, using surface and subsoil texture plus fertility modifiers.

G

- **GAGAN.** The Indian GPS Aided GEO Augmented Navigation system, a space-based augmentation service that improves GPS accuracy over India.
- **Geographic Information System.** A computer-based system for capturing, storing, checking, integrating, manipulating, analysing and displaying spatially referenced data.
- **Geoinformatics.** The science and technology of acquiring, storing, processing, analysing, modelling and presenting geographic information by integrating remote sensing, GIS and GNSS.
- **Georeferencing.** Assigning real-world coordinates to a map or image so that it can be overlaid on other layers.
- **GNSS.** Global Navigation Satellite System, the general term for any satellite system providing global position and time.
- **GPS.** Global Positioning System, the American satellite navigation system, and the most widely used GNSS.

- **Grid survey.** A method of soil survey in which the area is divided into a regular grid and the soil is examined at grid points.
- **Ground truth.** Field observation used to check and calibrate the interpretation of a remotely sensed image.

H

- **Horizon.** A layer within the soil profile, roughly parallel to the surface, differing from the layers above and below in one or more properties.
- **Hyperspectral.** A sensor recording many narrow contiguous bands, giving a near-continuous spectrum for each pixel.

I

- **Image interpretation.** Identifying features on an image and judging their condition, using the elements of tone, texture, pattern, shape, size, shadow, site and association.
- **Inclusion.** A soil or non-soil body inside a mapping unit, too small to be shown separately at the survey scale.
- **Inceptisols.** A soil order of soils with weak to moderate horizon development; common in the delta alluvium of Tamil Nadu.
- **Ionospheric delay.** The slowing and bending of a GPS signal by the ionosphere; the largest single source of GPS error.
- **IRS.** Indian Remote Sensing satellite series, operated by ISRO, providing the main satellite data used in India.

L

- **Land capability classification.** A system of grouping land into eight classes by the degree of limitation on sustained agricultural use.
- **Land irrigability classification.** A system of grouping land by its suitability for sustained irrigation development, in six classes.
- **Land suitability classification.** The assessment of land for a specified use by matching the requirements of the use against the qualities of the land; orders S and N, classes S1, S2, S3, N1 and N2.
- **Land use planning.** The systematic assessment of land and its resources and the choice and design of land uses and management practices that best meet people's needs while conserving the resource.
- **LISS.** Linear Imaging Self-Scanning sensor, carried on the IRS Resourcesat satellites; LISS-III at 23.5 m and LISS-IV at 5.8 m resolution.
- **LiDAR.** An active sensor that sends laser pulses and measures the return time, giving precise elevation and vegetation height.

M

- **Mapping unit.** A delineated area on a soil map, identified by a symbol and described in the legend, containing one or more soils within stated limits of purity.
- **Mie scattering.** Scattering by particles about the size of the wavelength, such as dust and smoke; dominant in the lower atmosphere.
- **Minimum legible area.** The smallest area that can be shown on a map at a given scale.
- **Miscellaneous land type.** A non-soil or unproductive area shown separately on a soil map, such as rock outcrop, gullied land or marsh.

- **Multipath.** A GPS error caused by the signal reaching the receiver by reflection as well as by the direct path.
- **Multispectral.** A sensor recording several separate wavelength bands, giving colour information.

N

- **NavIC.** Navigation with Indian Constellation, formerly IRNSS, India's own regional satellite navigation system.
- **NDVI.** Normalised Difference Vegetation Index, (NIR - RED) / (NIR + RED), a measure of the amount and vigour of green vegetation.
- **Near infrared.** The wavelength region of about 0.7 to 1.3 µm, where healthy vegetation reflects strongly and water absorbs strongly.
- **Non-selective scattering.** Scattering by particles much larger than the wavelength, such as cloud and fog droplets; it affects all visible wavelengths equally.

O

- **Orthorectification.** Correction of an image for terrain displacement so that positions are accurate on hilly ground.
- **Overlay.** A GIS operation combining two or more layers so that the result carries the properties of all of them.

P

- **Panchromatic.** A single broad band giving a black-and-white image of high spatial detail.
- **Passive remote sensing.** Remote sensing that measures radiation from an external source, usually the sun, or radiation emitted by the object.
- **Path radiance.** Scattered radiation reaching the sensor without touching the ground; it adds a haze to the image.
- **Pedon.** The smallest three-dimensional body of soil that can be classified, about 1-10 m2 in area; the unit of sampling and classification.
- **Phase (soil).** A subdivision of a soil series made on a feature that affects use but is not part of the soil itself, such as slope, erosion, depth or salinity.
- **Polypedon.** A group of contiguous similar pedons, about 1-10 ha in area; the unit of mapping.
- **Precision agriculture.** Management in which inputs are varied within a field using GPS, sensors and prescription maps.
- **Pseudorange.** The satellite-to-receiver distance computed from the signal travel time, uncorrected for the receiver clock error.
- **Pushbroom scanner.** A scanner using a linear array of detectors that records a whole line at once as the platform moves.

R

- **Radar.** An active microwave sensor that sends a pulse and measures the strength and time of the echo; it works through cloud.
- **Raster.** A data model in which the area is divided into a regular grid of cells, each carrying a single value.
- **Rayleigh scattering.** Scattering by gas molecules much smaller than the wavelength, affecting short wavelengths most and making the sky blue.

- **Reconnaissance soil survey.** A rapid small-scale survey, usually 1:100,000 to 1:250,000 or smaller, mapping associations and broad groups.
- **Red edge.** The steep rise in reflectance between about 0.68 and 0.75 µm, sensitive to chlorophyll content and plant stress.
- **Reflectance.** The ratio of the energy reflected by a surface to the energy incident on it, at a stated wavelength.
- **Relief displacement.** The leaning of tall objects away from the centre of a vertical aerial photograph.
- **Resolution (sensor).** Spatial, spectral, temporal or radiometric - the smallest ground area, the number and width of bands, the revisit period, or the number of intensity levels.
- **RTK.** Real-Time Kinematic GPS, using carrier phase with a base station to give centimetre accuracy in real time.

S

- **Scattering.** The redirection of radiation by atmospheric particles and molecules, without absorption.
- **Segments of GPS.** The space segment (satellites), the control segment (master control and monitor stations) and the user segment (receivers).
- **Semi-detailed soil survey.** A survey of medium to large scale, usually 1:20,000 to 1:50,000, mapping series, associations and complexes for project planning.
- **Sensor.** The instrument that detects and records electromagnetic radiation.
- **Simpson's rule.** A numerical method of estimating the area of an irregular shape from equally spaced ordinates.
- **Soil association.** A mapping unit of two or more soils in a regular, landscape-related repeating pattern.
- **Soil complex.** A mapping unit of two or more soils in an irregular, unpredictable pattern.
- **Soil correlation.** See correlation (soil).
- **Soil individual.** The smallest naturally occurring three-dimensional body of soil that can be described and classified as one kind of soil.
- **Soil profile.** A vertical section of the soil from the surface to the parent material, showing all its horizons.
- **Soil resource inventory.** The systematic survey, identification, classification, mapping and evaluation of the soils of an area.
- **Soil series.** A group of soils with similar horizons, similar control-section properties and a similar parent material; the basic taxonomic and mapping unit, named after a locality.
- **Soil survey.** The systematic examination, description, classification and mapping of soils.
- **Soil systematics.** The naming, describing, classifying and correlating of soils.
- **Spatial data.** The data that record where a feature is - its position, shape and extent in coordinates.
- **Spectral signature.** The characteristic pattern of reflectance or emission of a surface across wavelengths.
- **Standard soil survey.** A soil survey carried out by a uniform accepted procedure, producing comparable maps and reports.
- **Storie index.** A numerical rating of the suitability of a soil for general agricultural use, computed as A x B x C x X x 100.
- **Subclass.** A subdivision showing the kind of limitation, in land capability, irrigability and suitability classification.

- **Sun-synchronous orbit.** A near-polar orbit in which the satellite crosses each latitude at about the same local solar time.
- **SWIR.** Short-wave infrared, about 1.3 to 3.0 µm, sensitive to soil and plant moisture and to clay and carbonate minerals.

T

- **Tone.** The brightness of a feature on an image, from black through grey to white.
- **Topographic sheet.** A Survey of India map showing relief, drainage, roads and settlements at a stated scale, with contours.
- **Topology.** The stored record in a GIS of how spatial objects are related - adjacency, connectivity and containment.
- **Traversing.** The planned movement of a surveyor across an area, examining the soil at intervals to fix soil boundaries.
- **Trilateration.** The principle of fixing a position from distances to known points, used by GPS.
- **Vari.** Variant, a local subdivision of a soil series based on a use-related difference too small to define a new series.
- **Vector.** A data model in which features are represented as points, lines and polygons defined by coordinates.
- **Vertisols.** A soil order of clayey soils that swell and shrink, forming deep cracks on drying; the black cotton soils of Tamil Nadu.
- **Whiskbroom scanner.** A scanner in which a rotating mirror sweeps across the ground line by line.
- **Wien's displacement law.** The wavelength of peak emission of a blackbody is inversely proportional to its absolute temperature.

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Where this book gives a numerical range rather than a single value, the range is the standard published range from the sources listed above, and the exact value for any particular survey is fixed by the specification of that survey. For any examination, the definitions and the classification tables given in the lecture chapters are the items to reproduce.